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Performance, Safety, and Engineering Trade-offs of Rust and C++ in Nokia RAN Control-Plane Systems

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Context. Radio Access Network (RAN) control-plane software is written mostly in C++ for performance, but C++ carries well-known memory-safety and concurrency risks. Rust promises comparable performance with safety guaranteed at compile time, yet its fit for latency-critical telecom control planes is unproven. The work is done at Nokia on a subsystem modelled on a real gNB-CU-CP component.

Objective. The thesis compares modern C++ and Rust on a UE Context Management service. As similar average throughput is already known, it targets four less-explored dimensions: tail latency and jitter under overload; which C++ defect classes Rust prevents at compile time; the overhead of Rust's safety model; and the cost of incremental adoption in a C++ codebase.

Methods. A language-agnostic specification is frozen first; two functionally equivalent implementations are then built with identical algorithms, concurrency model, allocator, and build settings. A synthetic workload is swept to overload, both versions pass the same parity tests, and evaluation adds a fault-injection study and an FFI prototype.

Results. All five injected memory-safety and data-race defect classes are prevented at compile time in safe Rust; in C++ each compiles and is caught, if at all, only at runtime under the right sanitiser. Two boundary classes, a lock-order deadlock and a lost update, misbehave in both. The FFI seam keeps the core `unsafe`-free at a cost bounded at 10.91% of one operation. On performance the answer divides at saturation. Below it the two are equivalent where it matters, and an end-to-end operation is only 0.250% language work and the rest cross-thread handoff, so the sweep bounds the design rather than the language; measured apart, Rust's operation is 1.6× faster and its process a few MiB larger. Past saturation they diverge sharply and in the other direction: C++ sustains 1.8× the throughput of Rust, in 20/20 interleaved repeats under both allocators, while drawing more CPU per completed operation. The far tail below saturation remains unresolved on a virtualised host.

Conclusions. The thesis delivers a reproducible benchmarking framework, a parity-tested pair of implementations, a defect-class taxonomy with its boundary, and a quantified migration cost. The barrier to adopting Rust here is engineering effort and the learning curve rather than run-time performance at nominal load, and what is gained is freedom from data races rather than from concurrency defects generally. The overload result qualifies that: for a service that must degrade gracefully, the deciding factor is not the safety model but the standard library's synchronisation primitives, which a comparison stopping at nominal load would never see.

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Rust, C++, RAN control plane, fault injection, FFI

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1 Introduction

The mobile network is undergoing a profound softwarization. Functions that were once delivered by fixed-function, vertically integrated hardware are being re-implemented as software running on commodity servers, and the Radio Access Network (RAN), long the most specialised part of the system, is no longer an exception. In the 5G NG-RAN architecture the base station (gNB) is disaggregated into a central unit and one or more distributed units, and the central unit is split again into control- and user-plane functions [1, 2]. Open-RAN initiatives push this disaggregation further, turning the RAN into virtualised, multi-vendor software connected by open interfaces [3]. The resulting control plane, the gNB-CU-CP, is a large, stateful, highly concurrent software system. It manages the lifecycle of every connected device, drives the RRC signaling procedures [4], and coordinates mobility and handovers, all under strict latency budgets. The rates involved are high: a single busy site drives on the order of thousands of connection-management and mobility procedures per second, each bounded by a budget measured in milliseconds [5], while a deployed control plane coordinates millions of simultaneously connected devices. As the RAN moves to cloud-native, disaggregated deployments, the correctness, reliability, and timing behaviour of this software increasingly determine the quality of the network as a whole. An outage or a tail-latency regression here is not a localised defect; it is a degradation felt across every device the site serves [6, 7].

This control-plane software is, today, written predominantly in C++. The choice is a rational one: C++ offers the low-level control, mature tooling, and performance that latency-critical systems demand. But that performance comes with a well-known cost. C++ places the burden of memory and concurrency management on the programmer, and decades of experience show that this burden is routinely mishandled, producing use-after-free errors, double-frees, data races, and related defects that are expensive to find, dangerous in production, and responsible for a large share of security vulnerabilities in industry practice [8, 9]. Rust has emerged as the first mainstream systems language to promise an escape from this trade-off: memory safety and data-race freedom enforced at compile time, with performance intended to be competitive with C++ [10, 11]. Whether that promise holds for a telecom-grade, latency-critical control plane, where predictability under overload matters as much as raw speed, is not well

understood. This thesis, carried out at Nokia, addresses the question directly: it develops two functionally equivalent implementations of a representative RAN subsystem, a UE Context Management service modelled on a real Nokia gNB-CU-CP component, in modern C++ and in Rust, and compares them under a realistic synthetic signaling workload.

1.1 Key terms and abbreviations

For readers outside the telecom domain, Table 1.1 fixes the abbreviations and terms used throughout the thesis. Three of them recur constantly: the *gNB-CU-CP* is the control-plane software this work is about; *UE Context Management (UECM)* is the specific subsystem reimplemented and measured; and *tail latency* (the p99/p99.9 percentiles rather than the mean) is the property that, more than average speed, decides whether such a system meets its deadlines [6, 7].

1.2 Motivation and industrial context

Two trends make this comparison timely. The first is the cloud-native evolution of the RAN itself. Moving control-plane functions onto general-purpose servers raises, rather than lowers, the premium on software quality: the software is longer-lived, evolves continuously, runs at higher concurrency, and must meet its deadlines without the determinism that purpose-built hardware once provided. This shift is accelerated by Open-RAN disaggregation, which turns the base station into virtualised, multi-vendor software connected by open interfaces [3], so that the control plane a vendor ships increasingly competes on software-engineering quality (reliability, predictable timing, security, and the pace at which a large codebase can be evolved safely) rather than on bespoke hardware. In this setting a latency regression or a memory-corruption defect is not a localised bug but a network-wide reliability and security risk, and a language-selection question stops being academic and becomes a concrete engineering decision. The second trend is an industry-wide shift toward memory-safe languages. Large software organisations report that the majority of their serious security vulnerabilities stem from memory-safety errors, and that adopting memory-safe languages measurably reduces them [8, 9, 12, 13]. The case no longer rests on vendor reporting alone: a group spanning academia and industry has argued that memory safety

Table 1.1: Abbreviations and key terms used throughout the thesis.

Term	Meaning
RAN	Radio Access Network; the part of the mobile network connecting devices to the core
NG-RAN	Next-Generation RAN, the 5G radio-access-network architecture
gNB	5G base station (“next-generation NodeB”)
gNB-CU / DU	Central Unit / Distributed Unit; the disaggregated halves of a gNB
gNB-CU-CP	Control-plane part of the central unit; the system class studied here
UE	User Equipment; an end device (phone, modem) attached to the network
UECM	UE Context Management; the per-UE state-management subsystem reimplemented here
RRC	Radio Resource Control; the control-plane signaling protocol between UE and gNB
DRB	Data Radio Bearer; a per-UE user-plane data channel set up during a session
Control plane	The signaling and management path, as opposed to the user-plane data path
Tail latency	High-percentile latency (p99, p99.9), the worst-case-oriented timing metric
Jitter	How much latency varies from one operation to the next, as opposed to its typical value
Saturation	The offered load beyond which the service can no longer keep up and requests start queueing
Shard	One of several independent slices of the UE state, each owned by exactly one worker thread
FFI	Foreign Function Interface; calling Rust from C++ across a C ABI
HDR histogram	High-Dynamic-Range histogram, used for accurate latency capture

now needs standardised terminology and requirements if it is to be specified and procured at all [14]. This concern has risen to the level of national policy: cybersecurity agencies now urge software manufacturers to publish "memory-safe roadmaps" and to migrate away from memory-unsafe languages such as C and C++ [15, 16]. In the market this subsystem is sold into it has also become a regulatory matter. The EU Cyber Resilience Act places secure-by-design and lifecycle vulnerability-handling obligations on manufacturers of products with digital elements, with reporting obligations applying from September 2026 and the remainder from December 2027 [17]. It names no programming language, and it would be wrong to read it as requiring one; what it changes is that the recurring defect classes of a memory-unsafe language become a compliance exposure as well as an engineering one. Rust in particular has moved from a niche language into operating-system kernels, cloud infrastructure, and increasingly into telecommunications; production hypervisors such as AWS Firecracker, which underpins its serverless platform, are already written in Rust [18]. Yet the evidence base for Rust's suitability rests largely on general-purpose software and on average-case performance benchmarks. The specific question a telecom vendor must answer, whether Rust can meet the tail-latency and predictability requirements of a RAN control plane, whether it genuinely prevents the defect classes that plague the existing C++ code, and what it costs to adopt it incrementally inside a large C++ codebase, has not been answered with direct, controlled evidence. Providing such evidence for a representative subsystem is the motivation for this work.

Industrial setting. This thesis is carried out at Nokia, one of the small number of vendors that build, maintain, and evolve production RAN software end to end. The subsystem it studies, UE Context Management (UECM), is modelled on a real gNB-CU-CP control-plane component rather than a textbook example.

The gNB-CU-CP is not a minor component. In the 3GPP disaggregated architecture it is the control-plane half of the central unit: it hosts RRC and manages the UE context lifecycle, delegating user-plane bearer handling to the CU-UP over the E1 interface [1, 2, 19, 4]. It does so under the millisecond-scale procedure budgets the 5G system places on control signaling [5], and at the very high event rates and connected-device counts a deployed network sustains. Because it is written in C++, it inherits both that language's performance and its memory-safety and concurrency hazards. That is precisely where the two trends above converge.

Software of this class is shipped, evolved, and operated across live operator networks. A language-selection decision here is therefore a concrete production concern rather than a hypothetical one, which is what makes a controlled comparison on a representative subsystem valuable to a vendor.

UECM is a deliberate choice of study object within that setting. It is small enough to reimplement faithfully and to reason about in full, yet it exercises exactly the properties that make RAN control-plane code hard: per-UE mutable state, a strict operation lifecycle, high concurrency across many UEs, and hard tail-latency expectations. Conclusions drawn from it therefore transfer to the broader class of stateful, concurrent control-plane services. They also remain reproducible by readers outside Nokia, because the implementation is driven entirely by a public, frozen specification rather than by proprietary code.

A concrete failure mode. To make the stakes concrete, consider a mobility storm in which many UEs hand over at once. A single mishandled pointer, a UE context freed on release while an in-flight signaling message still references it, produces a use-after-free. In C++ this compiles cleanly and may pass ordinary testing, only to corrupt an unrelated UE's state or crash the worker thread under precisely the high-concurrency conditions the system exists to handle; the same latent race can equally show up as an unpredictable tail-latency spike rather than an outright crash. In safe Rust the same design does not compile: ownership and borrowing reject the dangling reference at build time. This one scenario captures why the thesis measures both *safety* (which defects are prevented, and when) and *predictability* (tail latency under overload) rather than average throughput alone.

1.3 Problem statement

Two properties of this domain make the language choice consequential. First, manual memory and concurrency management in C++ exposes the software to defects such as use-after-free, double-free, data races, and iterator invalidation, which are silent at compile time and costly to find and fix at telecom scale. Second, what governs whether RAN deadlines are met is not average throughput but worst-case and tail behaviour under load and overload [6], a property that is easy to regress and hard to measure correctly. Rust promises to remove the first class of defect at compile time

while claiming performance competitive on the second, but that claim rests mostly on general-purpose benchmarks rather than on stateful, concurrent, latency-critical telecom workloads.

Bringing these threads together, the concrete problem this thesis addresses is the following. Telecom vendors are under growing pressure, both technical and increasingly regulatory, to adopt memory-safe languages, yet for the RAN control plane they lack *direct, controlled* evidence on which to base that decision. It is not established whether a Rust implementation of a stateful, concurrent control-plane subsystem can match modern C++ in *tail latency and predictability under overload* (as opposed to average throughput), whether Rust's compile-time guarantees actually eliminate the specific defect classes that affect the existing C++ code, and what incremental adoption *costs* inside a large C++ codebase. The available evidence is largely general-purpose and average-case [20, 21], leaving these three questions (predictability, prevented-defect scope, and migration cost) open for a representative subsystem. Answering them under controlled, reproducible conditions is the problem this thesis sets out to solve.

1.4 Research questions

The thesis is organised around four research questions, ordered to put predictability and migration ahead of raw average performance:

RQ1 How do Rust and modern C++ compare in latency predictability (mean, p99, p99.9 tail latency, and jitter) for a RAN control-plane workload under increasing load and overload?

RQ2 Which classes of memory-safety and concurrency defects that occur in C++ are eliminated at compile time in the Rust implementation, and at what stage and cost are the remaining ones detected?

RQ3 What is the overhead, if any, of Rust's safety model (bounds checks, ownership-driven copies, abstraction cost) on throughput, CPU utilisation, and memory behaviour?

RQ4 What are the practical implications (effort, FFI interoperability, and risk) of adopting Rust incrementally within an existing C++ RAN software ecosystem?

1.5 Research approach at a glance

The four questions are answered constructively rather than by literature synthesis. The core artifact is a pair of functionally equivalent UECM implementations, one in modern C++ and one in Rust, built from a single frozen specification, so that any measured difference is attributable to the language and its idioms rather than to divergent designs. RQ1 and RQ3 are answered by driving both implementations with an identical, deterministic synthetic signaling workload in a load sweep from light load through saturation, capturing latency with matched high-dynamic-range histograms and resource use through the operating system, and comparing them with nonparametric confidence intervals [22]. RQ2 is answered by controlled fault injection: a catalogue of representative C++ defects is written deliberately and each is confronted with its Rust counterpart to record whether it is rejected at compile time, caught only at runtime, or missed entirely. The catalogue deliberately spans both defect classes Rust is expected to prevent and classes it is expected not to, so that the study can locate the boundary of the guarantee rather than only confirm it. RQ4 is answered by building a working C++-to-Rust FFI prototype and measuring the size of the interoperability seam, the scope of `unsafe`, and the per-call crossing cost. The full protocol (testbed, workloads, and statistical treatment) is given in Chapter 5.

1.6 Contributions, results, and limitations

The thesis contributes (i) a reproducible, telecom-relevant benchmarking framework for control-plane workloads; (ii) a frozen shared specification and two parity-tested implementations of a RAN control-plane subsystem in C++ and Rust; (iii) a quantitative comparison emphasising tail latency and predictability alongside average throughput; (iv) a defect-class taxonomy obtained through controlled fault injection, including the boundary at which Rust's compile-time guarantee stops; and (v) evidence-based engineering and migration guidelines for language selection in cloud-native RAN evolution (Section 8.4). The headline findings are as follows. On *safety* (RQ2), every injected memory-safety and data-race defect class is prevented at compile time in safe Rust, whereas in C++ each compiles and is caught, if at all, only at runtime and only with the right sanitiser, while two deliberately chosen boundary classes are prevented by neither language; this result is final. On *migration* (RQ4), the FFI seam is small, keeps the core

free of `unsafe`, and costs at most 10.91% of one operation when measured against that same operation called natively, so the real adoption cost is API design, not run-time overhead; this result is also final. On *performance* (RQ1, RQ3), the answer divides at saturation, and getting that division right is itself part of the contribution. Below saturation the two implementations are indistinguishable end to end at the median and their far tail is unresolvable on the available host. That null is also less informative than it looks: an end-to-end operation here is 0.250% the work the language performs and the rest a cross-thread handoff the operating system performs, so the sweep bounds the design rather than the language. Measured separately, the operation itself is 1.6× faster in Rust, for reasons traceable to the standard library's hash-map layout rather than to the safety model, while the Rust process carries a few MiB more resident memory under both allocators. Past saturation the two are not equivalent at all, and the ordering reverses: the C++ build sustains 1.8× the throughput of the Rust build, reproducibly and while consuming more CPU per completed operation, for reasons that point at the two standard libraries' blocking queue and completion primitives rather than at either safety model. The main limitations are the use of a synthetic workload, a single simplified subsystem, single-author implementation bias, and a virtualised measurement host; these are treated in detail in Chapter 8.

1.7 Significance and beneficiaries

The results are intended to be directly actionable. For RAN vendors and cloud-native network operators, the thesis supplies controlled evidence on the question that gates any language decision. That question is not "is Rust fast on average?" but "does Rust hold its tail latency and predictability under overload, and what does incremental adoption cost?", answered for a subsystem representative of production control-plane code. For the standards and policy discussion around memory-safe roadmaps [15, 16], it grounds a general mandate in a concrete telecom data point. For the research community, the frozen specification, the parity-tested implementations, the reproducible benchmarking framework, and the defect-class taxonomy form a reusable baseline for further language and systems studies. And for practising engineers, the fault catalogue and the migration prototype double as worked examples of what Rust's guarantees do and do not buy in this domain.

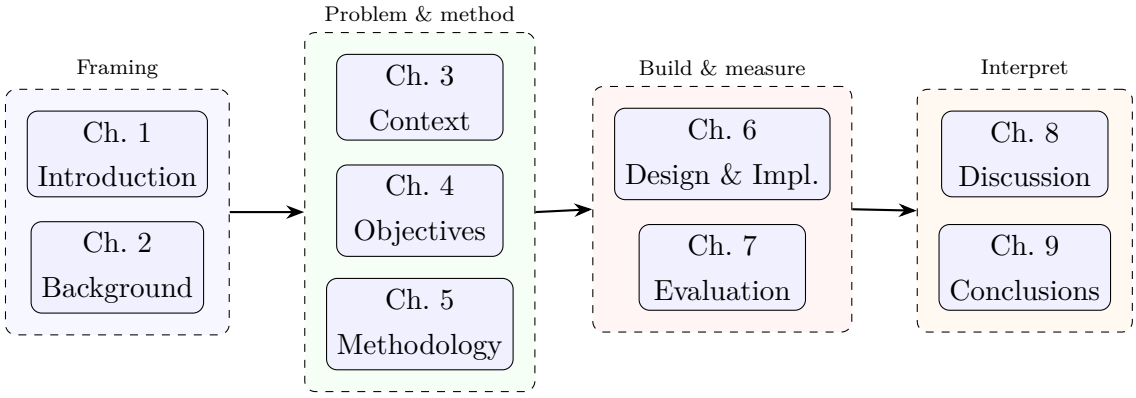


Figure 1.1: Roadmap of the thesis. The chapters progress from framing the problem, through the problem context, the objectives it sets, and the method by which they are pursued, to the constructed artifacts and their evaluation, and finally to interpretation and conclusions.

1.8 Thesis outline

Chapter 2 gives the background on RAN control-plane software, C++, and the Rust safety model. Chapter 3 describes the problem context: the UE Context Management subsystem, the relevant defect classes, and tail-latency predictability. Chapter 4 sets the objectives, hypotheses, scope, and boundaries, and Chapter 5 states the research approach by which they are pursued. Chapter 6 presents the design and implementation of both versions. Chapter 7 describes the experimental setup and reports the evaluation. Chapter 8 discusses the findings, related work, and threats to validity. Chapter 9 concludes and outlines future work. Figure 1.1 summarises this structure at a glance.

2 Background: Systems Languages for Latency-Critical Software

2.1 RAN control-plane systems and the gNB-CU-CP

The NG-RAN architecture splits the gNB into a central unit (gNB-CU) and distributed units, with the control plane (gNB-CU-CP) handling RRC and connection management for each UE [1, 2, 4]. Such control-plane systems must satisfy demanding non-functional requirements: very high event rates (UE mobility, signaling storms), strict and predictable latency bounds that the 5G service specifications quantify as KPIs [5], long-lived software with continuously evolving complexity, and heavily concurrent workloads. In these systems the worst-case and tail behaviour, rather than the average, determines whether deadlines are met, which motivates the predictability focus of this thesis.

Functionally, the gNB is disaggregated along two independent axes. The first is a *central/distributed* split: a central unit (gNB-CU) hosts the higher layers of the protocol stack, while one or more distributed units (gNB-DU) host the lower, more time-critical layers close to the radio. The second is a *control/user-plane* split inside the central unit, separating the control-plane function (gNB-CU-CP) from the user-plane function (gNB-CU-UP). These parts communicate over standardised interfaces: NG towards the 5G core, F1 between the CU and the DU, and E1 between the CU-CP and the CU-UP [1, 2, 19]. Open-RAN initiatives standardise these interfaces further so that each part can be supplied by a different vendor and run as software on commodity hardware [3]. Figure 2.1 places the subsystem studied in this thesis, UE Context Management (UECM), inside the gNB-CU-CP: it owns the per-UE control-plane state and mediates the RRC signaling and mobility procedures, which is why its correctness and timing behaviour are representative of the control plane as a whole.

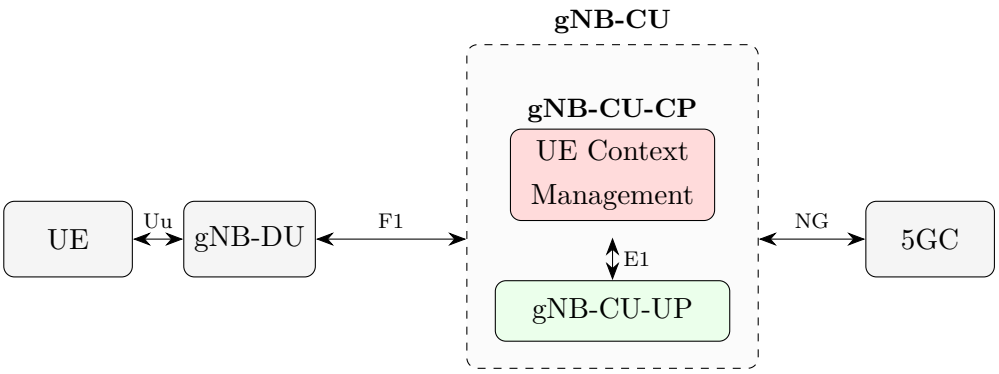


Figure 2.1: Where the studied subsystem sits in the 5G NG-RAN. The gNB is split into a distributed unit (gNB-DU) and a central unit (gNB-CU), and the central unit is split again into control-plane (gNB-CU-CP) and user-plane (gNB-CU-UP) functions, connected by the standardised F1, E1, and NG interfaces. UE Context Management (highlighted) is the per-UE control-plane state manager inside the gNB-CU-CP that this thesis reimplements and measures.

2.2 Latency, throughput, and predictability in real-time software

Latency-critical systems are judged less by how much work they do on average than by how *predictably* they do it. Three metrics must be distinguished. *Throughput* is the sustained rate of completed operations; *latency* is the time an individual operation takes; and *jitter* is the variability of that latency. Because a system can have excellent mean latency and still miss deadlines often, the meaningful figures are the high-percentile *tail* latencies, the 99th (p99) and 99.9th (p99.9) percentiles, rather than the mean. At the scale of a network, tail latency dominates user-visible behaviour: when a single user action fans out into many internal operations, the slowest of them sets the response time, so rare per-operation delays become common per-request delays [6]. Modern datacentre and networking software increasingly operates in a regime where the relevant latencies are measured in microseconds, a range in which conventional operating-system and software overheads are no longer negligible [7]. Fine-grained measurement studies show that, even at moderate utilisation, hardware, operating-system, and application-level effects such as background interference, scheduling, interrupt routing, and power-saving states inflate the tail far beyond what queueing models predict [23]. Figure 2.2 sets the two effects side by side: the distance between the median and the high percentiles, and the fan-out that turns a delay which is rare per operation into one that is common per procedure.

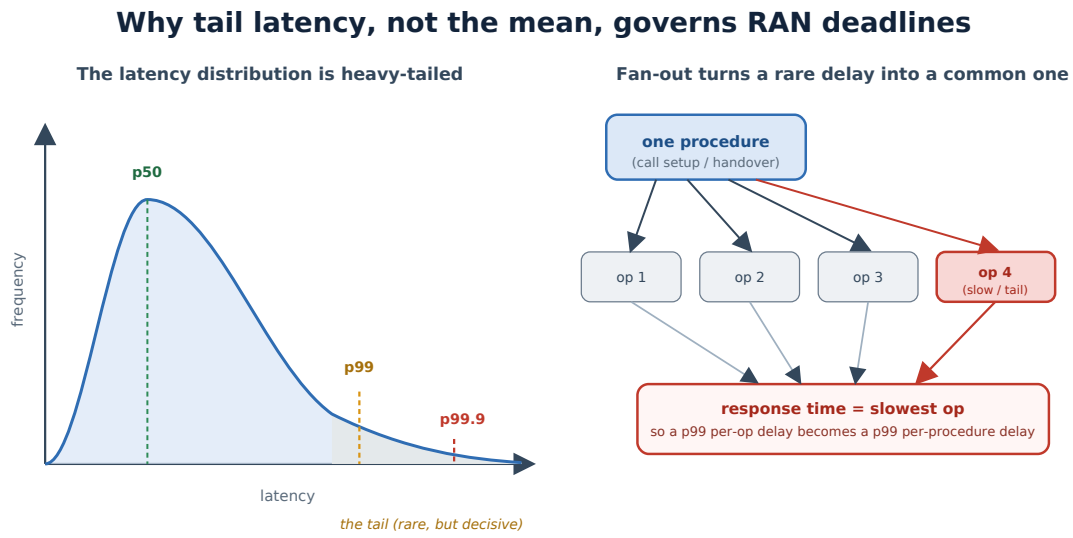


Figure 2.2: Why the tail, not the mean, governs RAN deadlines. The latency distribution is heavy-tailed (left), so the p99/p99.9 percentiles lie far from the median. Because a single procedure fans out into many dependent operations whose slowest member gates completion (right), a delay that is rare per operation becomes common per procedure; at fleet scale a one-in-a-thousand tail event is routine somewhere at every instant.

Real-time systems are conventionally graded by the consequence of a missed deadline: in *hard* real-time a miss is a failure, in *firm* real-time a late result is useless but not catastrophic, and in *soft* real-time a miss merely degrades quality. A RAN control plane is best seen as soft-to-firm real-time: an occasional late signaling response degrades service rather than endangering it, but at fleet scale and under signaling storms such degradations accumulate into a network-wide reliability problem. This is compounded by queueing behaviour: as offered load approaches service capacity, queueing delay grows sharply and the latency tail explodes near saturation. Characterising a system therefore requires driving it not just at nominal load but *into overload*, which is why the evaluation in this thesis sweeps load up to and beyond the saturation point.

Measuring tail latency correctly is itself a pitfall. A naive closed-loop load generator that waits for each response before issuing the next request suffers *coordinated omission*: when the system stalls, the generator stalls with it and simply fails to record the many requests that would have arrived during the stall, systematically hiding the very tail it is meant to expose. Figure 2.3 contrasts the two generator models. The distinction beneath this is a modelling one rather than an implementation detail. In a closed system a new request is triggered only by the completion of a previous one, whereas in an open

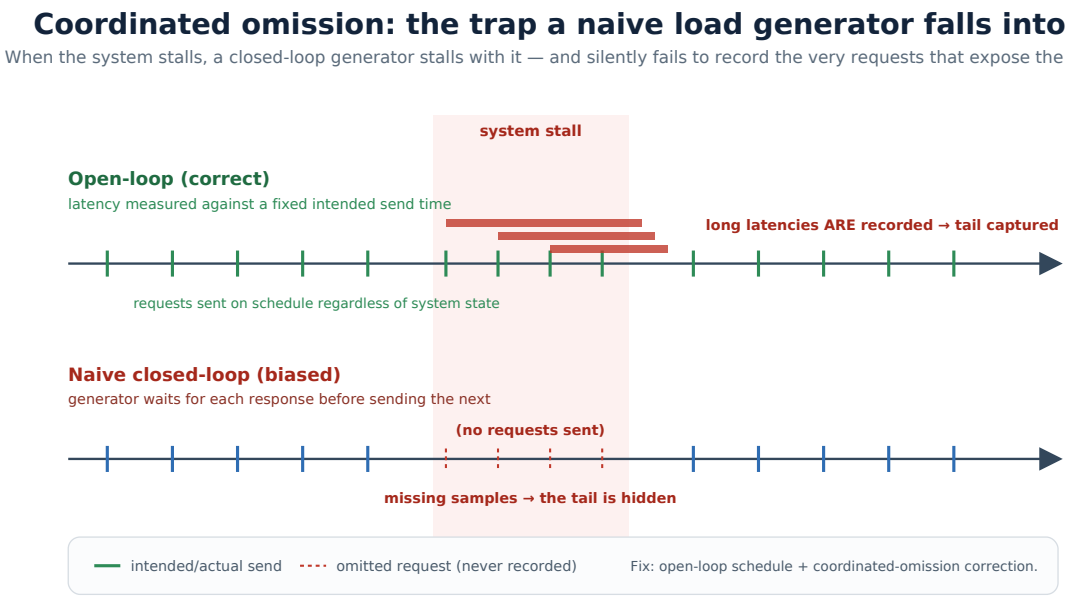


Figure 2.3: Coordinated omission. An open-loop generator (top) sends on a fixed intended schedule regardless of system state, so requests issued during a stall are recorded with their full latency and the tail is captured. A naive closed-loop generator (bottom) waits for each response before sending the next, so when the system stalls it stalls too and never issues the delayed requests, silently omitting exactly the samples that would expose the tail.

system arrivals are independent of completions, and the two produce systematically different response-time behaviour on the same service process, with the closed model reporting the lower figures [24]. Sound measurement therefore requires an open-loop model with an intended schedule, correction for coordinated omission, high-dynamic-range latency histograms that retain resolution across many orders of magnitude, and statistically rigorous reporting with repeated runs and confidence intervals rather than single point estimates [22]. These techniques are applied in Chapter 7.

2.3 C++ for latency-critical systems

C++ is the default language for RAN control-plane software for good reasons. It offers zero-overhead abstractions, direct control over memory layout and allocation, deterministic destruction through the Resource Acquisition Is Initialization (RAII) idiom, and mature compilers, libraries, and profiling tools, all without a garbage collector that could inject unpredictable pauses. Modern C++ (C++11 and later) adds smart pointers (`std::unique_ptr`, `std::shared_ptr`) that automate ownership, move se-

mantics that avoid needless copies, and a formal memory model with `std::atomic` for lock-based and lock-free concurrency. These features make it possible to write fast and, with discipline, reasonably safe systems code.

The difficulty is that safety in C++ remains the programmer's responsibility rather than a property the compiler enforces. RAII and smart pointers reduce but do not eliminate temporal errors: raw pointers and references can outlive the objects they point to, iterators can be invalidated by container mutation, objects can be used after a move, and nothing in the type system prevents a mutable object from being aliased and mutated concurrently from several threads. The result is a well-known catalogue of defects (use-after-free, double-free, buffer overflow, data races, deadlocks, and iterator invalidation) that are easy to introduce and expensive to find at telecom scale. Memory-corruption errors of this kind have been a persistent security problem for decades, and deployed mitigations remain only partial defences [25]. Their prevalence is well documented in industry: memory-safety issues account for roughly 70 % of the security vulnerabilities reported at Microsoft [8], and Google reports a sharp drop in such vulnerabilities as memory-safe languages are adopted in Android, to below a fifth of that platform's total vulnerabilities by 2025 [9, 12]. This specific catalogue is developed in detail in Chapter 3 and forms the basis for the fault-injection study in Chapter 7.

C++ practice mitigates these defects with tooling rather than with language guarantees, and the tooling is inherently partial. Dynamic sanitizers such as AddressSanitizer (ASan), ThreadSanitizer (TSan), and UndefinedBehaviorSanitizer detect many errors, but only on the code paths that a given test actually executes with the right inputs, and they impose runtime and memory overheads that make them unsuitable for production; TSan in particular reports only those data races that happen to occur during a run, so a race that manifests only under a rare interleaving can pass unnoticed. Static analysers help but suffer false negatives and cannot reason about the whole program. Detection is therefore best-effort and after the fact, which is precisely the situation a language with compile-time guarantees is meant to change.

That account describes C++ as it can be deployed today, and it would misrepresent the situation to leave the impression that the language is standing still. Since 2023 the standards committee has pursued *profiles*: named sets of rules, covering type, bounds, lifetime, and arithmetic safety, that a conforming implementation must enforce wherever they are enabled [26]. That direction was chosen over *Safe C++*, a proposal for a rigorously safe subset with borrow checking and an accompanying standard library,

which is essentially Rust's model expressed in C++ [27]. Three things nonetheless keep the comparison drawn here live. Profiles are not yet standardised: in January 2026 their own authors were still calling for implementations, describing the state of the art as a specification together with "an experimental implementation ... being conducted in a major C++ compiler", and treating the design and specification of the initial profiles as work still to be done [28]. The concurrency profile, which is the one that would bear on the concurrency defect classes catalogued in Chapter 3, is placed after that initial set rather than within it: "more profiles should be defined and implemented, notably a concurrency profile" [28]. And a rule that is enforced when switched on is a different kind of object from a property the type system carries unconditionally, particularly where the enforcement is a run-time check rather than a compile-time rejection. The implementations compared in this thesis are therefore built against C++23 as specified and shipped, and Section 8.6 records what that pins the comparison to.

2.4 The Rust ownership and safety model

Rust offers memory safety and data-race freedom enforced at compile time through its ownership and borrowing model [10], together with a strong concurrency story and modern tooling (Cargo, integrated testing). It is the first industry-supported language to overcome the traditional trade-off between the safety of high-level languages and the low-level control of C and C++. It achieves this with a type system based on ownership and borrowing that statically forbids the mutation of shared state, confining the unavoidable exceptions to `unsafe` code encapsulated behind safe APIs [29].

These guarantees rest on more than assertion. The soundness of the safe fragment of the language, including the role of `unsafe` as an explicitly marked escape hatch, has been established formally by the RustBelt project [11], and the aliasing discipline that even `unsafe` code must respect has been given a precise operational semantics [30]. Empirical evidence supports the model in practice. In real codebases `unsafe` is used sparingly and in localised ways [31, 21], and a study of all Rust memory-safety CVEs found that every one of them involved `unsafe` code [32]. Safe Rust, in other words, keeps its promise.

The caveat is that unsafety propagates through dependencies. Fewer than a third of libraries use `unsafe` directly, but a much larger fraction cannot be fully statically checked because `unsafe` hides somewhere in their call chain [33]. That is a further

argument for keeping the safety-critical core of a subsystem `unsafe`-free, as this thesis does.

One property matters crucially in a latency-critical setting: this safety is enforced by the type system and borrow checker rather than by runtime instrumentation, and is intended to be *zero-cost*. Systems research argues Rust attains safety without the overhead of garbage collection or runtime checks, and even enables efficiency techniques that are impractical in C [20]. Whether that zero-cost claim holds for a telecom control-plane workload is precisely the property this thesis evaluates.

Concretely, the borrow checker rejects at compile time the very use-after-free sketched in the introduction. The following program does not build, because `ctx` is moved (dropped) while a reference to it is still live:

```
let ctx = UeContext::new(42);
let r = &ctx;      // immutable borrow of ctx
drop(ctx);         // move/free ctx ...
use_ctx(r);        // error[E0505]: cannot move out of ‘ctx‘
                   //              because it is borrowed
```

Figure 2.4 puts the same defect in its UE Context Management form beside the C++ version, which compiles and is caught, if at all, only at runtime.

The same discipline extends to concurrency. Rust encodes thread-safety in the type system through the `Send` and `Sync` marker traits: a value may cross a thread boundary only if its type is `Send`, and may be shared by reference across threads only if it is `Sync`. Sharing a value that permits unsynchronised mutation is therefore a compile error, not a latent race; this is the property Rust markets as “fearless concurrency.” Error handling is likewise explicit: fallible operations return a `Result` value that the caller must handle, rather than raising an exception that unwinds through hidden control flow. Both the cost and the control flow of the failure path therefore stay visible, which is an advantage in latency-sensitive code where surprises in the tail are what matter. Table 2.1 sets the two languages’ mechanisms side by side. The distinction it is built around is not which hazards each language names, since both name much the same ones, but *when* and *by what* each is ruled out, and the last two rows mark where the answer stops being “by the compiler” for either of them.

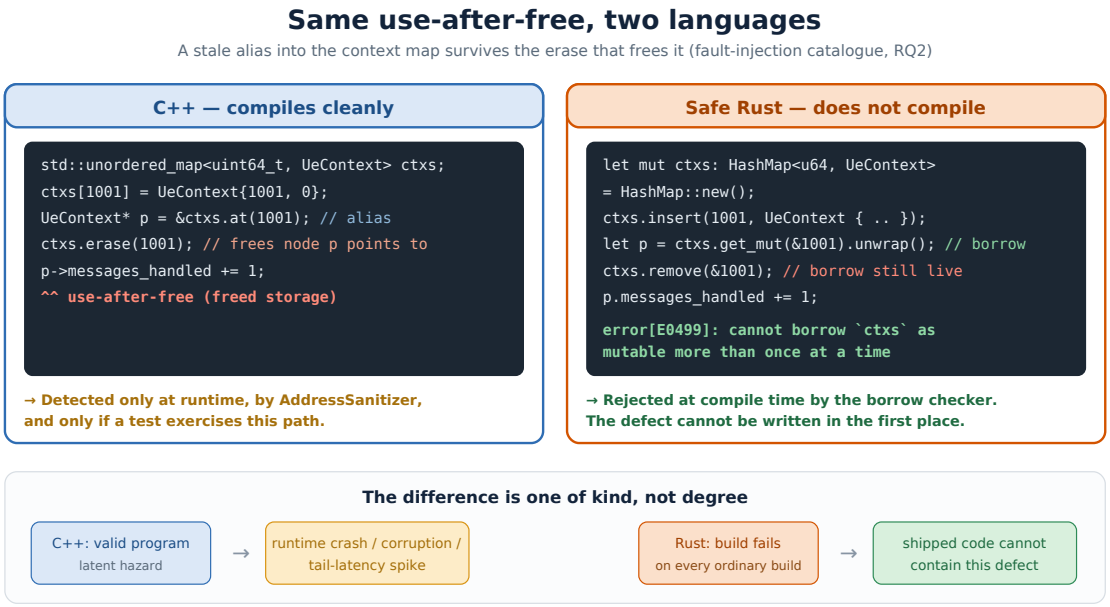


Figure 2.4: The same use-after-free defect in C++ and in safe Rust. The stale alias into the context map survives the `erase`/`remove` that frees it: in C++ this compiles and is caught only at runtime (and only under AddressSanitizer, on a path a test exercises), whereas in safe Rust the borrow checker rejects it at compile time (`E0499`), so the defect cannot be written at all. The difference is one of kind, not degree.

Table 2.1: How each language enforces the properties this thesis measures. The rows above the rule are the ones on which the two differ in kind rather than in degree: a compiler that rejects a program rules the defect out of every execution of it, whereas a sanitiser rules it out only of the executions an instrumented test happens to take. The rows below the rule are the boundary developed in Chapter 3 and measured in Chapter 7, where neither language offers a static guarantee.

Concern	Modern C++	Rust
Object lifetime	RAII and smart pointers, applied by convention; a raw pointer or reference may outlive its referent	Ownership: each value has one owner and is dropped exactly once
Aliasing	Unrestricted; any number of mutable aliases may coexist	Borrowing: either one mutable reference or many shared ones, never both
Bounds	Unchecked on `operator[]`, checked only through `at()`	Checked by default, elided where the index can be proved in range
Thread safety	Not expressed in the type system; sharing a mutable object across threads compiles	`Send` and `Sync` marker traits; unsynchronised sharing is a type error
Error signalling	Exceptions or error codes; the failure path can be invisible at the call site	`Result` values the caller must handle; the failure path is explicit
Escape hatch	The whole language	`unsafe` blocks, delimited and mechanically countable
Stage of detection	Run time, under the right sanitiser, on a path a test exercises	Compile time, on every ordinary build, whatever the test coverage
Lock-acquisition order	Not checked	Not checked
Atomicity of a composed operation	Not checked	Not checked

2.5 The systems-language landscape

Rust is not the only memory-safe language, but it occupies a distinctive position for this problem. Managed languages such as Java, C#, and Go achieve memory safety through a runtime and garbage collection. That is an excellent trade for much software, but it is a poor fit for a tail-latency-critical control plane: garbage collection introduces pauses and timing variability that are a recognised source of tail latency in large systems [6], and a managed runtime also cedes the low-level control over memory layout and scheduling that this domain relies on. At the other extreme, C is as memory-unsafe as C++ while offering fewer abstractions for building large systems. Rust is the first widely supported language to combine memory and thread safety, enforced at compile time, with the absence of a garbage collector and performance intended to be competitive with C++ [29, 20]; cross-language benchmark surveys place Rust's throughput and memory use alongside C and C++ while it retains its safety guarantees [34]. That combination is exactly why the meaningful comparison for RAN control-plane software is C++ against Rust, and why managed languages are out of scope here. Figure 2.5 places the candidates on the two axes that decide it, low-level control and compile-time safety.

2.6 Concurrency models for control-plane software

How concurrency is structured shapes both correctness and tail latency. The dominant model is *shared-memory* concurrency, in which threads mutate common state guarded by mutexes and atomics; it is flexible but is the source of races, deadlocks, lock contention, and priority inversion, and its worst-case timing is hard to reason about. The alternative is *message-passing* or *shared-nothing* concurrency, in which state is partitioned so that each piece is owned by a single thread and all interaction happens through messages, as in the actor model and in sharded designs. Shared-nothing trades some flexibility for scalability and far simpler reasoning, and it maps naturally onto per-UE partitioning in a control plane, where each UE's state is largely independent. The choice is not neutral with respect to the quantity this thesis measures: thread-per-core structuring has been shown to change application tail latency in its own right [35], which is one reason the model is held fixed across both implementations rather than left to each language's idiom. Figure 2.6 contrasts the two models and marks what

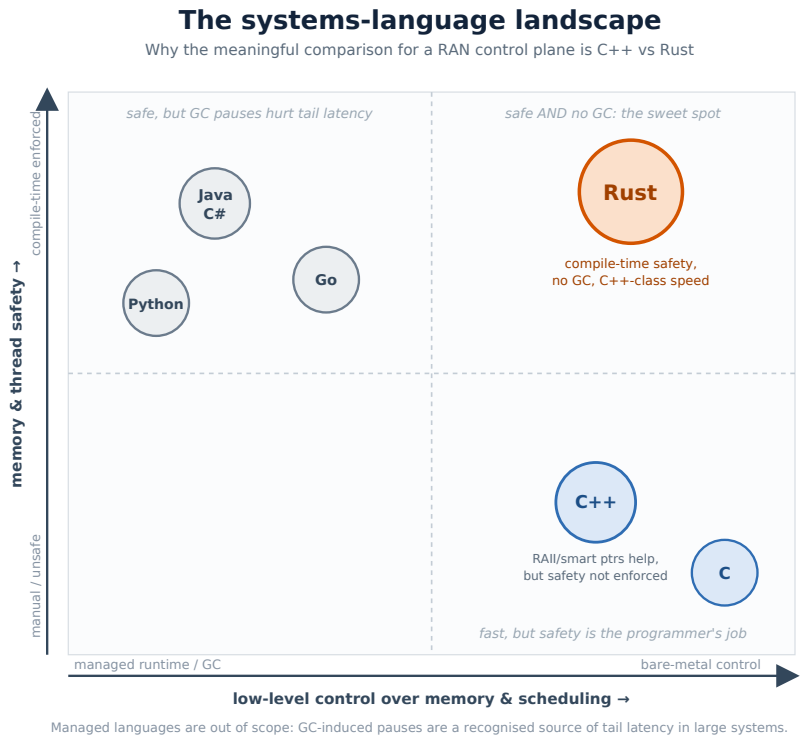


Figure 2.5: The systems-language landscape along two axes: low-level control over memory and scheduling (horizontal) and memory/thread safety (vertical). Managed languages (Java, C#, Go, Python) are safe but pay a garbage-collection cost that hurts tail latency; C and C++ give bare-metal control but leave safety to the programmer. Rust occupies the sweet spot of compile-time safety with no garbage collector, which is why the meaningful comparison for a RAN control plane is C++ against Rust.

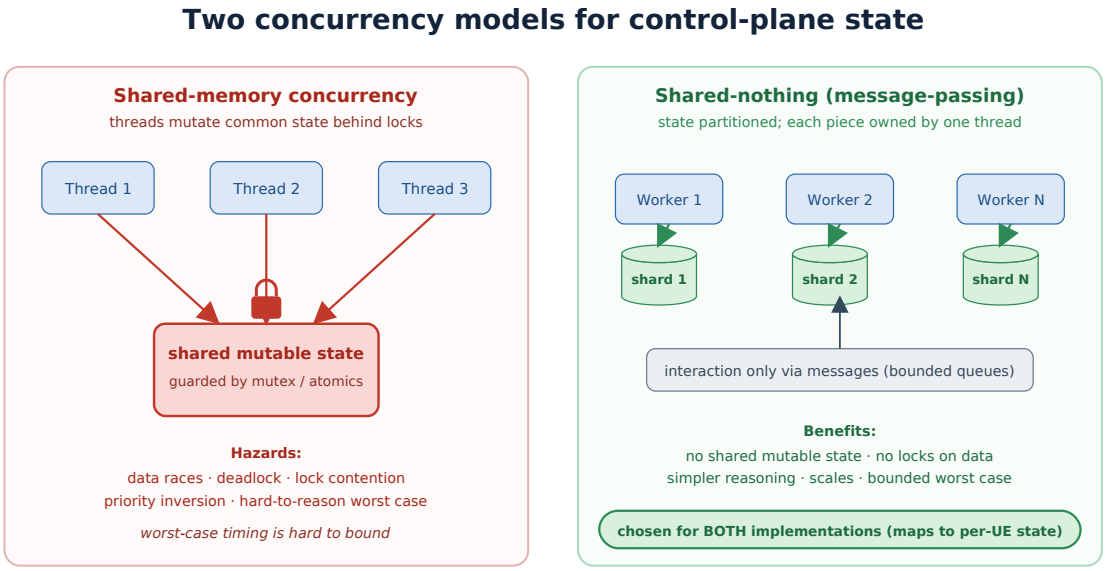


Figure 2.6: Two concurrency models for control-plane state. Shared-memory concurrency (left) lets threads mutate common state behind locks and is the source of data races, deadlock, and lock contention, with a worst case that is hard to bound. Shared-nothing message-passing (right) partitions state so each shard is owned by a single worker and all interaction happens through bounded queues; this eliminates locks on context data and maps naturally onto per-UE partitioning, and is the model adopted for both implementations.

each does to the hazards catalogued above.

A third structure, an asynchronous runtime over a work-stealing scheduler, is idiomatic in both languages and is deliberately used in neither. Its timing behaviour belongs to the runtime rather than to the language hosting it: a real-time study of asynchronous Rust finds that one and the same application meets or misses its deadlines depending on which runtime and thread-priority mapping it is given, and that the configuration meeting them is not the idiomatic work-stealing one [36]. Adopting such a runtime would therefore introduce a variable that this study is built to exclude, and Section 8.5 returns to a comparison in which it was not excluded.

A second, independent choice is how work is queued between producers and consumers. Unbounded queues mask overload and let latency and memory grow without limit; *bounded* queues with either blocking (backpressure) or shedding bound worst-case memory and make overload visible, which is essential for predictable tail latency. Lock-free structures can reduce contention but are notoriously difficult to implement correctly. These considerations directly motivate the architecture adopted for both implementa-

tions in Chapter 6: a shared-nothing, thread-per-shard design. The UE state is divided into independent slices called *shards*, and each shard is owned by exactly one worker thread, so no two threads ever touch the same context. Requests reach a shard through a bounded multi-producer, single-consumer queue that blocks when full. This model is chosen so that any measured latency difference reflects the language rather than a divergent concurrency strategy.

2.7 Interoperability between C++ and Rust

Adopting a new language in a large existing system is necessarily incremental: a Rust component must coexist with, and be callable from, a substantial C++ codebase. The common denominator for such interoperability is the C application binary interface (ABI). Rust can expose functions with `extern "C"` linkage and hand out *opaque handles* to Rust-owned objects, which C++ then calls through a C header; tooling automates parts of this, including `bindgen` (generating Rust bindings from C headers), `cbindgen` (generating C headers from Rust), and the `cxx` crate (a safer bidirectional bridge). Interoperability is not free: crossing the boundary steps outside the checked language, so the seam must contain `unsafe` code; passing anything beyond simple values requires marshalling; and the shape of the foreign interface has to be designed deliberately. The size of that seam, how much `unsafe` it concentrates, and the per-call crossing cost are exactly the quantities measured for RQ4 in Chapter 7.

A distinction has to be drawn at this point, because it bounds what any of those measurements can establish. Concentrating `unsafe` in the seam is a statement about what the compiler checks, not about what can happen at run time. Once a Rust component is linked into a C++ host they share one address space, and a defect on the C++ side can corrupt Rust-owned data irrespective of how much of the Rust side the borrow checker verified; this has been demonstrated experimentally, and the proposed remedies rely on hardware or runtime isolation of the two heaps rather than on the discipline of the interface [37, 38]. The defect classes arising specifically at Rust/C boundaries have been catalogued separately from those of either language [39], and the binding generators themselves are known to carry bugs [40]. The consequence for this thesis is stated once here and honoured throughout: an `unsafe`-free core means the core's own defects are ruled out at compile time, and it does not mean the core is protected from its host.

2.8 Research gap

The preceding sections establish three things. First, the RAN control plane is a soft-to-firm real-time, highly concurrent, latency-critical domain in which tail behaviour under overload, not average throughput, governs the quality experienced by the network. Second, C++ supplies the control and performance this domain demands but leaves whole classes of memory and concurrency defects to best-effort, after-the-fact detection. Third, Rust promises to prevent those classes at compile time at no runtime cost, with a credible incremental adoption path through C-ABI interoperability.

The domain is not entirely unstudied. Two vendor-hosted master's theses have compared candidate languages for RAN and 5G control-plane software, one weighing runtime cost against development efficiency across C++, Rust, and Go [41], the other measuring median and 99th-percentile latency for control-plane procedures across five asynchronous programming frameworks [42]. Both are directly relevant, and the second reports a tail difference unfavourable to Rust. Neither, however, isolates the language: in both, the language varies together with the runtime, the concurrency model, and the implementation, so a measured difference cannot be attributed to the language rather than to the framework built on it. Figure 2.7 sets the neighbouring studies against the properties a controlled answer would require, and the pattern it shows is that each holds some of them and none holds all.

What is therefore missing from the literature is *direct, controlled* evidence for a representative control-plane subsystem: whether Rust actually holds its tail latency and predictability under overload against modern C++ *when the concurrency model, algorithms, and data structures are held fixed*, exactly which defect classes it eliminates and at what stage the remainder are caught, what its safety model costs in throughput and resources, and what incremental adoption costs in practice. Closing this gap is the purpose of the four research questions. The next chapter grounds them in the specific subsystem, defect classes, and timing properties at stake; Chapter 4 turns them into objectives and testable hypotheses; and Chapter 5 sets out the methodology by which they are pursued.

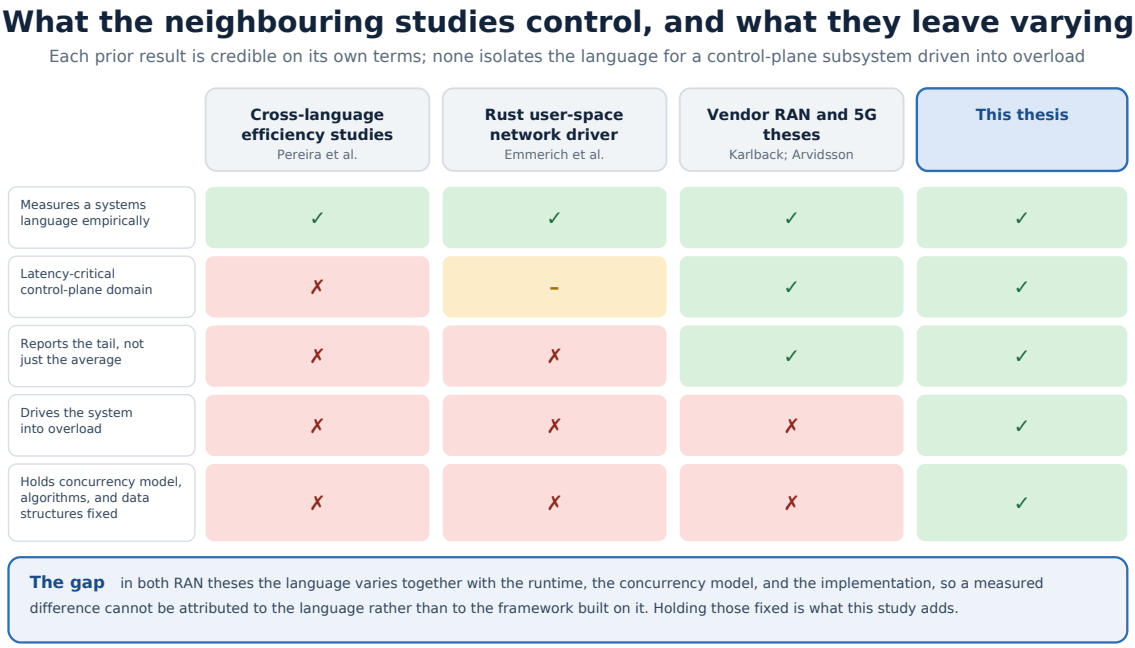


Figure 2.7: What the neighbouring studies control, and what they leave varying. Each row is a property that a controlled answer to the research questions would need; each column is a study. The general cross-language comparisons measure the right kind of object but the wrong regime, reporting averages on general-purpose workloads. The two vendor RAN theses reach the right domain and do report the tail, but in both the language varies together with the runtime, the concurrency model, and the implementation, which is where the bottom two rows stop them. Holding those three fixed and driving the system past saturation is what this thesis adds; the contribution is the control, not a new measurement technique.

3 Problem Context: UE Context Management and Predictability

3.1 The UE Context Management subsystem

The subsystem under study is a UE Context Management service. It is modelled on a real Nokia gNB-CU-CP service domain that owns the UE context lifecycle, RRC procedure execution [4], and mobility/handover. Those are the very functions the 3GPP architecture assigns to the control-plane part of a disaggregated gNB: the CU-CP hosts the RRC and the control-plane part of PDCP and manages the UE context, delegating user-plane bearer handling to the CU-UP over the E1 interface [1, 2, 19].

It is a good representative on four counts. It is core RAN L3 functionality, it is performance- and latency-sensitive, it is inherently stateful and concurrent, and it is self-contained enough to specify precisely and benchmark fairly. The modelled functionality includes UE context creation and deletion, state transitions (e.g. idle to connected), signaling message handling, and concurrent context lookup under load.

Figure 3.1 shows the UE-context lifecycle state machine that both implementations encode identically. Illegal transitions, for example signaling an `Idle` context, return a typed error rather than mutating state. This state machine is exactly what the mixed workload of Section 7.2 walks each UE through.

Representativeness and tractability. Two properties make UECM a good study object for a language comparison. It is *representative*: per-UE mutable state, a strict operation lifecycle, high inter-UE concurrency, and the hard latency and reliability targets the 5G system places on control-plane procedures [5] are exactly the properties that make production RAN control-plane code difficult, so conclusions about how a language handles them transfer to the broader class of stateful, concurrent control-plane services. It is also *tractable*: the subsystem is small enough to specify precisely, implement faithfully in two languages, and reason about in full, which is what makes a controlled, parity-tested comparison possible at all. A larger or less well-bounded subsystem would improve external validity at the cost of the internal validity on which

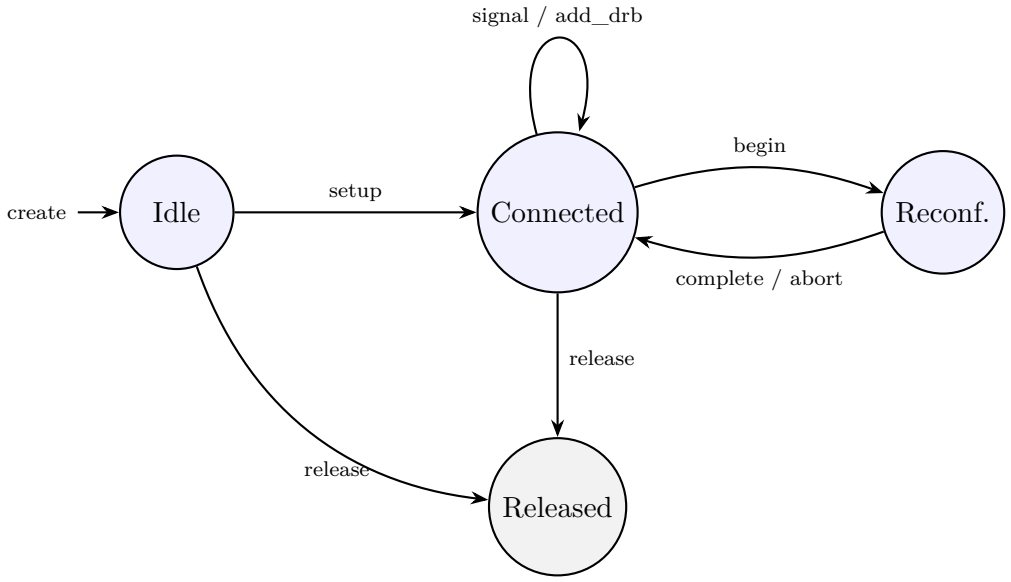


Figure 3.1: UE-context lifecycle state machine. `create` allocates a context in `Idle`; `setup` connects it; `signal` and `add_drb` are handled in place while `Connected`; a reconfiguration/handover moves through `Reconfiguring` and back on `complete` or `abort`; `release` tears the context down. Operations invalid for the current state return an `OpError` instead of changing state.

a fair language comparison depends; UECM is chosen as the point that keeps both within reach.

Operation set. The interface exercised by the study is deliberately small and lifecycle-oriented: create and release a context; set up a connection (`Idle`→`Connected`); handle an in-session signaling message; add a data radio bearer; begin, complete, or abort a reconfiguration/handover; and look up a context's state under concurrent load. Every operation is defined for every state of the machine in Figure 3.1: an operation that is illegal in the current state returns a typed error and leaves the context unchanged, rather than throwing or corrupting state. This closed, well-typed operation set is what lets the same test vectors and the same workload drive both implementations along identical paths (Section 6.7).

3.2 Memory-safety and concurrency defect classes

The defects that affect C++ control-plane code of this kind fall into two families. The *memory-safety* family concerns the lifetime and bounds of objects: a *use-after-free* accesses storage that has already been released, a *double-free* releases the same

storage twice, and a *buffer overflow* reads or writes past the end of an allocation. The *concurrency* family concerns unsynchronised or ill-ordered access from multiple threads: a *data race* is a pair of unsynchronised accesses to the same location of which at least one is a write, a *deadlock* is a cyclic wait for locks, and *iterator invalidation* is the use of an iterator or reference into a container after a mutation has moved or freed the underlying storage. What these share is that they are all *silent* in C++: each is accepted by the compiler and, on the paths ordinary tests happen to exercise, may appear to work, surfacing only later as corruption, a crash, or an unpredictable latency spike. These are the same memory-corruption classes that decades of security research have struggled to contain in C and C++ [25].

Each class has a natural incarnation in UE Context Management, which is why the subsystem is a good vehicle for studying them:

- **Use-after-free.** A signaling message holds a raw pointer or reference to a UE context that a concurrent `release` has already erased from the context map; the in-flight handler then dereferences freed storage.
- **Double-free.** A release path runs twice for the same UE, for example because a retransmitted release is not made idempotent, freeing the context's owned resources a second time.
- **Iterator invalidation.** Code iterates a context's data-radio-bearer list while adding or removing a bearer, so the iterator refers into storage the container has since reallocated.
- **Data race.** Two workers update the same context field, an activity counter or a state flag, without synchronisation, so the accesses race.
- **Unguarded shared mutation.** A context is shared by reference across threads and mutated without a lock or other discipline, a race whose root cause is the absence of any ownership rule for the shared state.

These five classes lie squarely within the guarantees Rust claims to provide, which makes them the natural test of whether that claim holds. Testing only them, however, would build the expected answer into the choice of test cases, so two further classes are carried into the study for the opposite reason, because there is good reason to expect that safe Rust does *not* prevent them:

- **Lock-order deadlock.** A handover locks the source UE context and then the target while a concurrent handover in the opposite direction locks target then source, and the two wait for each other indefinitely. Nothing about object lifetimes or aliasing is violated; the fault is in the order of acquisition.

- **Lost update.** Two workers increment a context's activity counter through a read-modify-write split across two separate critical sections. Every individual access is correctly synchronised, so this is a *race condition* rather than a *data race*: the atomicity violation lives in the composition of the operations, not in the memory accesses themselves, and it corrupts a value silently instead of crashing.

Together these seven form the catalogue injected in Chapter 7, and the split between the two groups is what lets the study report a boundary rather than a verdict. Buffer overflow is described above for completeness but is not injected, because neither implementation performs raw pointer arithmetic on the context store, so an injected instance would be an artificial construct rather than a plausible bug in this subsystem. Table 3.1 collects the catalogue with the outcome expected of each case *before* the study was run, which is the form in which the runner records it: an expectation written down in advance can be violated in either direction and reported as a discrepancy, whereas one formed after the fact cannot.

3.3 Tail latency and predictability in RAN

Section 2.2 set out why the tail rather than the mean governs a latency-critical system, and what measuring it correctly demands. What that section left general, this one makes specific, because two properties of the RAN setting are what turn a methodological preference into the governing metric.

The first is *fan-out*: a single user-visible action, a call setup or a handover, decomposes into many dependent control-plane operations, and because the slowest of them gates completion, a latency that is rare per operation becomes common per procedure [6]. The second is *scale*: a busy site drives thousands of such procedures per second across millions of connected devices, so a tail event occurring once in a thousand operations is not an edge case but a routine occurrence somewhere in the fleet at every instant. Taken together, the p99.9 of a single UE Context Management operation is not a curiosity at the margin of a distribution; it is a rate of degraded procedures the network sustains

Table 3.1: The injected defect catalogue: each class in the form it would plausibly take in UE Context Management, and the outcome recorded for it in advance of the study. The last column is a prediction rather than a result; Table 7.9 reports what was measured, and the fault runner flags a mismatch in either direction rather than absorbing it into the conclusion. Buffer overflow is listed for completeness and is not injected, because neither implementation does pointer arithmetic on the context store.

Defect class	Incarnation in UE Context Management	How it fails	Expected in safe Rust
Memory-safety family: object lifetime and bounds			
Use-after-free	An in-flight signaling handler dereferences a context that a concurrent `release` has already erased from the map	Corruption or crash	Prevented
Double-free	A retransmitted release is not made idempotent, so the release path frees the context's resources a second time	Corruption or crash	Prevented
Buffer overflow	No plausible incarnation here: the context store is reached only through container APIs	—	Not injected
Concurrency family: unsynchronised or ill-ordered access			
Iterator invalidation	A bearer is added while the context's data-radio-bearer list is being iterated	Corruption or crash	Prevented
Data race	Two workers update the same context field, a counter or a state flag, without synchronisation	Torn or lost value	Prevented
Unguarded shared mutation	A context is shared by reference across threads and mutated with no ownership rule governing it	Torn or lost value	Prevented
Boundary group: included because safe Rust is expected not *to prevent them*			
Lock-order deadlock	Opposing handovers lock source-then-target and target-then-source	Hang	Not prevented
Lost update	A counter's read-modify-write is split across two separate critical sections, so each access is synchronised but their composition is not atomic	Silently wrong value	Not prevented

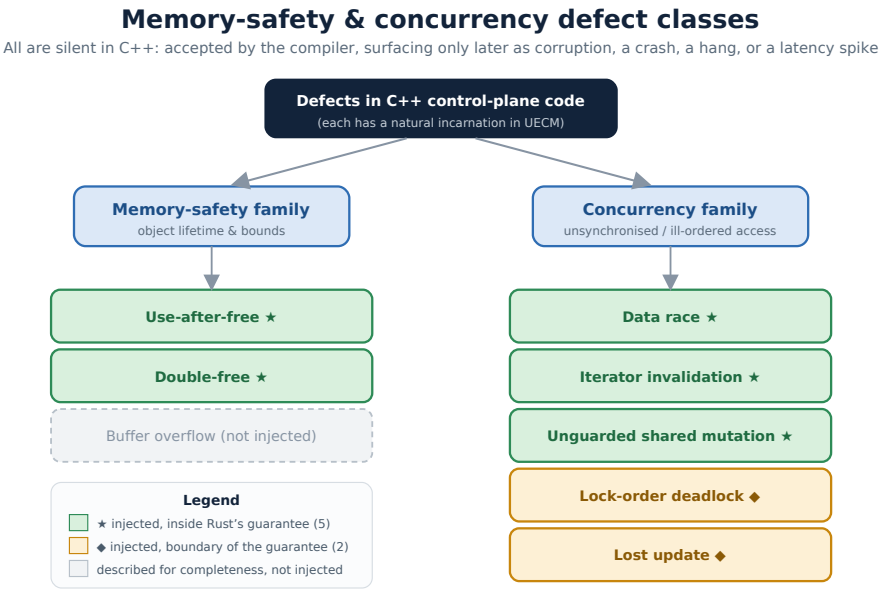


Figure 3.2: The defect classes that affect C++ control-plane code, split into a memory-safety family (object lifetime and bounds) and a concurrency family (unsynchronised or ill-ordered access). Starred classes are the five injected from inside Rust's guarantee; the two diamond-marked classes, lock-order deadlock and lost update, are injected precisely because they lie outside it, so the catalogue locates the boundary of the guarantee rather than only confirming it. Buffer overflow is described for completeness but is not injected. All are silent in C++, surfacing only later as corruption, a crash, a hang, or a latency spike.

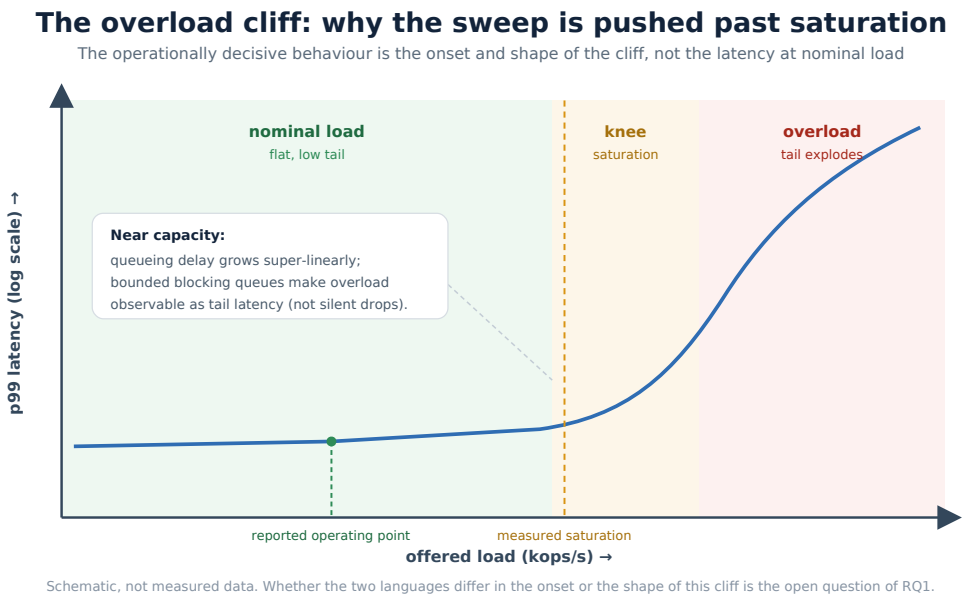


Figure 3.3: The overload cliff. Tail latency stays low and flat while offered load is below service capacity, then climbs steeply once the system is driven past saturation and requests queue behind the bounded, blocking queues. The operationally decisive behaviour is the onset and shape of this cliff, which is why the load sweep is pushed beyond the saturation point rather than stopping at nominal load. The figure is a schematic of the phenomenon and not a plot of this study's data: whether the two languages differ in where the cliff begins or how steeply it rises is exactly what RQ1 asks, and Section 7.5.1 reports that the development host cannot resolve it.

continuously.

The scale at which those percentiles matter is not hypothetical. A measurement study of a disaggregated 5G RAN times the CU-CP-to-DU UE Context Setup procedure over F1-C, and the CU-CP-to-CU-UP bearer context setup over E1, under different interface-protection schemes, and finds the choice worth hundreds of microseconds: on F1-C, IPsec adds on the order of 120 μs on average to UE context setup, against roughly 430 μs for DTLS [43]. Overheads of that size are treated as consequential for precisely the procedure family this thesis reimplements, which places these operations in a regime where tens of microseconds are an engineering quantity rather than noise.

The study therefore characterises the full latency distribution, focusing on the high percentiles (p99, p99.9, p99.99), the maximum, and jitter, and on how these degrade as offered load approaches and then exceeds saturation (Chapter 7). Stopping at nominal load would miss the regime that decides service quality, since queueing delay grows super-linearly near capacity; it is also the regime in which a difference between two implementations of one design, if there is one, has room to become visible.

4 Objectives

4.1 Deficiencies in existing comparisons

The gap argued in Section 2.8 is only as good as the search that looked for it, so the search is stated rather than assumed. Candidate literature was gathered from the ACM Digital Library, IEEE Xplore, USENIX, and Google Scholar using combinations of the terms *Rust*, *C++*, *memory safety*, *tail latency*, *systems programming*, *performance comparison*, and *RAN / 5G control plane*, restricted to peer-reviewed venues except where an industry or government report is itself the object of study (the memory-safety mandates and vendor vulnerability statistics). The result set was then extended by backward and forward citation chaining from the core Rust-safety papers, and filtered to work that either measures a systems language empirically, formalises Rust's guarantees, or reports adoption experience. The domain literature (3GPP specifications and the O-RAN survey) was added separately to fix the architectural context. The search returned no study that measures the tail-latency behaviour of a Rust and a C++ implementation of the same stateful, concurrent control-plane subsystem under overload.

What it did return is the baseline this thesis works against, and two results deserve naming because they are the strongest evidence that the answer may simply be "no difference". Cross-language studies report that compiled systems languages achieve broadly similar time and energy efficiency, well ahead of managed and interpreted ones [44]. Closer to this domain, an idiomatic Rust 10 Gbit/s user-space network driver executes many more instructions per packet than its C reference yet runs only about 4% slower [45], which suggests Rust's safety checks are largely absorbed by out-of-order execution rather than being literally free. Both are average-case results on general-purpose or packet-forwarding workloads, so neither addresses the four questions Section 2.8 left open. They establish that this comparison is worth making under controlled conditions, not that it has been made.

4.2 Solution objectives

The overarching objective is a fair, reproducible comparison of modern C++ and Rust for a RAN control-plane subsystem, together with reusable artifacts and evidence-based guidance. It decomposes into concrete objectives, each aligned with a research question so that progress is traceable:

O1 (predictability, RQ1) Quantify and compare the latency distributions of the two implementations, focusing on the p99/p99.9 tail and jitter, across a load sweep that reaches overload, and identify where along that sweep, if anywhere, the languages diverge.

O2 (defect prevention, RQ2) Determine, for a catalogue of defect classes representative of the subsystem, which are prevented at compile time in safe Rust and, for those that are not, at what stage and cost they are detected in C++.

O3 (overhead, RQ3) Measure the resource cost, throughput, CPU utilisation, and memory behaviour, of each implementation under matched conditions, isolating any systematic penalty attributable to Rust's safety model.

O4 (migration, RQ4) Build and measure a realistic incremental-adoption prototype, quantifying the size of the interoperability seam, the scope of `unsafe` it requires, and the per-call cost of crossing the language boundary.

O5 (artifacts and guidance) Deliver a frozen specification, a parity-tested pair of implementations, a reproducible benchmarking framework, a defect-class taxonomy, and engineering guidelines that practitioners can reuse for language-selection decisions in cloud-native RAN evolution.

4.3 Hypotheses

The objectives are approached with three prior expectations, stated so that the evaluation can confirm or refute them rather than merely describe results after the fact. **H1:** for this workload the two languages are performance-equivalent in the operationally relevant region below saturation, with any difference confined to the far tail. **H2:** safe Rust prevents at compile time the memory-safety and concurrency defect classes that

C++ leaves to best-effort, after-the-fact detection. **H3:** the run-time cost of C++-to-Rust interoperability is negligible relative to the cost of the operations invoked, so that the true cost of incremental adoption is engineering effort rather than performance. These hypotheses are revisited against the evidence in Chapter 8.

4.4 Success criteria

The study is successful to the extent that it produces *decision-grade* evidence, evidence on which a practitioner could base a language choice, rather than merely a set of measurements. Concretely this requires: that any performance comparison be reported with repeats and confidence intervals so that a claimed difference is distinguishable from run-to-run noise; that the defect-prevention result be demonstrated on concrete, compilable programs rather than argued from language semantics alone; that the migration cost be read off a working prototype rather than estimated; and that the whole be reproducible from published artifacts and a public specification. Meeting these criteria is what distinguishes the contribution from the general-purpose, average-case comparisons whose deficiencies motivate the work.

The four questions, the five objectives, and the three hypotheses are three views of the same plan, and Table 4.1 lines them up so that a reader can follow any one question from the objective it sets, through the method that pursues it and the section that reports its evidence, to the place where its verdict is recorded. Every claim in Chapter 8 is reachable from this table, which is the practical form of the "decision-grade" requirement above: a result a practitioner cannot trace back to the measurement that produced it is not one they can act on. Figure 4.1 adds what a row-per-question table cannot: grouped by hypothesis rather than by question number, it exposes the two places where the mapping is not one-to-one, H1 carrying two questions and O5 sitting beneath all four.

4.5 Scope and boundaries

The work focuses on a single representative subsystem studied under controlled, synthetic conditions, and its boundaries are drawn deliberately to protect the internal validity of the language comparison. In scope are the UE Context Management functionality of Chapter 3, the shared-nothing concurrency design, the load sweep to overload,

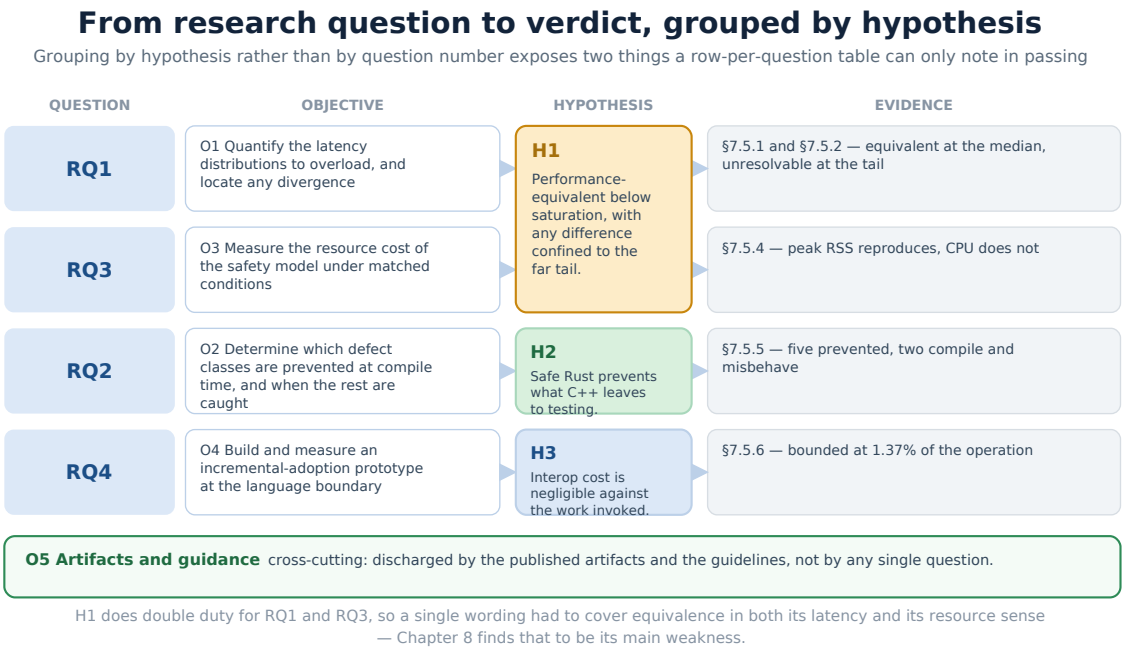


Figure 4.1: From research question to verdict, grouped by hypothesis rather than by question number. The regrouping is what makes two structural facts legible that Table 4.1 can only note in passing. The first is that H1 is shared by RQ1 and RQ3, so a single wording had to cover equivalence in both its latency and its resource sense; Section 8.3 finds that to be its main weakness. The second is that O5 sits underneath all four questions rather than beside any one of them, and is discharged by the published artifacts and the adoption guidelines instead of by a measurement.

Table 4.1: Traceability from each research question to the objective it sets, the hypothesis it is tested against, and the sections in which it is pursued, evidenced, and judged. RQ3 shares H1, which was worded to cover performance equivalence in both its latency and its resource sense; Chapter 8 finds that wording to be the hypothesis's main weakness. O5 (artifacts and guidance) is cross-cutting and is discharged by Section 5.6 and Section 8.4 rather than by any single question. Table 5.1 gives the complementary view, what each method actually does and which metric its conclusion rests on.

RQ	Objective	H	Method	Evidence	Verdict
RQ1	O1 tail latency and jitter to overload	H1	§5.3.1	§7.5.1, §7.5.2	§8.2, §8.3
RQ2	O2 which defect classes are prevented, and when	H2	§5.3.2	§7.5.6	§8.2, §8.3
RQ3	O3 cost of the safety model in resources	H1	§5.3.1	§7.5.5, §7.5.2	§8.2, §8.3
RQ4	O4 seam size, `unsafe` scope, and per-call cost	H3	§5.3.3	§7.5.7	§8.2, §8.3

the fault-injection catalogue, and the FFI prototype. Explicitly out of scope are the full gNB system and its remaining control-plane domains, PHY/L2 processing and the data-heavy user plane, real captured traffic traces (as opposed to a spec-justified synthetic workload), managed-language alternatives (excluded on the tail-latency grounds argued in Chapter 2), and production deployment and operational concerns such as upgrade, observability, and multi-tenancy. These exclusions trade external validity for the controlled conditions a fair language comparison needs, and each is revisited as a limitation in Chapter 8.

5 Research Methodology

The previous chapter set out what the study must achieve; this one sets out how. It states the overall research strategy, the method chosen for each research question and why it is appropriate, the controls that make a language comparison fair, the statistical treatment of the measurements, and the reproducibility and ethical commitments under which the work is carried out.

5.1 Research approach

The thesis follows an empirical, experimental research approach with a constructive (design-science) element. A software artifact, two functionally equivalent implementations of the subsystem together with a benchmarking framework, is built against a single frozen specification and then evaluated under controlled experiments. Evidence is triangulated from three complementary methods: quantitative performance measurement under a swept load (addressing RQ1 and RQ3), a controlled fault-injection study of defect detection (RQ2), and a qualitative-plus-quantitative FFI migration case study (RQ4). Fairness is treated as a first-class concern: because the goal is to isolate language effects, the study fixes the specification, algorithms, concurrency model, allocator, and build settings across both implementations, and reports variance over repeated runs.

Table 5.1 maps each research question to the method, artifact, data, and metric used to answer it, making the chain from question to evidence explicit and traceable, and Figure 5.1 draws the same chain as a flow, which is where the split between the questions the measurement host bears on and those it does not becomes visible.

5.2 A constructive, comparative experiment

The four research questions are causal and comparative: each asks how an outcome (tail latency, prevented defects, resource use, or migration cost) depends on a single independent variable, the implementation language. Answering such questions credibly

Table 5.1: Research design: how each research question is answered. Every question is tied to a concrete method, a published artifact, the data it produces, and the metric on which the conclusion rests. RQ1 and RQ3 carry two methods rather than one: the sweep measures the round-trip a deployment would experience, and the component decomposition measures the part of it the language is responsible for. Section 5.3.1 states why one cannot stand in for the other.

RQ	Method	Artifact / data	Primary metric
RQ1	Load sweep to overload, interleaved repeats	Benchmark harness; latency histograms	p99/p99.9 tail latency and jitter vs. offered load
RQ3	Same sweep, resource capture	`getrusage` samples per run	peak RSS, CPU utilisation, throughput
RQ1, RQ3	Component decomposition: the core called directly, then through the service	Matched microbenchmark pair, interleaved repeats	per-operation cost split into the operation and the cross-thread handoff
RQ2	Controlled fault injection	Defect catalogue; ASan/TSan and `rustc` logs	defect-class taxonomy (compile-time / runtime / undetected)
RQ4	FFI migration prototype	C++ host + Rust C-ABI seam	seam size, `unsafe` scope, per-call interop cost

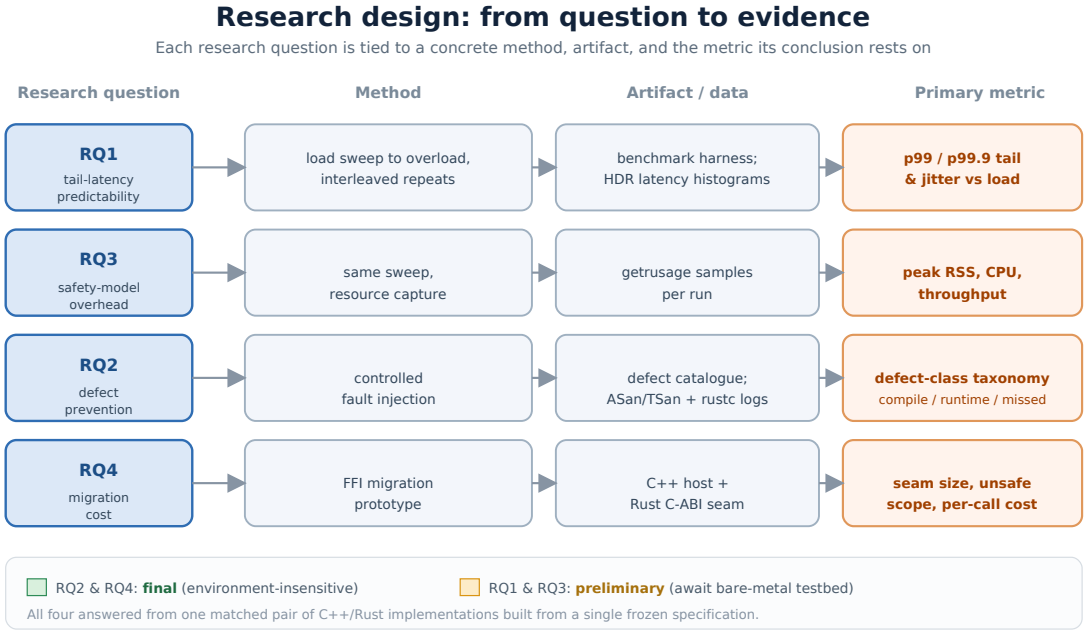


Figure 5.1: The research design as a flow, showing for each research question the method, the artifact and data it produces, and the primary metric its conclusion rests on. All four questions are answered from one matched pair of C++/Rust implementations built from a single frozen specification; RQ2 and RQ4 are environment-insensitive, while RQ1 and RQ3 depend on the measurement host and are reported against what it can carry (Section 7.1).

requires that the language be the *only* systematic difference between the two objects under test. This rules out the most common form of language comparison in the literature, in which two independently written programs, or two unrelated projects, are measured and their differences attributed to the language; such comparisons confound the language with the design, the libraries, and the intent and skill of their authors. The thesis therefore adopts a constructive (design-science) strategy: rather than locating two pre-existing systems, it *builds* the object of study as a matched pair of implementations derived from one frozen specification, so that the comparison is between two encodings of the same design rather than between two designs. The artifact, the specification, the two implementations, and the benchmarking harness, is thus simultaneously the contribution of the thesis and the instrument through which it is measured.

The experiment is empirical and quantitative where the questions admit numbers (RQ1, RQ3, and the per-call cost of RQ4) and systematic-but-qualitative where they do not (the kind-of-guarantee distinction in RQ2 and the API-design effort in RQ4). Evidence is deliberately triangulated across three complementary methods rather than resting on a single campaign, so that a weakness in any one method, for example the environment sensitivity of tail-latency measurement, does not undermine the conclusions as a whole.

5.3 Methods for the four research questions

Table 5.1 pairs each question with a method; this section states what each method does and why it is appropriate.

5.3.1 Performance measurement under a load sweep (RQ1, RQ3)

RQ1 and RQ3 are answered by driving both implementations with one deterministic synthetic workload across a sweep of offered load, from light load through saturation and into overload, and recording latency and resource use at each level. Two design decisions make the measurement sound. First, the load generator follows an *open-loop* schedule with a fixed intended send time per request, and latency is measured against that intended time rather than against the moment the request is actually admitted; this is what corrects for coordinated omission, the bias by which a naive

closed-loop generator stalls together with the system and silently fails to record the very requests that would have exposed the tail (Chapter 2). Second, the sweep is pushed past saturation deliberately: the operationally decisive behaviour of a control plane is not its latency at nominal load but the onset and shape of the overload cliff, and that is visible only once offered load exceeds service capacity. Latency is captured in high-dynamic-range histograms configured identically on both sides, and resource use (peak resident set size, CPU utilisation, throughput) is sampled through the operating system, so that the same measurement apparatus is applied to both languages.

Measuring the component as well as the round-trip. A sweep of this kind measures a round-trip, and a round-trip bundles two costs of quite different character. One is the operation the language executes inside a single thread: a hash lookup, a state-machine guard, and a field update, all of it code the compiler generated. The other is the cross-thread submit-and-complete handoff around it: an enqueue, a futex wake, a context switch, and a completion, none of it language-specific. Only the first can plausibly differ between C++ and Rust, and no end-to-end measurement can separate them. The method therefore operates at two levels. The single-threaded operation processor is called directly, with no queue in the path, and the same operation is then called through the full threaded service; the difference between the two is the handoff. Both are batch means over a long loop rather than per-call medians, deliberately: a clock read costs tens of nanoseconds, the same order as the operation itself, so per-call timestamps would measure the instrument rather than the code. The two languages are interleaved across repeats and compared with the same exact paired test as the sweep, so the two levels differ in what they measure rather than in how they are analysed.

Reporting both is what lets the study separate two questions that the research questions as originally worded did not distinguish: what a deployment of this shape would experience, which is the round-trip, and what the language is doing, which is the component. Section 7.5.2 finds the two to be more than two orders of magnitude apart, so the distinction is not a refinement but the difference between an interpretable result and an uninterpretable one. This second level was added after a first pass of the campaign rather than designed in from the start, and Section 8.6 records that revision instead of presenting the design as though it had always been complete.

5.3.2 Controlled fault injection (RQ2)

RQ2 asks which defect classes a language prevents, and it cannot be answered by measuring defect-free code. The absence of a bug in a passing test says nothing about whether the language would have caught it. The method is therefore adversarial, following the established practice of software fault injection, in which representative faults are deliberately introduced to assess how a system behaves in their presence [46].

A catalogue of representative C++ defects is written *deliberately*, drawn from the classes that affect this subsystem (Chapter 3). Each is a small self-contained program flavoured with the domain: a stale pointer into the context map, a double release, a mutation during iteration, an unsynchronised shared update. Each C++ defect is then built and run under AddressSanitizer and ThreadSanitizer, recording which tool, if any, detects it and at what stage. The analogous construct is written in safe Rust and compiled, recording the compiler's outcome and, where it compiles, the behaviour of the resulting program.

Why the C++ side is tooled with sanitisers only. No static analyser is run against the C++ catalogue, and that is a decision rather than an oversight. RQ2 does not ask which tool finds the most defects; it asks at what *stage*, and with what force, each language rules a defect out, and the taxonomy's three outcomes, compile-time rejection, runtime detection, and a silent miss, are the categories that question needs. A static analyser belongs to none of them. It is not the compiler refusing to build a program, and it is not an observation of an execution that occurred; it is an opt-in pass whose output is a warning a project can configure away, and which Section 2.3 has already noted suffers false negatives and cannot reason across a whole program.

What the choice costs is worth stating plainly, because it makes the C++ side of the resulting taxonomy narrower than the best a C++ project could manage. The memory-safety cases are small and intra-procedural, which is the shape at which static analysis is strongest, so a well-configured analyser could plausibly reach some of them; the concurrency cases, whose manifestation depends on an interleaving, are the shape at which it is weakest. The study therefore reports the tooling it ran, which is the sanitiser-based practice described in Section 2.3, rather than a ceiling on what C++ tooling can achieve. What no analyser would change is the property the comparison turns on: a warning a build can be configured to ignore is a different kind of object

from a compilation that cannot succeed, and it is that difference, rather than a count of detected defects, that the method measures.

The comparison is controlled in the same way as the performance one. Because the independent variable is again only the language while the injected fault is held fixed, the difference in outcome is attributable to the language's guarantees rather than to the specific bug. That outcome is one of three: compile-time rejection, runtime detection, or a silent miss. The result is a taxonomy rather than a performance number: the method establishes the *kind* of protection each language offers.

One design decision determines whether that taxonomy can be trusted. A catalogue drawn only from classes the language is known to prevent would produce a foregone conclusion: the method would be capable of confirming the hypothesis but not of disturbing it, and the result would describe the author's choice of cases as much as the language. The catalogue is therefore constructed in two groups. The first contains classes that lie inside Rust's stated guarantee; the second contains classes deliberately chosen from just outside it, where ownership and the `Send`/`Sync` discipline give no protection. Each case carries a recorded expectation, prevented or not prevented, and the runner reports a violation in either direction, so an unexpected outcome would surface as a discrepancy rather than being absorbed into the narrative. What this buys is a boundary rather than a score: the study can state not only which defect classes the language removes but which comparable ones it leaves in place, and the second half of that statement is the part a practitioner planning a migration needs. Figure 5.2 shows the procedure end to end, with the two groups on the same footing.

5.3.3 Foreign-function-interface migration case study (RQ4)

RQ4 concerns the practical cost of adopting Rust inside an existing C++ system, which is an engineering question as much as a performance one. It is answered by constructing a working prototype in the realistic direction of migration: an existing C++ host calling a Rust component across a C application binary interface. Three quantities are then measured on it. The *size* of the interoperability seam counts its entry points, lines of code, and dependencies. The *scope* of `unsafe` records where the checked language is left and whether the safety-critical core is affected. The *per-call cost* is the price of crossing the boundary, measured against the same operation invoked natively so that the comparison isolates what the seam adds rather than what

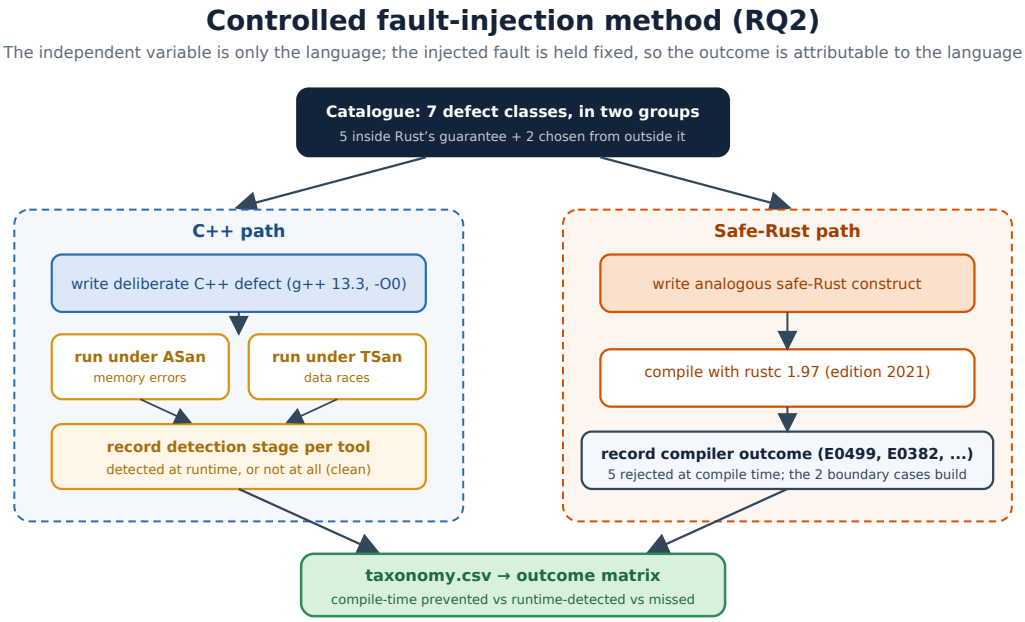


Figure 5.2: The controlled fault-injection method for RQ2. A fixed catalogue of representative defect classes is confronted in two languages: each C++ defect is built and run under AddressSanitizer and ThreadSanitizer to record the detection stage, while the analogous safe-Rust construct is compiled to record the compiler's outcome and, where it compiles, run to record its behaviour. Because only the language varies while the injected fault is held fixed, the difference in outcome is attributable to the language's guarantees. The catalogue spans classes inside and outside those guarantees so that the method locates their boundary rather than only confirming them.

the operation costs. The first two are structural and are read off the code; the third is a matched pair of microbenchmarks. This combination of a structural account and a quantitative cost measurement is what lets the study separate the run-time cost of interoperability from its true cost, which lies in API design and human learning.

5.4 Fairness as a controlled-variable strategy

Because every research question attributes a difference to the language, the validity of the whole study depends on holding everything else equal, and fairness is treated as the study's central experimental control rather than as an afterthought. It is enforced at several levels. The *specification* is frozen before any code is written and fixes the public API, the state machine, the data structures and their algorithmic strategy, and the sharding and concurrency model (Section 6.1). The *run-time substrate* is equalised by linking the same memory allocator on both sides and matching build-optimisation levels as closely as the two toolchains allow. The *workload* is produced by a driver that is identical in structure across languages and, for a given seed and active-UE population, drives both implementations through exactly the same sequence of operations, so that any latency difference reflects how each language executes the same work rather than a difference in what work is done. Where the two languages genuinely cannot express the same construct, the deviation is not hidden but recorded in a forced-idiom log so it can be weighed explicitly in the discussion. What remains uncontrolled, principally single-author skill asymmetry, is acknowledged as a threat to validity (Chapter 8) rather than claimed to be eliminated.

Table 5.2 states the strategy variable by variable, and its third column is the part that matters most. A controlled experiment is credible in proportion to the honesty of its account of what it failed to control, and three of the rows below record a residual rather than a clean fix: the languages cannot always express the same construct, they do not ship the same hash-map layout, and one person wrote both implementations. Figure 5.3 traces the strategy as a flow instead, which is what makes the two escape valves legible as parts of the design rather than as exceptions to it: the forced-idiom log, where the languages cannot express the same construct, and the one variable left with no control at all.

Table 5.2: The controlled-variable strategy: what is held fixed so that the language remains the only systematic difference, and what each control does not reach. The third column is not a list of oversights but the study's own account of its limits; each entry is picked up again as a threat to validity in Section 8.6. Only the last row names a variable for which there is no control at all.

Variable	How it is held fixed	What the control does not reach
Design	One specification, frozen before any code is written, fixing the public API, the state machine, the data structures and their algorithmic strategy, and the sharding granularity (§6.1)	Where the two languages cannot express the same construct, the divergence is logged rather than removed (Table 6.2)
Concurrency model	Thread-per-shard over bounded blocking MPSC queues on both sides, with no asynchronous runtime on either	—
Hashing	SplitMix64 for shard selection and FNV-1a inside the shard, pinned identically, so neither standard library's default hasher becomes an accidental advantage	The container itself is still each language's own hash map, and the two differ in layout (§7.5.2)
Allocator	The same `jemalloc` linked on both sides, with a system-allocator build kept on each as a control	—
Optimisation	`-O2` against `opt-level = 2`	Nominal parity is not identical code generation
Workload	One deterministic driver; for a given seed and active-UE population both implementations execute the same operation sequence, checked by per-kind counts	Synthetic and spec-justified rather than replayed from production traffic
Instrumentation	The same HDR histogram algorithm and configuration, the same open-loop schedule, the same repeat and interleaving discipline	—
Behaviour	A language-neutral conformance suite whose canonical transcripts are diffed op for op (§6.7)	Parity bounds behaviour, not performance
Author	No control	One person wrote both implementations and is not equally expert in both

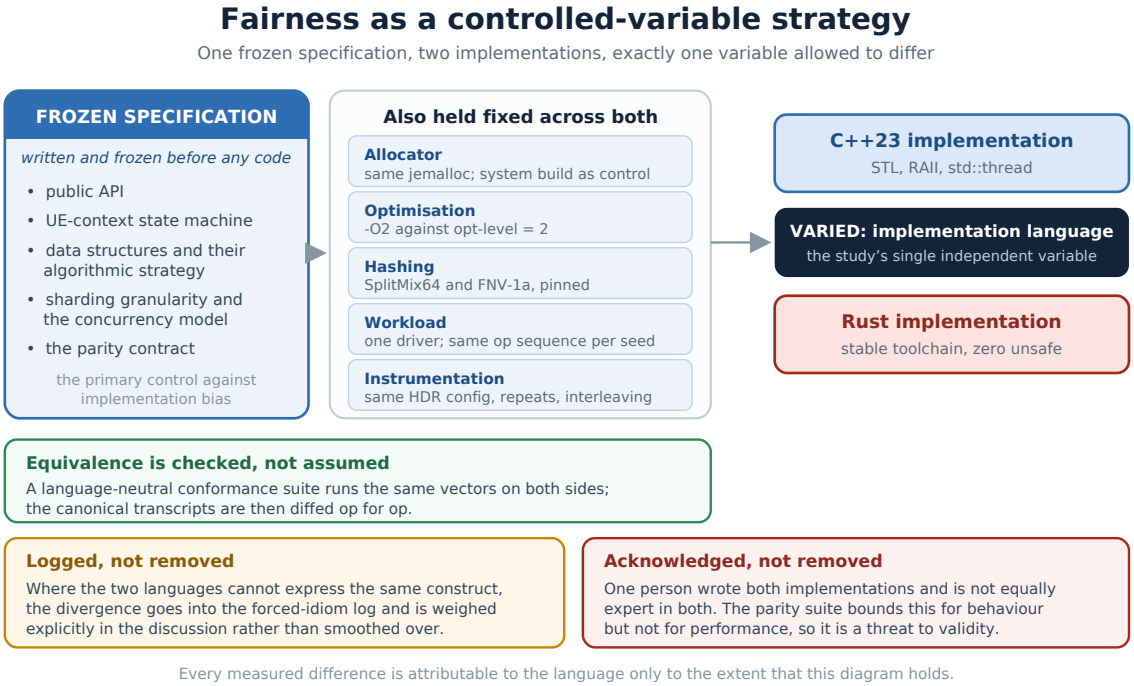


Figure 5.3: Fairness as a controlled-variable strategy. A single specification is frozen before any code is written and fixes everything the two implementations share, so that exactly one variable, the implementation language, is allowed to differ. The two escape valves are shown rather than hidden: constructs the languages cannot express identically are recorded in the forced-idiom log and weighed in the discussion, and the one variable with no control at all, single-author skill asymmetry, is carried forward as a threat to validity.

5.5 Statistical treatment of measurements

Performance measurements are non-deterministic, so single-run numbers are avoided throughout. Each configuration (language × allocator × load level) is measured over many independent repeats, and the four variants are *interleaved* within each repeat so that slow drift, thermal effects, and background activity perturb them equally rather than biasing one. Results are summarised by the median with a nonparametric bootstrap 95% confidence interval rather than by the mean with a standard deviation, because latency distributions are heavy-tailed and skewed and a distribution-free interval on the median makes no normality assumption. This follows established guidance on statistically rigorous performance evaluation, which warns that reporting a single number invites conclusions that run-to-run variation does not support [22]. Tail percentiles (p99, p99.9, p99.99) are read from the HDR histograms, whose logarithmic bucketing retains three significant figures of resolution across the whole range so that high percentiles are not quantised away. Intervals are reported in full rather than as a symmetric half-width, because a bootstrap interval on the median of a heavy-tailed sample is markedly asymmetric and a half-width wide enough to cover its upper bound would place the lower bound below zero.

Interval estimates alone, however, are a weak basis for a comparison. Two intervals can overlap while one variant is ahead in every single repeat, and they can separate on a difference that does not reproduce. The study therefore also tests the comparison directly, and the interleaved design is what makes this possible: because the four variants are measured back to back inside each repeat, the runs form a matched design in which the interference common to a block cancels in the pairwise difference. Language is compared against language with the allocator held fixed, using an exact two-sided paired permutation test on the repeat-wise differences. The null hypothesis is that a pair differs only by run-to-run noise, under which the sign of each paired difference is exchangeable; for the repeat counts used here every one of the 2^n sign assignments is enumerated rather than sampled, so the resulting p-value is exact and, like the bootstrap, assumes nothing about the shape of the distribution. Because the same comparison is made for several metrics and both allocators, the family of tests is corrected with the Holm step-down procedure, which controls the family-wise error rate without assuming independence between the tests. Figure 5.4 gathers this treatment together with the two decisions set out next.

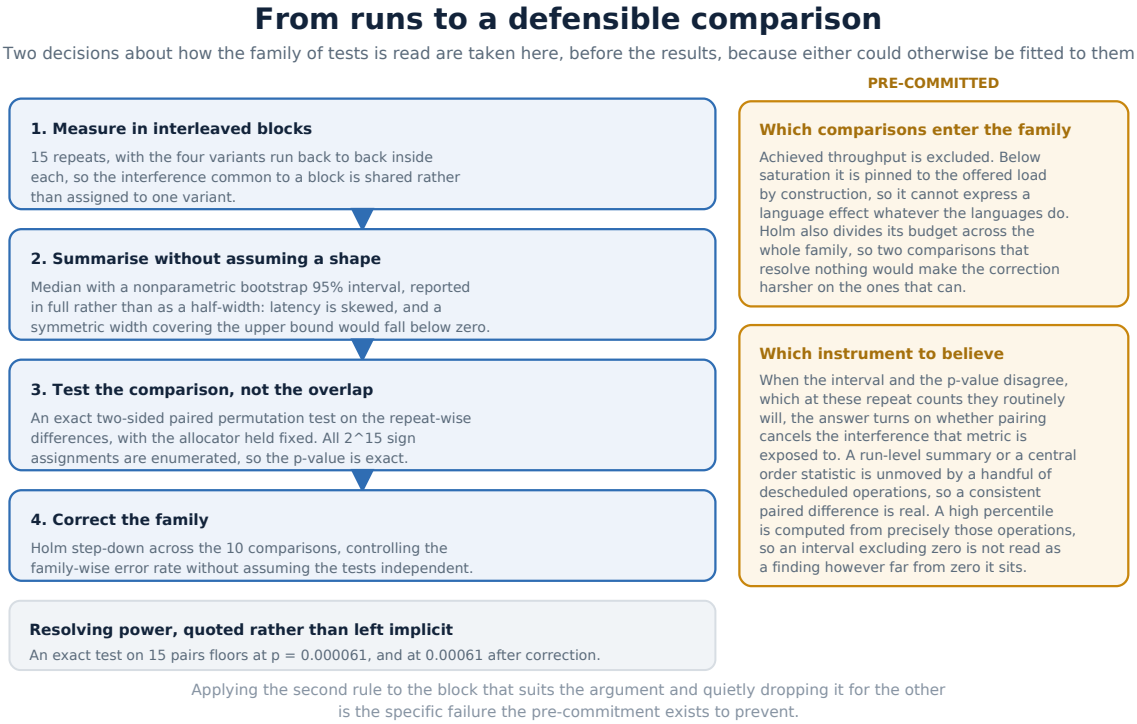


Figure 5.4: From runs to a defensible comparison. The numbered column is the analysis applied to every configuration, each step chosen so that it assumes nothing about the shape of a heavy-tailed distribution. The column beside it records the two decisions about how the resulting family of tests is read; both are taken before the results because either could otherwise be made to suit whatever the data turned out to be. The floors quoted at the foot are what let a reader tell a null result from an unresolvable one.

Two decisions about how that family is read are taken here rather than after the results, because either could otherwise be made to suit whatever the data turned out to be.

The first is which comparisons enter it, and the answer turns on the operating point rather than on the metric. The representative operating point is chosen below saturation, where every variant completes work as fast as it is offered, so achieved throughput there is pinned to the offered load by construction and cannot express a language effect whatever the languages do. Testing it would re-measure the operating point rather than the code. It would also not be free: Holm divides its budget across the whole family, so two comparisons that cannot resolve anything would make the correction harsher on the ones that can. Below saturation, therefore, throughput is reported with its interval in the resource table, where its job is to confirm that the operating point is sustained rather than nominal.

Above saturation that reasoning inverts, and the design has to follow it. Once the offered rate is no longer being met, what each variant still sustains is pinned to nothing and becomes the one quantity that can separate them; excluding it there would discard the only observable the regime offers. A second operating point is therefore fixed in the overload region, defined as the deepest swept load at which *every* variant has stopped meeting the offered rate, and sustained throughput and CPU utilisation are tested at it (Section 7.5.4). Two commitments go with that, both taken here for the same reason as the rest of this section. The overload comparisons join the same Holm family rather than forming a second one, so the correction covers everything the thesis reports and the division into two operating points cannot be read as a split chosen once the p-values were known. And no latency percentile is tested above saturation at all: with a fixed intended send schedule, a producer that cannot keep up falls behind cumulatively, so the percentiles there measure accumulated backlog against a measurement window that truncates it, which makes them a lower bound rather than an estimate. Attaching a p-value to a lower bound would test the harness rather than the software.

The second is what to conclude when the interval and the p-value disagree, which at modest repeat counts they routinely will: a paired interval can sit wholly on one side of zero while the corrected p-value stays far above any conventional threshold. Which instrument to believe is not a free choice, and it does not depend on whether the metric happens to be interesting. It depends on whether pairing removes the interference that particular measurement is exposed to. Pairing cancels whatever an interleaved block shared; it cannot cancel interference that landed on one run of a pair and not on the

other. A run-level summary such as peak resident set size, and a central order statistic such as the median of tens of thousands of per-operation samples, are both unmoved by a handful of descheduled operations, so a paired difference that is consistent in sign across repeats is a difference in the software. A high percentile is the opposite case: p99 and p99.9 are computed from precisely the operations that were disturbed, so an interval excluding zero is as consistent with one run of a pair having been unlucky as with a real effect. Table 7.4 is therefore blocked by that distinction, and intervals in its second block are not read as findings however far from zero they sit. Applying the rule to the block that suits the argument and quietly dropping it for the other is the specific failure this pre-commitment exists to prevent.

The resolving power the design can reach is bounded by the number of repeats, and is stated rather than left implicit. An exact test on n pairs cannot return a two-sided p below $2/2^n$, and after correcting a family of m comparisons the smallest attainable adjusted value is m times that. With 20 repeats and 14 comparisons, those two floors are 0.0000019 and 0.000027. Quoting them is what lets a reader tell a null result from an unresolvable one. A campaign whose corrected floor lies above the threshold being applied cannot demonstrate significance however large the effect, so reporting "not significant" from such a campaign would say more about its repeat count than about the languages. Choosing the repeat count so that the corrected floor sits below the conventional 5% is therefore part of the measurement protocol, alongside pinning the clock and isolating the cores (Section 7.1).

5.6 Reproducibility and research artifacts

The study is designed so that its results can be regenerated by a third party without access to Nokia's proprietary code. Three things make that possible. The implementations are driven entirely by the public frozen specification. The benchmarking harness, the fault catalogue and its runner, and the FFI prototype are published together with the thesis. And every figure and table in the evaluation is produced by a single script directly from the raw measurement logs, so that rerunning the campaign on a different host regenerates the artifacts verbatim.

The execution environment is captured automatically alongside each result set: CPU model, kernel, compiler and toolchain versions, and the frequency and scheduling settings of the testbed. The conditions of a measurement therefore travel with its data.

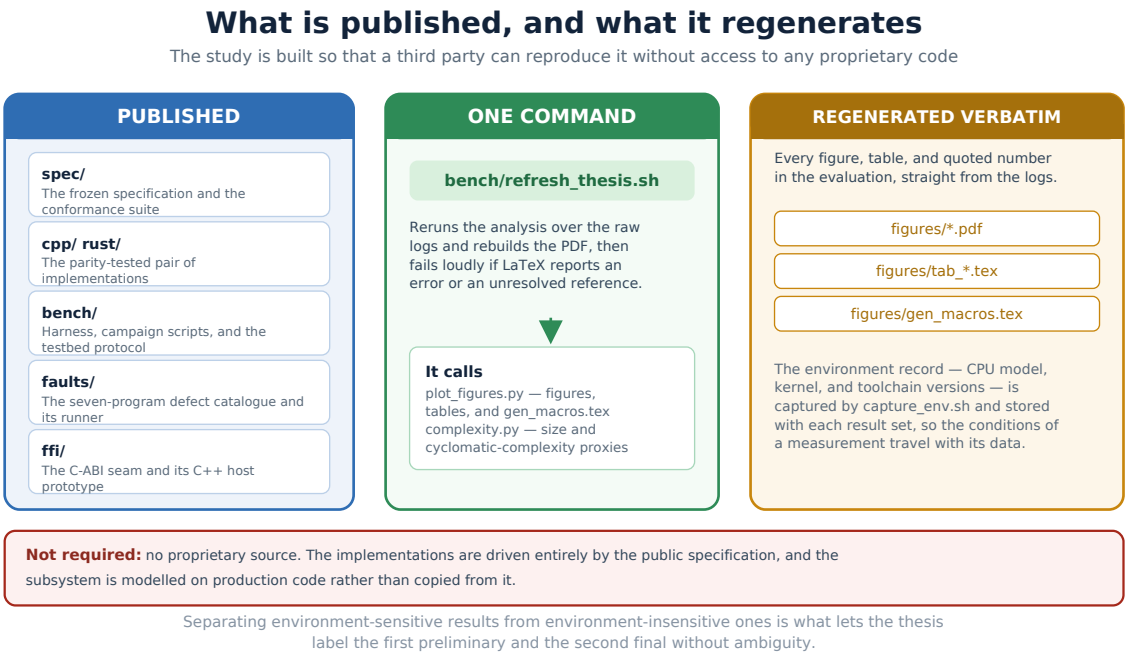


Figure 5.5: What is published, and what it regenerates. The left column is the artifact set released with the thesis, the middle the single command that reruns the analysis over the raw logs, and the right what that command reproduces verbatim. The environment record travels with each result set, so a measurement carries the conditions under which it was taken. Nothing on the path requires proprietary source, which is what makes third-party reproduction possible at all rather than merely permitted.

That matters for one distinction in particular. Separating environment-sensitive results (the performance sweep) from environment-insensitive ones (parity testing, fault injection, and the FFI structure) is what lets the thesis state, per result, which of the two a reader is looking at. Figure 5.5 sets out the published artifact set, the single command that regenerates the evaluation from it, and what that command reproduces.

5.7 Ethical and data-protection considerations

The study processes no personal, subscriber, or operator-confidential data. All traffic is synthetic and generated by the deterministic driver, the UE contexts contain only fabricated identifiers and a fixed dummy security field, and the subsystem is modelled on, rather than copied from, production code. There are therefore no privacy, consent, or data-retention concerns, and the resulting artifacts can be released openly for reproduction.

6 Design and Implementation

6.1 Shared specification: the fairness baseline

Before any code is written, a single language-agnostic specification is frozen: the public API, the UE state machine, the data structures and their algorithmic strategy (e.g. a sharded hash map keyed by UE identifier), the locking and sharding granularity, and the concurrency model. Both implementations conform to this specification so that the comparison measures language differences rather than design or skill asymmetry. The specification is itself a deliverable and is the primary control against implementation bias.

6.2 Conceptual model and concurrency design

Two functionally equivalent implementations are developed against the shared specification: a modern C++ version (C++23, STL, RAII, standard threading) and a Rust version (stable toolchain, ownership-based memory model, safe concurrency). A single concurrency model is used for both: a *shared-nothing, thread-per-shard* model that mirrors how real RAN L3 partitions UE state, and mixing of async and thread-based models is intentionally avoided.

The UE-identifier space is partitioned into N_{shards} shards (eight by default; the measurement campaign pins seven so that each worker and each producer owns one of the pinned cores, as Section 7.1 records), each owned by exactly one worker thread. Only that worker reads or mutates its shard's contexts, so there is no shared mutable state between workers and no locks on context data. Callers (load-generator threads) never touch context state directly; they submit request messages to a shard's bounded multi-producer, single-consumer (MPSC) queue and wait on a single-use completion handle to which the worker writes the result. The queue is *bounded and blocking*: when a shard's queue is full, submission blocks until space frees up, which produces realistic queueing latency under overload and makes tail latency the observable of the study (a non-blocking drop policy would hide overload). Requests to the same shard are

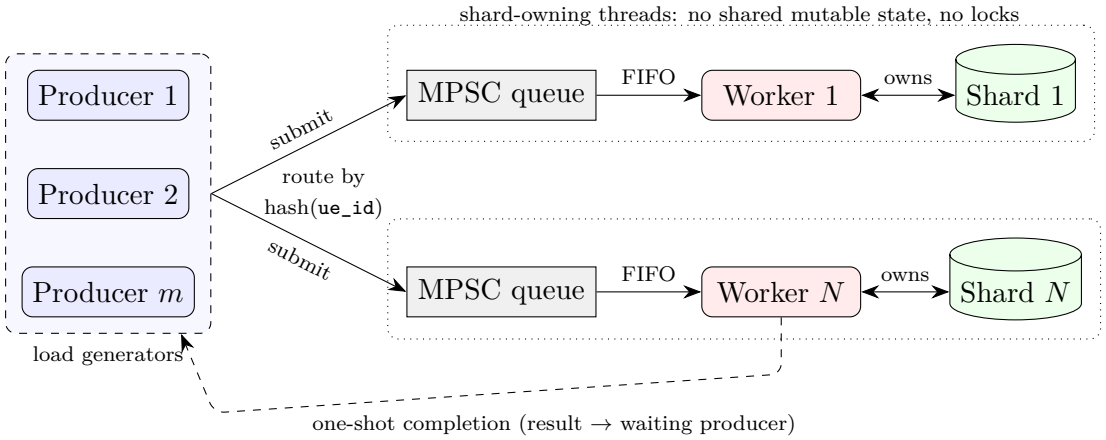


Figure 6.1: Shared-nothing, thread-per-shard architecture. Producers submit request messages to a shard's bounded, blocking MPSC queue (selected by a pinned hash of the `ue_id`) and wait on a single-use completion handle; each shard is read and written by exactly one worker thread, so there are no locks on context state. Blocking on a full queue is what makes overload observable as tail latency.

processed strictly first-in-first-out. This structure is what makes the safe-concurrency comparison meaningful: the interesting machinery is the queue, the completion hand-off, and the ownership transfer of requests, rather than lock contention. The only shared mutable state permitted is a single atomic counter used to stamp a per-context activity sequence number, which is write-only bookkeeping that does not affect control flow. Figure 6.1 shows this data flow.

6.3 Architectural choices and technology stack

The C++ side uses CMake with Ninja, the standard library, and `std::thread`; the Rust side uses Cargo and the stable toolchain. To keep the comparison a language comparison rather than an allocator comparison, both link the same memory allocator: C++ links the system `jemalloc` and Rust installs a `jemalloc` global allocator (from `tikv-jemallocator`), with a system-allocator build retained on each side as a control. Both record latency through the same high-dynamic-range (HDR) histogram algorithm [47], configured identically (range $[1\,\text{ns}, 3.6\,\text{s}]$, three significant figures); Rust uses the canonical `hdrhistogram` crate and C++ uses a header-only port of the same bucketing algorithm, validated against a known distribution. Build optimisation is matched as closely as the two toolchains allow: the Rust release profile is pinned to `opt-level = 2` to match the C++ `-O2` build.

6.4 Implementation details

Both implementations follow the frozen specification field-for-field, and both are small enough to be read end to end. The C++ core is 649 lines across eight headers and two translation units, the Rust core 736 lines across eight modules. Discarding blank lines and comments leaves 516 and 602 respectively.

The Rust version is therefore the larger of the two, by 13% in raw lines and 17% in code lines. That is worth stating plainly rather than leaving it to a footnote, for two reasons. Line count is one of the software-quality proxies this study set out to collect, and the difference is not incidental verbosity: it is the net of several forced idioms pulling in opposite directions, which Section 6.4.7 decomposes file by file. The remainder of this section works outward from the data model to the concurrency machinery, and reports the measured consequences of each language-forced divergence as it arises.

6.4.1 The UE context and its memory layout

Each UE context stores its identifier, lifecycle state (a four-value enumeration `Idle`, `Connected`, `Reconfiguring`, `Released`), serving cell, RRC transaction id, a fixed 32-byte dummy security context, a dynamic list of up to eight data radio bearers, and two counters (messages handled and an activity sequence number). The data radio bearer list is deliberately a growable container (`std::vector` / `Vec`) rather than a fixed array, so that the allocator study (RQ3) observes realistic per-context allocation. Reads return an immutable snapshot (a `UeContextView`) copied out of the worker rather than a reference into live shard state, which preserves the shared-nothing guarantee and keeps the two implementations semantically identical. Figure 6.2 gives the record and the read path in full.

Identical field declarations do not, however, produce identical objects. Measured on the evaluation host, a `UeContext` occupies 96 bytes in C++ but only 88 bytes in Rust. The difference is entirely layout policy. C++ guarantees that non-static data members are laid out in declaration order, so the one-byte `state` enumeration sitting between the eight-byte identifier and the four-byte cell id forces three bytes of padding, and the 32-byte security key ending at offset 50 forces a further six before the eight-byte-aligned bearer vector. Rust, absent a `#[repr(C)]` annotation, is free to reorder fields and does so, packing the small scalars together and eliminating both holes. The speci-

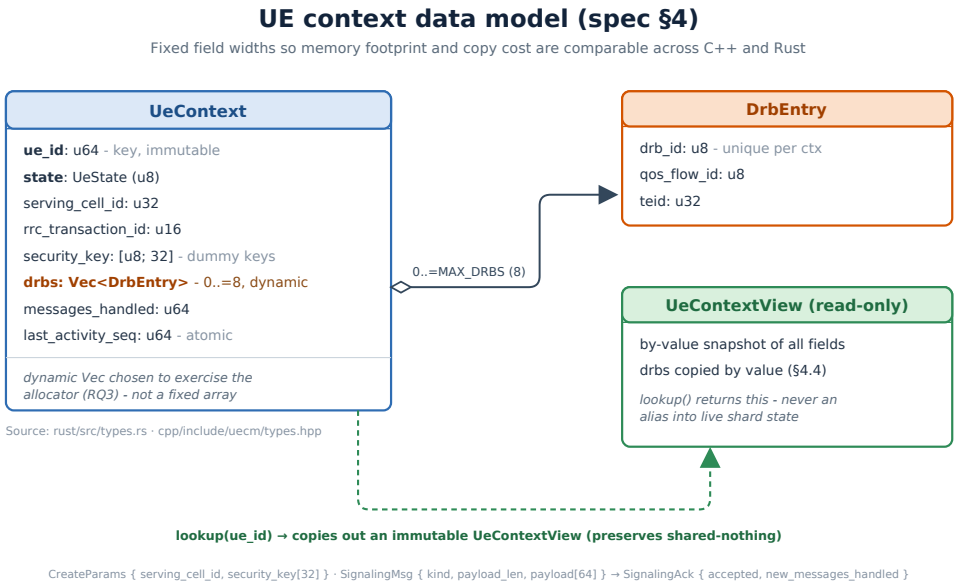


Figure 6.2: The UE-context data model shared by both implementations. Field widths are fixed so that memory footprint and copy cost are comparable across languages; the data-radio-bearer list is a dynamic `Vec`/`std::vector` on purpose, to exercise the allocator for RQ3. `lookup` returns an immutable `UeContextView` copied out by value rather than an alias into live shard state, preserving the shared-nothing guarantee.

fication deliberately does not try to erase this difference: pinning the Rust layout with `#[repr(C)]` would penalise Rust for nothing but conformity to a C++ padding rule, while hand-reordering the C++ declarations would be an optimisation the specification asks of neither side. It is instead recorded as a language-level effect and bounded, at the 5 000 active UEs used in the sweep the eight-byte gap accounts for roughly 40 kB, against resident set sizes measured in megabytes, so it is immaterial to the RQ3 comparison.

6.4.2 The state machine as implemented

The lifecycle of Figure 3.1 is realised as a single guarded transition per operation, and Table 6.1 gives the implemented table in full. Two properties hold uniformly and are what make the two implementations comparable op for op. First, every operation validates before it mutates: the guard sequence checks the identifier, then locates the context, then tests the state, and returns the corresponding `OpError` before touching any field, so a rejected operation leaves the context bit-identical to what it was. Second, no operation can partially succeed, because each performs at most one mutation after

Table 6.1: The implemented state-transition table, identical in both languages. Every operation additionally returns `InvalidArgument` for a zero `ue_id`, and a rejected operation leaves the context unmodified. `Released` is never stored: release erases the entry.

Operation	Required state	Resulting state	Errors
`create_context`	context absent	`Idle`	`AlreadyExists`, `CapacityExceeded`
`setup_connection`	`Idle`	`Connected`	`NotFound`, `InvalidState`
`handle_signaling`	`Connected`	`Connected`	`NotFound`, `InvalidState`
`add_drb`	`Connected`	`Connected`	`NotFound`, `InvalidState`, `CapacityExceeded`, `AlreadyExists`
`begin_reconfigure`	`Connected`	`Reconfiguring`	`NotFound`, `InvalidState`
`complete_reconfigure`	`Reconfiguring`	`Connected`	`NotFound`, `InvalidState`
`abort_reconfigure`	`Reconfiguring`	`Connected`	`NotFound`, `InvalidState`
`release_context`	any live state	entry erased	`NotFound`
`lookup`	any live state	unchanged	`NotFound`

the last guard has passed.

Two behaviours deserve explicit mention because the table alone hides them. The `Released` state is never actually stored: `release_context` erases the entry from the shard map outright, so `Released` is a transient terminal label in the specification rather than a resting value an implementation can be found in, and a second release therefore returns `NotFound` rather than `InvalidState`. And `abort_reconfigure` restores the *state* to `Connected` but does not restore the serving cell that `begin_reconfigure` overwrote with the handover target. This is a deliberate simplification, the model carries no saved pre-handover cell to roll back to, but it is a semantic commitment rather than an oversight, and both implementations make it identically, which is what the parity suite of Section 6.7 checks.

6.4.3 Shard selection and the intra-shard store

To prevent the standard library hashers from becoming an accidental language advantage, both sides pin their hashing. Shard assignment applies the SplitMix64 finalizer to the `ue_id`, three xor-shift and multiply rounds with the constants `0xBF58476D1CE4E5B9` and `0x94D049BB133111EB`, before reducing modulo the shard count. The finalizer matters because the workload draws sequential identifiers, which

a naive modulo would map onto shards in a fixed repeating pattern. Diffusing the identifier first makes the offered load per shard uniform in both languages by construction rather than by luck.

The per-shard container is each language's standard hash map (`std::unordered_map` / `std::collections::HashMap`), but the default hashers are bypassed. They differ too much to leave in place: C++'s `std::hash<uint64_t>` is typically the identity function, while Rust's default `SipHash-1-3` is cryptographically strong and correspondingly slower. Keeping them would have measured a standard-library choice rather than a language. Both are therefore keyed through a pinned FNV-1a 64-bit hash over the eight little-endian bytes of the identifier, and a test checks that the Rust `BuildHasher` produces a digest bit-identical to the reference `fnv1a_64`.

Both maps reserve an initial capacity of 4 096 entries at start-up so that steady-state measurements are not disturbed by rehashing. The remaining configuration is likewise shared: the default eight shards, a queue bound of 1 024 requests per shard, at most eight bearers and a 64-byte signaling payload per context, and a ceiling of 200 000 contexts per shard. Only the shard count is varied from these defaults, and only by the campaign, which sets it to match the cores it is given.

6.4.4 The bounded queue

The bounded blocking queue is where the two standard libraries diverge most sharply, and Listing 6.3 shows why. Rust's standard library provides the primitive the specification calls for directly: `sync_channel(capacity)` is a bounded multi-producer, single-consumer channel whose `send` blocks when the queue is full and whose `recv` blocks when it is empty, which is precisely the required backpressure semantics in a single call. The C++ standard library, through C++23, offers no bounded blocking queue at all, so the same semantics must be assembled by hand from a `mutex`, two `condition_variable`s, and a `std::queue`, and the result is the 45-line class reproduced on the left.

The point of the comparison is not that the C++ code is badly written; it is short, it uses the predicate form of `wait` so spurious wakeups are handled correctly, and it releases the lock before notifying. The point is that it exists at all. Every one of those lines is a place where a concurrency defect could be introduced, and the correctness argument for it, that the two condition variables are never confused, that the predicates

C++: hand-written, 45 lines

```
template <class T> class BoundedQueue {
 public:
  void push(T value) {
    std::unique_lock lk(m_);
    not_full_.wait(lk, [&]{
        return q_.size() < capacity_; });
    q_.push(std::move(value));
    lk.unlock();
    not_empty_.notify_one();
  }
  T pop() {
    std::unique_lock lk(m_);
    not_empty_.wait(lk, [&]{
        return !q_.empty(); });
    T value = std::move(q_.front());
    q_.pop();
    lk.unlock();
    not_full_.notify_one();
    return value;
  }
 private:
  const std::size_t capacity_;
  std::mutex m_;
  std::condition_variable not_full_;
  std::condition_variable not_empty_;
  std::queue<T> q_;
};
```

Rust: standard library

```
let (tx, rx) =
    sync_channel(cfg.queue_capacity);

// tx.send(req) blocks when full
// rx.recv()    blocks when empty
```

Figure 6.3: The bounded blocking MPSC queue in both languages (C++ abridged only by the omission of the constructor). The semantics are identical; the C++ standard library simply does not supply the primitive, so it is assembled by hand from a mutex and two condition variables, and its correctness is argued rather than guaranteed.

match the notifications, that no wakeup is lost, is carried by the author rather than by the compiler or the library. This is the same asymmetry the fault-injection study of Chapter 7 measures, appearing here in ordinary, non-defective code rather than in a deliberately injected bug.

6.4.5 Requests, responses, and the cost of a tagged struct

The message type carried through that queue is the second forced divergence, shown in Listing 6.4. Rust models a request as an `enum` with one variant per operation; each variant carries exactly its own arguments plus a `Sender` parameterised by exactly that operation's result type. C++ has no equivalent of a payload-carrying enumeration in the standard idiom used here, so a request is a tagged struct: an `OpKind`

discriminant alongside *every* argument field any operation might need, paired with a single `std::promise` whose value type must therefore be a `std::variant` over all four possible result types.

This has three measurable consequences. The first is size: a `Request` is 160 bytes in C++ against 96 in Rust, because the C++ struct always carries the create parameters, the 68-byte signaling message, the bearer entry, and the reconfiguration parameters, whether or not the operation in flight uses them, while the Rust enum is sized by its largest variant alone. With the default eight shards holding up to 1 024 requests each, that is roughly 1.3 MB of queue-resident request storage at full occupancy against 0.8 MB. The second is a copy cost paid on every submission, since the unused fields are default-initialised and moved along with the used ones. The third, and the most interesting for the thesis's argument, is a type-safety gap: the C++ caller must recover its typed result from the variant with `std::get<OpResult<T>>`, which throws `std::bad_variant_access` if the worker ever writes an alternative the caller did not expect. That exception cannot fire in the present code, because each `OpKind` is handled by exactly one branch that stores the matching alternative, but the invariant guaranteeing it is maintained by inspection across two files. In Rust the same invariant is discharged by construction: because the sender embedded in each variant is already typed, there is no response union, nothing to unwrap, and no runtime dispatch on the way back.

6.4.6 Service lifecycle, shutdown, and ownership

Starting the service allocates the shards, spawns one worker thread per shard, and hands each worker exclusive ownership of its store; the request path is then as described in Section 6.6. Shutdown is the more delicate half, because it must terminate workers that are blocked on an empty queue without abandoning requests that are still in flight. Both implementations solve it the same way, with an in-band `Stop` sentinel: one `Stop` request is pushed onto each shard's queue, and because the queue is strictly FIFO, every request already enqueued ahead of it is executed and completed before the worker observes the sentinel and returns. No separate closed flag, timeout, or drain API is required, and no completion handle is left unfulfilled. The worker threads are then joined, so shutdown returns only once every shard has actually stopped. Figure 6.5 shows the three phases and why the sentinel alone is enough.

C++: tagged struct + variant (160 B)

```
enum class OpKind : std::uint8_t {
  CreateContext, SetupConnection,
  HandleSignaling, AddDrb, /* ... */
  Lookup, Count, Stop };

using Response = std::variant<
    OpResult<void>,
    OpResult<SignalingAck>,
    OpResult<UeContextView>,
    OpResult<std::uint64_t>>;

struct Request {          // move-only
  OpKind kind{};
  UeId ue_id = 0;
  CreateParams create_params{};
  SignalingMsg signaling{};
  DrbEntry drb{};
  ReconfigParams reconfig{};
  std::promise<Response> completion;
};
```

Rust: enum + typed sender (96 B)

```
pub enum Request {
    Create { ue_id: UeId,
        params: CreateParams,
        resp: Sender<OpResult<()>> },
    Signal { ue_id: UeId,
        msg: SignalingMsg,
        resp: Sender<
            OpResult<SignalingAck>> },
    Lookup { ue_id: UeId,
        resp: Sender<
            OpResult<UeContextView>> },
    // ... one variant per operation
    Stop,
}
```

Figure 6.4: The request message in both languages (abridged: C++ elides four `OpKind` values, Rust seven variants). The C++ tagged struct carries every argument field on every request and returns a union that the caller must unwrap; the Rust enum carries only the arguments of the operation in flight and a sender already typed to that operation's result.

Both languages tie this to object destruction, and the mechanism is one of the few places where the two are near-identical: the C++ destructor calls `shutdown()`, and Rust's `Drop` implementation does the same, so a service that goes out of scope cannot leave detached threads behind. Both guard against double invocation with a `running` flag, making shutdown idempotent. The remaining ownership discipline is what upholds the shared-nothing invariant end to end: a request is moved into the queue rather than copied, carrying its completion handle with it, so at no point do the producer and the worker both hold a usable reference to the same request. In C++ this is enforced by making `Request` move-only, a property it acquires automatically from the move-only `std::promise` it contains; in Rust it follows from ordinary move semantics, and the compiler rejects any attempt to use the request after submission. The only shared mutable state in either implementation is the atomic activity counter, held by reference in C++ and as a cloned `Arc<AtomicU64>` in Rust, and incremented with relaxed ordering because nothing branches on its value.

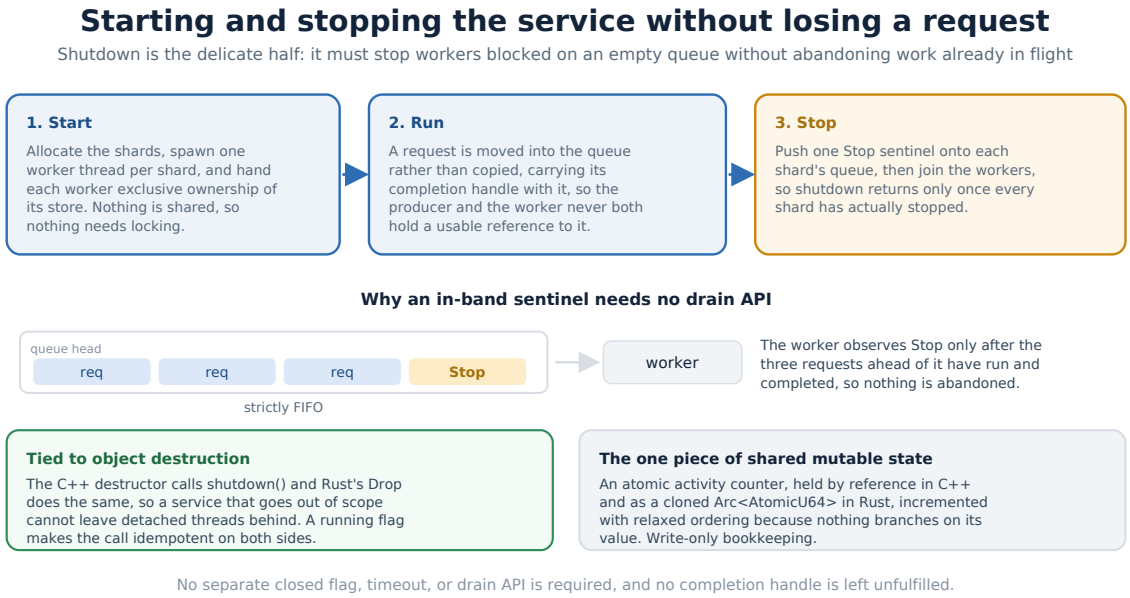


Figure 6.5: Starting and stopping the service. Startup hands each worker exclusive ownership of one shard, which is what removes locks from the request path; the run phase moves a request into the queue rather than copying it, so producer and worker never both hold a usable reference to the same request. Shutdown pushes a single `Stop` onto each shard's queue and relies on strict FIFO ordering to guarantee that everything enqueued ahead of it is executed and completed first. That is why no drain API, timeout, or closed flag is needed, and why no completion handle is left unfulfilled.

Table 6.2: The forced-idiom log: every point at which the two implementations diverge in order to satisfy the same specification requirement, with the measured consequence where there is one.

Concern	C++	Rust
Result type	`std::expected<T, OpError>`	`Result<T, OpError>`
Request	tagged struct: discriminant plus every argument field (160 B)	`enum`, one variant per operation, carrying only its own arguments (96 B)
Response	`std::variant` of four result types, unwrapped by the caller with `std::get`	none needed; each variant carries a sender of its own result type
Bounded queue	hand-written `mutex` plus two `condition_variable`s (45 lines)	`sync_channel(capacity)`
Completion	`std::promise` / `std::future`	a fresh `mpsc::channel()` per request
Per-shard map	`std::unordered_map` with an `FnvHash` functor	`HashMap` with an `FnvBuildHasher`
Activity counter	`std::atomic<uint64_t>` held by reference	`Arc<AtomicU64>` cloned into each worker
Teardown	destructor calls `shutdown()`	`Drop` calls `shutdown()`
Optimisation level	`-O2`	`opt-level = 2`

6.4.7 Forced idioms and what they cost

Where the two languages must diverge to satisfy the same requirement, the deviation is recorded in the specification's forced-idiom log so that it can be weighed rather than silently absorbed into the results. Table 6.2 collects the log in full, and the preceding subsections have already quantified the two entries with measurable consequences.

The log also explains the line-count difference reported at the head of this section, and it is more informative than the net figure suggests, because the contributions do not all point the same way. Table 6.4 decomposes the 87-line gap by concern. Rust is substantially *shorter* in exactly one place, the bounded queue, where it needs no code at all against 45 lines of C++, and longer almost everywhere else, with three contributions accounting for most of the excess.

The largest, and the one that best illustrates the trade-off, is hashing. C++ satisfies `std::unordered_map`'s hasher requirement with a five-line functor wrapping a call to `fnv1a_64`, whereas Rust must supply the whole trait apparatus: a `FnvHasher` holding

the running state, a `Default` implementation seeding it with the offset basis, a `Hasher` implementation providing `write`, `write_u64`, and `finish`, a unit `FnvBuildHasher`, and a `BuildHasher` implementation to manufacture hashers. Every line of it is boilerplate, and it buys a statically dispatched, type-checked hasher rather than a conventionally shaped callable. The service module costs a comparable amount for the reason developed in Section 6.4.5: because the C++ response is a single `variant`, all ten public operations funnel through one shared `submit` helper and each typed wrapper shrinks to two or three lines, while Rust's per-variant typed sender admits no such helper without generics, so each operation repeats the sequence of creating a completion channel, constructing its variant, sending, and receiving. The third, `lib.rs`, is the module tree and public re-export list, work that C++ gets implicitly from its header structure and `#include` graph.

None of this is a defect on either side, and none of it is visible in the performance figures. It is exactly the kind of engineering cost the forced-idiom log exists to surface: Rust spends source lines on trait machinery and call-site repetition to obtain guarantees the compiler can check, and recovers a large part of that spend wherever its standard library happens to provide a primitive that C++'s does not.

Counting lines, however, answers only half the question a reviewer of either codebase would ask, because it treats a line of boilerplate and a line of branching as the same thing. Table 6.3 therefore adds cyclomatic complexity, measured by `lizard` [48] over both cores. Applying one tool to both languages is what makes the figures comparable: it applies the same definition of a decision point on both sides, whereas each language's native tooling would count somewhat different things. The two measures point in opposite directions. Rust is 19% longer than C++ by code lines, but its total cyclomatic complexity is 15% *lower*, 79 against 93, spread over slightly more functions (40 against 37). Both cores are undemanding by the usual thresholds and both have a median per-function complexity of one, so the aggregate is carried by a small number of functions at the top.

That is where the difference sits, and it traces to a divergence already described rather than to anything diffuse. The most complex function in the C++ core is `worker_loop` at 13, more than twice its Rust counterpart's 6: it switches on the `OpKind` discriminant and then stores the matching alternative into the response `variant`, so the tagged-struct design of Section 6.4.5 concentrates the whole dispatch, and the correctness argument that goes with it, into one branch-heavy function. Rust's per-variant typed

Table 6.3: Size and cyclomatic-complexity proxies for the two cores, measured with `lizard` [48] applied identically to both languages. Benchmark drivers, tests, conformance runners, and the FFI seam are excluded; only the code the parity contract constrains is counted. Complexity is one plus the number of decision points in a function. The line count is NLOC as `lizard` defines it, which also discounts brace-only and preprocessor lines and therefore runs eight below the non-blank count quoted at the head of Section 6.4 for C++; the Rust figure agrees exactly.

Measure	C++	Rust
Files	10	8
Functions	37	40
Lines of code (blank and comment lines excluded)	508	602
Total cyclomatic complexity	93	79
Mean complexity per function	2.5	2.0
Median complexity per function	1	1
Most complex function	13 (`worker_loop`)	6 (`add_drb`)

sender removes the return-path dispatch entirely, which is why its worker has nothing comparable to unwrap. The trade is therefore sharper than the line count alone suggested: the extra lines Rust spends buy back branching in the one place where branching is hardest to review, and the `std::bad_variant_access` invariant that Section 6.4.5 noted is maintained by inspection across two files lives inside precisely the function this measurement flags.

6.4.8 The safety argument for the two cores

The Rust library contains *no* `unsafe` code at all, a property verified mechanically over the crate. Safe Rust therefore guarantees data-race freedom for this core at compile time through the `Send` and `Sync` traits: the shard store is owned by a single worker and never shared, the request type is moved across the thread boundary, and the one piece of genuinely shared state, the activity counter, is an atomic. No runtime sanitizer is needed to establish any of this, and no test coverage is required for it to hold. The C++ implementation performs no manual memory management beyond STL containers and no explicit locking outside the queue class, and it was verified data-race free under ThreadSanitizer.

The two statements are not, however, evidence of the same kind, and the difference is the thesis's central safety claim in miniature. The Rust property is a theorem about the program, discharged by the compiler on every ordinary build and covering every

Table 6.4: Where the 87-line difference between the two cores comes from, by concern (raw lines, including comments; C++ figures merge header and translation unit). Rust is shorter only where its standard library supplies a primitive C++ lacks.

Concern	C++	Rust	Δ
Bounded queue	45	0	−45
Hashing (functor vs. trait pair)	36	70	+34
Service (queues, workers, API)	221	255	+34
Module tree and re-exports	0	23	+23
Data model	73	85	+12
Message types	48	60	+12
Shard core (state machine)	176	188	+12
Configuration	18	29	+11
Error model	32	26	−6
Total	649	736	+87

execution. The C++ property is an observation about the executions a particular instrumented binary happened to take, and it would have to be re-established after any change, on a build most projects do not run by default. Quantifying exactly how far apart those two kinds of assurance are is the purpose of the fault-injection study in Chapter 7.

6.5 Public API and operation semantics

Both implementations expose the same small, lifecycle-oriented API against the shared specification, so that the surface a caller sees is identical up to each language's naming conventions. A service is created with a fixed shard count and a per-shard queue bound; thereafter callers invoke context operations, create, setup, handle-signaling, add-bearer, reconfigure begin/complete/abort, release, and state lookup, each identified by the target `ue_id`. Every operation is *total* and *typed*: it returns a success value or a typed `OpError` drawn from a closed enumeration (for example `NotFound`, `InvalidState`, `BearerLimit`), and an operation that is illegal for the context's current state (Figure 3.1) returns the corresponding error and leaves the context unchanged rather than mutating it or signaling failure out of band. This choice matters for fairness as much as for correctness: because neither implementation uses exceptions or error codes as a hidden control-flow channel, the cost and the control flow of the failure path

are visible and identical on both sides, and the same conformance vectors (Section 6.7) can assert op-for-op agreement on both the returned value and the resulting state.

6.6 Request lifecycle and the completion handoff

The end-to-end path of a single operation is the machinery the study actually measures, so it is worth stating precisely. A caller constructs a request carrying the operation, its arguments, and a single-use completion handle, and computes the target shard from a pinned hash of the `ue_id`. It submits the request to that shard's bounded MPSC queue: if the queue has space the request is enqueued and the caller blocks on the completion handle; if the queue is full the *submission* blocks until space frees, which is the point at which offered overload becomes observable as queueing latency rather than being silently dropped. The shard's single worker dequeues requests in strict FIFO order, executes the operation against its exclusively owned store, writes the result (a success value or an `OpError`, and for reads an immutable `UeContextView` copied out of live state) into the completion handle, and moves on to the next request. Ownership of the request, and of any heap data it carries, is transferred to the worker on submit and is never shared, which is what upholds the shared-nothing invariant end to end. Latency is measured across this entire path, from intended send time to completion, so that it includes queueing delay under load and reflects the same cross-thread round-trip on both sides. Figure 6.6 traces the same path as a sequence, and Section 7.5.2 later decomposes it into the part the language executes and the part the operating system does.

6.7 Parity verification

Because the whole comparison rests on the two versions being functionally identical, parity is tested rather than assumed. A language-neutral conformance suite expresses test vectors as sequences of operations with their expected results and an expected post-state summary; a runner in each language executes the vectors and emits a canonical transcript, and the two transcripts are diffed for op-for-op agreement. The seeded suite covers the full error model, the reconfiguration begin/complete/abort paths, data-radio-bearer fill and overflow, and shard-assignment determinism; both implementations currently pass and agree. The benchmark workloads add a second, independent

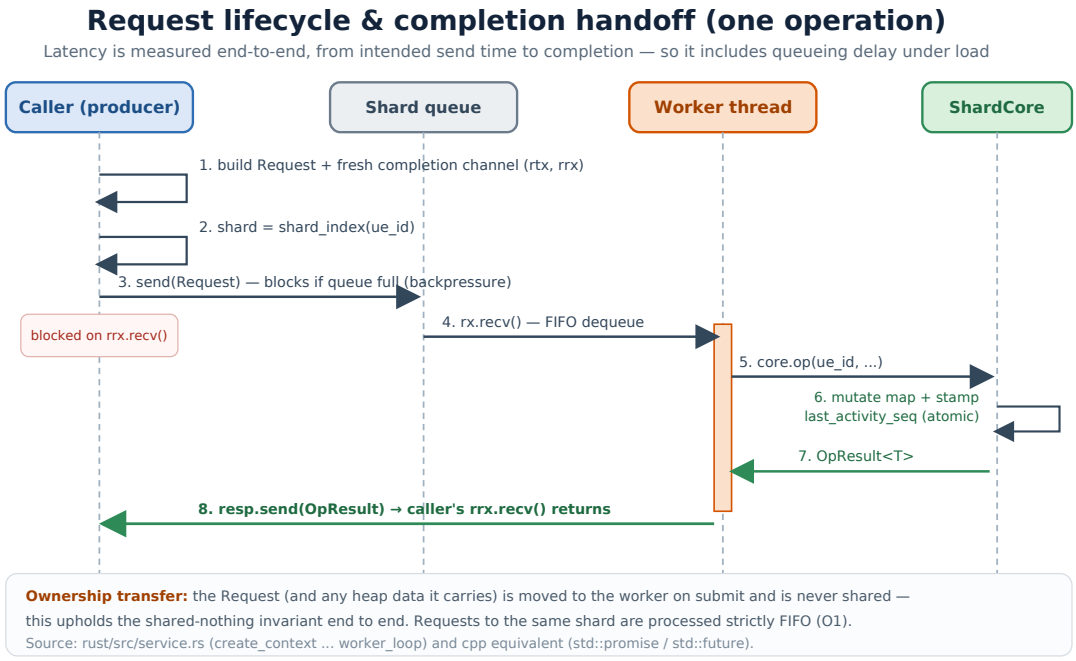


Figure 6.6: The end-to-end path of a single operation and the completion handoff. The caller builds a request with a single-use completion handle, routes it by `shard_index(ue_id)` to the owning shard's bounded queue (blocking if it is full), and waits; the shard's single worker dequeues in FIFO order, executes the operation against its exclusively owned store, and writes the result back to the completion handle. Ownership of the request is transferred to the worker on submit and never shared. Latency is measured across the whole path.

parity signal: the mixed workload's per-operation-kind counts are verified to be identical across languages for a given seed and active-UE population, confirming that both implementations are driven along exactly the same code paths.

7 Experimental Setup and Evaluation

7.1 Testbed configuration

Two execution environments must be distinguished, because the thesis deliberately reports results of two different evidential strengths and the distinction rests entirely on where they were produced.

The environment used. All performance results reported in this chapter were measured on a dedicated dual-socket server: two eight-core Intel Xeon E5-2680 v4 processors, sixteen physical cores on a single NUMA node with simultaneous multithreading absent, running Rocky Linux 8.10 with `g++ 15.1.1` and `rustc 1.97`. The benchmark is pinned with `taskset` to cores 2–15, and the service is configured with seven shards and seven producer threads so that every worker and every producer owns a core and none contends with the operating system. Every run captures its own environment record (CPU model and topology, kernel, compiler and toolchain versions, and the frequency, turbo, and multithreading settings) alongside the measurement data, so the conditions of a measurement travel with it.

One property of this host has to be stated plainly because it bounds the tail claims below: it is a virtual machine. The environment record identifies a KVM guest, and two consequences follow. The CPU frequency governor and turbo state are host-controlled and not merely unset but unobservable from inside the guest, so frequency stability is assumed rather than enforced. And the record shows non-zero CPU steal, on the order of ten milliseconds of vCPU time per second: small in aggregate, but the same order as the tail latencies being measured, and not removable from inside the guest. The cores were also not removed from the scheduler with `isolcpus`, so a benchmark thread can still lose its core to other work.

This is a substantially better instrument than a shared development laptop, which is what an earlier pass of this campaign used: the cores are homogeneous and server-class, nothing else of consequence runs on the machine, and the run-to-run dispersion at p99

fell by more than an order of magnitude between the two. It is not the bare-metal host the protocol below specifies.

The protocol the performance campaign still lacks. Testbed-grade tail figures require a dedicated bare-metal host, and the protocol for one is specified, scripted, and published so that it can be executed without further design work (`bench/TESTBED.md`): the benchmark process pinned to cores removed from the scheduler and the timer tick with `isolcpus` and `nohz_full`, the frequency governor set to `performance` and verified, turbo disabled, and hardware performance counters available. What executing it would add is narrower than it once was, because three of the four things it was going to buy have been obtained without it: the saturation knee, the shape of the overload cliff (Section 7.5.4), and the replication of the resource comparisons across the allocator control (Section 7.5.5). What remains is the far tail *below* saturation, where the confidence interval on a single variant's p99 is still several times the spread between all four variants, and where steal and scheduler preemption are the plausible cause rather than anything either language does. That one result is therefore reported as unresolved rather than as a null, and the mechanism to resolve it belongs to the machine rather than to the analysis.

Measurement discipline common to both. Independently of the host, each sweep point is repeated 20 times and reported as a median with a bootstrap 95% confidence interval rather than as a single run, and the four language×allocator variants are interleaved within each repeat so that thermal or background drift affects them equally rather than biasing one. That interleaving is what makes a contended host tolerable for a *comparative* conclusion even though it invalidates the absolute numbers: interference is shared, not assigned. Throughput is counted over the measured phase alone, with the warm-up iterations excluded from both the numerator and the denominator, so that achieved throughput is directly comparable with offered load. The full configuration and scripts are published for reproducibility.

Table 7.1 sets the host used against the protocol still to be executed, row by row, and Figure 7.1 carries the same line forward onto the results themselves.

Table 7.1: The host the reported results were measured on, against the protocol that testbed-grade tail figures require (`bench/TESTBED.md`). The rows where the two columns agree are properties of the source and the compilers; the rows where they differ are what a bare-metal campaign would still buy. Only the last of those differences now costs a result, because the saturation knee and the resource comparisons were resolved on this host: what remains is the far tail below saturation.

Aspect	Used for the results reported here	Required for testbed-grade tail figures
Host	Dedicated dual-socket server, 2× Xeon E5-2680 v4, 16 cores, one NUMA node	Same class of machine, without the hypervisor
Virtualisation	KVM guest; frequency and turbo host-controlled and unobservable	None; governor and turbo set and verified
CPU steal	Non-zero, order 10 ms of vCPU time per second	None
Kernel	Rocky Linux 8.10, distribution kernel	Same, with `isolcpus` and `nohz_full` on the benchmark cores
Toolchain	`g++ 15.1.1`, `rustc 1.97`	`g++ 15.1.1`, `rustc 1.97`
CPU pinning	Cores 2–15 via `taskset`; 7 shards, 7 producers	Same, on cores also removed from the scheduler
Simultaneous multithreading	Absent on this processor	Absent or disabled
Warm-up	Initial fraction of each run discarded	Initial fraction of each run discarded
Repeats and interleaving	20 repeats, four variants interleaved per repeat	20 repeats, four variants interleaved per repeat
Environment capture	`bench/capture_env.sh`, stored with each result	`bench/capture_env.sh`, stored with each result
Hardware counters	Unavailable in the guest	Available; the clearest instrumentation gap
Adequate for	Parity, fault injection, FFI structure, the saturation knee and the overload comparison, and the resource comparisons	Additionally the far tail below saturation, and the mechanism behind the resource and operation results

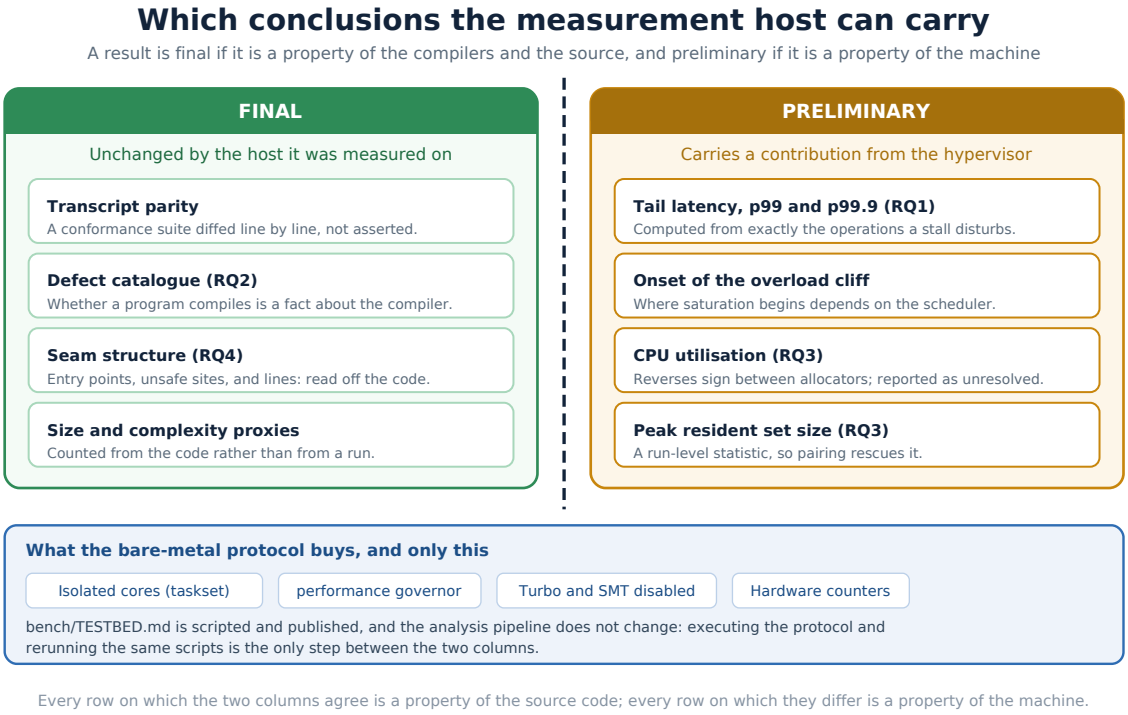


Figure 7.1: Which conclusions the measurement host can carry. The division is not between important and unimportant results but between those that are properties of the compilers and the source, which this host establishes as firmly as any would, and those that are properties of the machine, which it cannot. The band below is what executing the published bare-metal protocol adds; because the analysis pipeline is unchanged by it, rerunning the same scripts is the only step separating the two.

7.2 Workload and evaluation measurements

A synthetic but spec-justified telecom workload is generated by a deterministic driver shared in structure by both implementations, so that the two versions are exercised along identical code paths. Two workloads are provided. The *signal* workload preloads a population of connected UEs and then hammers the `handle_signaling` hot path, isolating steady-state per-message latency. The *mixed* workload is a pinned RAN lifecycle generator that walks each UE through a realistic procedure sequence, context creation, connection setup, signaling, bearer setup, reconfiguration/handover (begin/complete/abort), and release, using a fixed SplitMix64 draw order and a per-slot expected-state model so that every emitted operation is valid. Because the draw order is fixed, the per-operation-kind counts are identical in C++ and Rust for a given active-UE population and seed, which doubles as the parity signal described in Section 6.7. The control parameters are the target event rate and the number of active UEs, swept from nominal load through saturation and overload; a fixed send schedule with histogram correction is used so that coordinated-omission bias is avoided when the system stalls [24, 47]. Three measurement campaigns are run: a load sweep for performance, the fault-injection study, and an FFI migration prototype. Figure 7.2 shows the two workloads and the procedure sequence the mixed one walks.

What "spec-justified" claims, and what it does not. The term is used precisely, because the distinction bounds what the workload can support. The *sequence* of operations is justified: it is the RRC connection lifecycle the 3GPP specifications define for a UE, from establishment through in-session signaling and bearer setup to reconfiguration and release [4, 1], and the per-slot expected-state model guarantees that every operation the driver emits is legal for the state its context is actually in, so the workload never drives the service along a path the specification forbids. The *proportions* are not justified in the same way. How often a real cell hands over relative to how often it establishes a bearer depends on deployment, mobility, and time of day, and no public figure fixes it, so the mix used here is plausible rather than validated. That is reported as a threat to external validity in Section 8.6, and trace replay heads the future-work list for exactly this reason.

For a *comparative* question the distinction costs less than it would for an absolute one, and saying why is part of arguing that the data suits the task. Both implementations

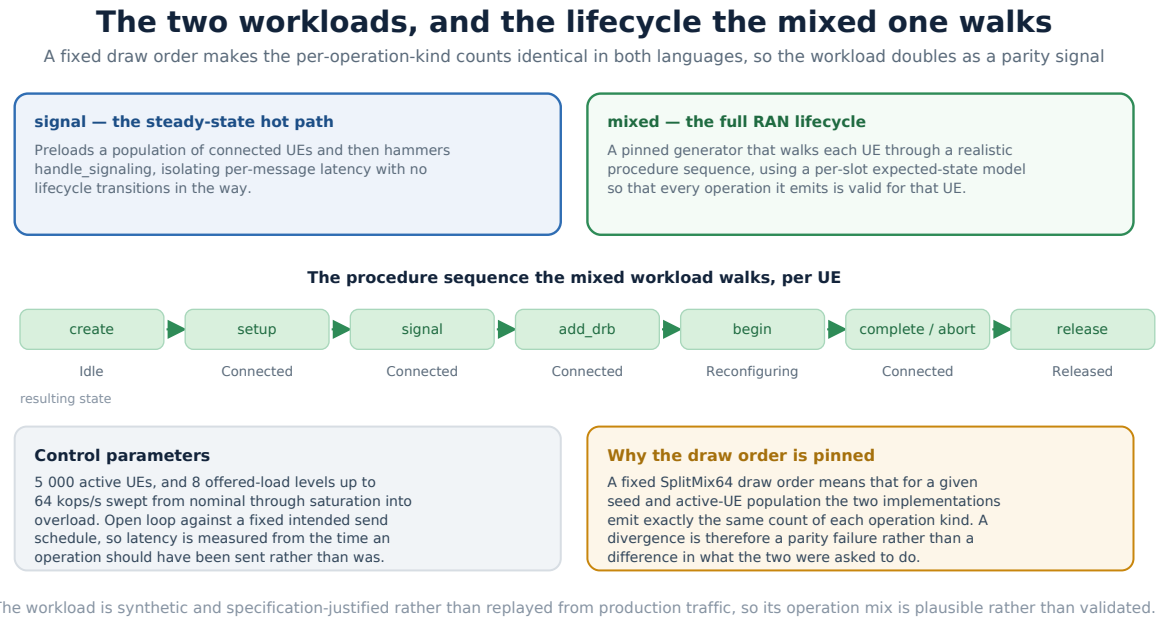


Figure 7.2: The two workloads. The *signal* workload isolates the steady-state hot path; the *mixed* workload walks each UE through the full RAN procedure sequence, with a per-slot expected-state model ensuring that every operation it emits is legal for the state its context is actually in. Because the SplitMix64 draw order is pinned, a given seed and active-UE population produce identical per-operation-kind counts in both languages, so a divergence is a parity failure rather than a difference in what the two implementations were asked to do.

execute the same sequence in the same order for a given seed and active-UE population, verified through the per-operation-kind counts, so an unrepresentative mix shifts both languages equally and cannot manufacture a difference between them. An unvalidated operation mix therefore bounds how far the numbers travel; it does not compromise the fairness of the comparison they are used to make.

Table 7.2 gathers the configuration of all of this in one place. The values are otherwise distributed across the specification, Chapter 6, and this chapter, which is convenient while reading each in turn and unhelpful to anyone trying to reproduce the campaign or judge whether a parameter was chosen to suit the result. Every entry in the campaign block is either fixed by the specification or generated from the measurement logs, so the table cannot drift out of step with the data reported below.

7.3 Data collection and metrics

The study records latency predictability (per-event mean, median, p99/p99.9/p99.99, and maximum latency, plus jitter), throughput (sustained events/sec and throughput at a fixed latency service-level objective (SLO)), resource use (CPU utilisation, memory over time, allocation patterns), concurrency robustness, safety (defect classes prevented at compile time vs runtime), and software-quality proxies. The last of these are reported where they are most legible rather than gathered into this chapter: lines of code and cyclomatic complexity alongside the forced-idiom log they explain (Table 6.3), and the count and scope of Rust `unsafe` blocks alongside the FFI seam that is the only place either implementation contains any (Table 7.10). Each configuration is measured and summarised under the statistical treatment of Section 5.5, and only synthetic data is used, so no personal, subscriber, or confidential data is processed. Figure 7.3 traces the whole path, from the specification both implementations are built against to the generated macros this prose cites.

7.4 Evaluation data

Three campaigns produce three datasets, and they differ in kind and not only in content. What follows describes each as captured, because a reader judging whether the analysis is sound needs to know what was recorded and at what granularity, not merely what was reported.

Table 7.2: Configuration of the service under test, the workload driving it, and the measurement campaign. Service parameters are fixed by the frozen specification, with one exception: the shard count is set to match the cores the host offers (Section 7.1). All of them, that one included, are identical in both languages. Campaign figures are generated from the measurement logs together with the tables that report them. The representative load is not chosen by eye but derived, as the busiest swept load at which all four build variants still hold their p99 below 5 ms.

Parameter	Value	Note
Service under test, identical in both languages		
Shards, and worker threads	7	one thread owns one shard; one of each per pinned core
Queue bound per shard	1 024 requests	bounded and blocking, so overload is observable
Initial map capacity	4 096 entries	reserved so steady state does not rehash
Bearers per context	at most 8	
Signaling payload	64 bytes	
Context ceiling per shard	200 000	
Shard selection	SplitMix64, then modulo	pinned; diffuses sequential identifiers
Intra-shard hash	FNV-1a, 64-bit	pinned on both sides and checked bit-for-bit
Workload		
Workloads	signal, mixed	mixed walks the full RAN lifecycle
Active UEs	5 000	
Offered-load levels	14	up to 784 kops/s
Representative load	56 kops/s	derived, see caption
Measured saturation	335 kops/s	tracks the offered rate to 336 kops/s
Load model	open loop, fixed intended schedule	latency measured from intended send time
Campaign		
Build variants	4	two languages × two allocators
Repeats per sweep point	20	variants interleaved within each repeat
Measured runs	1120	
Latency capture	HDR histogram, $[1\,\text{ns}, 3.6\,\text{s}]$, 3 s.f.	same algorithm and configuration on both sides
Summary statistic	median with bootstrap 95% CI	intervals reported in full, not as half-widths
Language comparison	exact paired permutation test	14 tests, Holm-corrected
Attainable p floors	0.0000019 / 0.000027	uncorrected / after correction

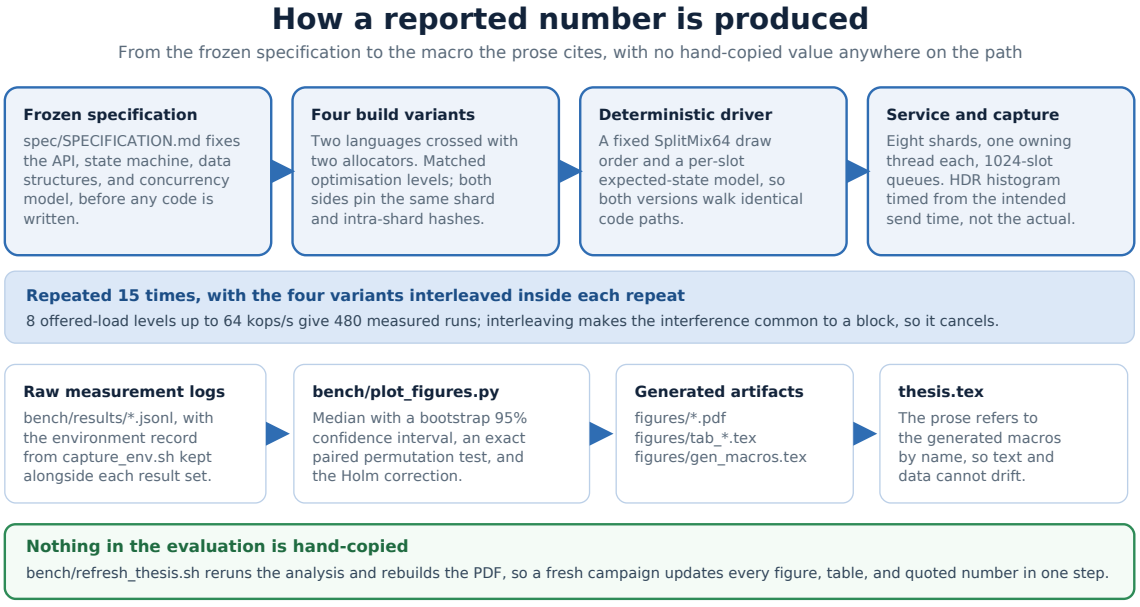


Figure 7.3: How a reported number is produced, from the frozen specification to the macro the prose cites. The upper row is the measurement path and the lower row the analysis path; the band between them is the interleaved repeat structure that turns the campaign into a matched design. No value in the evaluation is hand-copied: `bench/refresh_thesis.sh` regenerates every figure, table, and quoted number directly from the raw logs, so the text and the data cannot drift apart.

The performance sweep. The sweep is a full factorial design over two languages × two allocators × 14 offered-load levels × 20 repeats, or 1120 measured runs, captured on the host of Section 7.1. Each run contributes one record, written as newline-delimited JSON so that a campaign is an append-only log rather than a summary table: nothing is aggregated at capture time, and every statistic in this chapter is recomputed from the raw records. A record carries the run's configuration (language, allocator, workload, shard and producer counts, target rate), its outcome (operations completed, errors, wall-clock duration, achieved throughput), the latency distribution read out of the HDR histogram (mean, p50, p90, p99, p99.9, maximum, in nanoseconds), the resource counters taken through `getrusage` (peak resident set size, user and system CPU seconds, CPU utilisation), and the per-operation-kind counts. Retaining the kind counts in every record is what makes a divergence between the two languages visible as a parity failure (Section 6.7) rather than being mistaken for a performance difference.

The component decomposition. The second measurement level of Section 5.3.1 yields a much smaller dataset of a different shape: for each of 15 interleaved repeats, the batch-mean cost in both languages of the operation called directly and of the same operation called through the threaded service. The FFI comparison of Section 7.5.7 has the same structure over 15 repeats. Both are paired by construction, which is what the exact paired test of Section 5.5 requires.

The fault-injection catalogue. This dataset is categorical rather than numeric. Each of the seven defect cases yields three tool logs, the AddressSanitizer and ThreadSanitizer output for the C++ program and the `rustc` output for the safe-Rust counterpart, together with one row in a machine-readable taxonomy recording the two sanitiser outcomes, the detection stage, the compiler result with the diagnostic codes that fired, and the expectation registered before the study. Carrying the expectation in the data rather than in the prose is what allows the runner to report a mismatch in either direction; the table and figure in Section 7.5.6 are generated from that file rather than transcribed from it.

What travels with the data. Every campaign writes an environment record beside its results, capturing host and kernel, distribution, compiler and toolchain versions, CPU model and topology, and the frequency governor, turbo, and simultaneous-multithreading settings, *including* the fields a given host does not expose. That detail

matters more than it sounds. The record identifies the host as a KVM guest, and the governor and turbo state are stored as unavailable rather than omitted, so the limitation documented in Section 7.1 is a property of the dataset itself and not only of the prose describing it, and a reader can confirm the limitation without taking it on trust.

The performance dataset is complete and self-consistent as an analysis and is used below to answer RQ1 and RQ3 provisionally. Rerunning the same scripts over a campaign captured on a bare-metal host replaces these numbers with testbed-grade ones without any change to the analysis.

7.5 Evaluation results

The safety and migration results do not depend on the measurement host at all. The performance and resource results do, and Section 7.1 states what that host can and cannot carry.

Because this chapter reports resolved and unresolved outcomes side by side, it is worth saying at the outset which are which, so that a null result is not read as an absent one. Six findings are resolved: the defect-class taxonomy and its boundary (Section 7.5.6), the cost of the FFI seam (Section 7.5.7), the memory difference between the two builds (Section 7.5.5), the cost of the bare operation (Section 7.5.2), the equivalence of the four variants at the median below saturation, and their divergence past it (Section 7.5.4), which is the largest and most reproducible language effect in the study.

One thing is unresolved, and one is resolved but unexplained. The far tail *below* saturation is unresolved because of the host: run-to-run dispersion at p99 exceeds every difference between the variants, and no number of repeats on a virtualised machine with non-zero CPU steal closes that gap (Section 7.5.1). The overload divergence, by contrast, is not in doubt as a measurement; what is missing is its mechanism, which is narrowed to two standard-library primitives but not measured on either. The bare-metal protocol of Section 7.1 would answer the first. Only a contended-queue microbenchmark answers the second.

7.5.1 Performance (RQ1, RQ3)

The hypothesis under test is H1: that the two languages are performance-equivalent below saturation, with any difference confined to the far tail. This section reports

where, if anywhere, each wins and by how much, across the sweep up to and beyond saturation.

Everything in it, figures, tables, and the numbers quoted in the prose alike, is generated from the raw logs by `bench/plot_figures.py`, so the section can be regenerated verbatim if the campaign is rerun on the bare-metal testbed. The absolute figures belong to the host of Section 7.1. Each point aggregates 20 interleaved repeats; tabulated values and shaded bands are medians with nonparametric bootstrap 95% confidence intervals.

Table 7.3 summarises the per-operation latency distribution at a sustained offered load of 56 kops/s. That operating point is not chosen by eye: it is the highest swept load at which every one of the four build variants still holds its p99 below 5 ms, that is, the busiest point at which all four are simultaneously healthy, which makes it the fairest place to compare them.

The table must be read at two different levels of confidence, and separating them is the honest reading of this campaign. At the *median* the four variants sit within 2 μs of one another, or 2.1%, and each variant's confidence interval contains the others' point estimates, so the table on its own separates nothing. That is a statement about unpaired intervals rather than about the languages: the paired test below does resolve a small and consistent difference at the median, and the distance between those two readings is precisely why Section 5.5 tests the comparison instead of inspecting the intervals. *Above* the median the situation is different in kind, and it is not that the languages differ but that this host cannot say whether they do. The p99 values differ by 293 μs across variants, while the widest 95% confidence interval on a single variant's own p99 spans 1480 μs, many times the between-variant difference. When the uncertainty about one variant exceeds the gap between all of them, no ordering can be read off, and reporting one would be reporting the scheduler. The same holds with greater force at p99.9 and at the maximum.

This is the concrete form of the limitation stated in Section 7.1, and it is worth being precise about what it does and does not cost. It does not affect the throughput and saturation behaviour, the resource results below, or anything in the RQ2 and RQ4 chapters. It does mean that one half of RQ1, how the two languages behave in the high percentiles while the service is still meeting its offered load, is answered by this campaign only in the negative: that tail is not yet measurable. It leaves the other half untouched. How they behave once the service stops coping is resolved decisively

Table 7.3: Per-operation latency at a sustained offered load of 56 kops/s (mixed RAN-lifecycle workload, 7 shards). Values in microseconds: the median over 20 repeats above its nonparametric bootstrap 95% confidence interval. The interval is printed in full rather than as a symmetric half-width because it is strongly asymmetric at these percentiles, and a half-width wide enough to cover the upper bound would place the lower one below zero. Measured on the host of Section 7.1.

Variant	p50	p99	p99.9	max
C++ (system alloc)	79.9	4667	6222	7817
	[79.6, 80.7]	[4272, 4895]	[5501, 7242]	[6572, 8886]
C++ (jemalloc)	80.3	4506	5949	7472
	[79.4, 80.7]	[4289, 5216]	[5540, 7160]	[6470, 8826]
Rust (system alloc)	79.0	4530	6013	7932
	[78.7, 79.7]	[4197, 5677]	[5612, 7905]	[6527, 9327]
Rust (jemalloc)	78.6	4375	5624	6951
	[78.1, 79.7]	[4173, 4661]	[5476, 6425]	[6363, 8143]

on this same host, and Section 7.5.4 reports it. Closing the remaining gap needs the bare-metal protocol rather than more repeats on this host, because the dispersion is systematic interference rather than sampling noise, and Section 7.5.2 shows that the quantity a language could affect lies more than three orders of magnitude below the one this measurement resolves.

Testing the comparison rather than eyeballing it. Overlapping intervals are a weak instrument in both directions, so the comparison is also tested directly, by the procedure fixed in Section 5.5 and reported in Table 7.4. What licenses the pairing is a property of this campaign rather than of the statistic: because the four variants are interleaved inside every repeat, the i-th run of each was measured in the same block under the same ambient conditions, so the interference that block carried is common to the pair and cancels in their difference. That is the second thing the interleaving discipline buys, beyond the fairness argument it was adopted for.

Four things follow. First, 9 of the 14 tests in the family survive the Holm correction. At this operating point they are the two median-latency rows, peak resident set size under the system allocator, and both CPU rows; the four remaining survivors are the overload comparisons of Section 7.5.4. Peak resident set size under jemalloc is the one robust-block row that does not clear the correction despite an interval nowhere near zero, and Section 7.5.5 takes up why.

Second, the reading rule fixed in Section 5.5 has to be applied before those are inter-

Table 7.4: Exact paired permutation test of C++ against Rust at 56 kops/s, with the allocator held fixed. The effect is the median of the 20 paired differences (Rust − C++), shown above its bootstrap 95% confidence interval; the test statistic is their mean. p is exact over all 2^{20} sign assignments, and the final column applies the Holm correction across the whole family of 14 tests; the attainable floors are 0.0000019 uncorrected and 0.000027 after correction. The rows are blocked by the criterion set out in Section 5.5: pairing rescues the statistics in the upper block, so a consistent interval there is evidence, whereas the high percentiles below are computed from exactly the operations interference falls on, so an interval excluding zero is not read as a finding. Achieved throughput is not tested, for the reason given in the same section, and appears in Table 7.8 instead. Measured on the virtualised Xeon host of Section 7.1.

Metric	Allocator	Effect (Rust − C++)	p	Holm
Robust to interference on single operations				
p50 latency (µs)	system	−1.1 [−2.0, −0.4]	0.002	0.012
p50 latency (µs)	jemalloc	−1.6 [−2.0, −0.5]	0.000086	0.00069
Peak RSS (MiB)	system	+1.9 [+1.8, +2.0]	0.0000019	0.000027
Peak RSS (MiB)	jemalloc	+3.4 [+3.3, +3.5]	0.727	1.000
CPU utilisation (%)	system	−3.4 [−5.0, −2.4]	0.0000076	0.000069
CPU utilisation (%)	jemalloc	−1.9 [−3.1, −0.8]	0.005	0.029
Determined by it: high percentiles				
p99 latency (µs)	system	+12 [−356, +866]	0.433	1.000
p99 latency (µs)	jemalloc	+111 [−389, +304]	0.462	1.000
p99.9 latency (µs)	system	−133 [−508, +1495]	0.460	1.000
p99.9 latency (µs)	jemalloc	−141 [−1028, +508]	0.407	1.000

preted. 10 paired intervals exclude zero; in the tail block the intervals that do so are: none. The rule is stated in advance rather than at this point precisely because that outcome is not guaranteed, and an earlier seven-repeat pass of the same campaign did produce tail intervals excluding zero which more repeats then removed, which is the behaviour the rule anticipates.

Third, the allocator control supplies a second and independent check: every comparison is run twice under conditions that should not change a language effect, so an effect that is genuinely the language ought to keep its sign whichever allocator is linked. Here p50 latency, p99 latency, p99.9 latency, Peak RSS and CPU utilisation all do, and 0 reverse. That the control confirms rather than vetoes is not evidence that it was unnecessary: on an earlier campaign, run before the two `jemalloc` builds were held to one configuration, the CPU effect cleared the correction under one allocator and reversed sign under the other, and was reported as unresolved on exactly this rule. A control is worth keeping because of what it would have caught, not because of what it happens to catch.

Fourth, the median, which is now a resolved difference rather than a null: 1.6 μs in Rust's favour, consistent under both allocators. The interval on it reaches 2 μs against a median of roughly 80 μs, and the temptation is to report that as a bound on the language effect. That reading would be wrong, and the reason is the subject of the next section.

7.5.2 What the end-to-end number is made of

Every latency the sweep reports is an end-to-end round-trip, and Section 5.3.1 set out why that bundles two costs of quite different character: the operation itself, executed inside one thread and generated by the compiler, and the cross-thread submit-and-complete handoff around it, charged to the operating system. Only the first can plausibly differ between the languages, and the sweep cannot separate them.

That matters for the bound just quoted, because a bound is only as informative as the quantity it is a fraction of. The two are therefore measured directly by the two-level method described there, in both languages and at the sweep's own shard count and context population: `ShardCore` called with no queue in the path gives the operation, `Service` gives the operation plus the handoff, and their difference is the handoff. Table 7.5 gives the result, and it reframes the whole performance comparison.

The operation is 34.5 ns in C++ and 21.3 ns in Rust. The handoff is 13.58 μs and

Table 7.5: Where an end-to-end operation's cost actually goes, measured at 7 shards and 5 000 active UEs with a single uncontended caller. Each row is the median over 15 interleaved repeats above its bootstrap 95% confidence interval, in the unit its own magnitude calls for: the two terms differ by more than two orders of magnitude, which is the point of the table. p is the exact two-sided paired permutation test of C++ against Rust on the repeat-wise differences. The handoff is obtained as the difference between the two measured rows, so it carries the uncertainty of both. Measured on the host of Section 7.1.

Component	C++	Rust	p
Operation alone, no queue in the path (ns)	34.5	21.3	0.000061
	[32.4, 35.2]	[21.0, 21.5]	
Cross-thread submit-and-complete handoff (µs)	13.58	13.59	0.439
	[13.38, 13.83]	[12.69, 14.52]	
End-to-end round-trip, uncontended (µs)	13.62	13.61	0.421
	[13.41, 13.87]	[12.71, 14.54]	
Operation's share of the round-trip	0.250 %	0.160 %	

13.59 μs. The work the languages actually perform is therefore at most 0.250% of what the sweep measures, and better than 99.9% of every latency in Table 7.3 is the operating system moving a message between two threads.

That shape is not peculiar to this system or to this pair of languages. Revisiting a classic decomposition of transaction-processing cost on a shared-nothing database structured, as this service is, with one partition per worker thread, Zhou et al. report that once the per-operation work has been optimised the remaining cost is dominated by communication across isolation boundaries rather than by the operation itself [49]. A mature shared-nothing design tends to end up communication- and scheduler-bound, and the table above is that pattern appearing in a control-plane service. It is worth saying because it makes the result below a general property of the design rather than an artefact of this measurement.

Two conclusions follow, and they pull in opposite directions.

The first is that the sweep cannot bound the language effect, and the bound quoted above should not be read as though it does. An interval of 2 μs on the paired median difference is 58 times the operation's own cost. A language effect could be several hundred percent of the work being done and sit comfortably inside it. What the sweep establishes is the much weaker and still useful statement that neither language perturbs the scheduler-dominated round-trip, which is what a deployment of this shape would actually experience; what it cannot establish, at any repeat count, is anything about

the code the compilers emit. That is a property of the measurement's construction rather than of the host, so unlike the tail result it will not be fixed by moving to bare metal. It is fixed by measuring the component separately, which is what this section does.

The second is that once the component *is* measured separately, the languages are not equivalent. Rust runs the bare operation about 1.6× faster, the difference is present in the great majority of repeats, and the paired test puts it two orders of magnitude below the conventional threshold. This decomposition is a separate experiment from the sweep and is not part of its Holm family: it runs 15 repeats rather than 20, so its own attainable floor is 0.000061 rather than the 0.000027 quoted in Section 5.5, and no correction is applied to it. The same holds for the interoperability comparison of Section 7.5.7. This is the only place in the whole performance campaign where a language difference in *time* is resolved at all, and the sweep is blind to it by more than two orders of magnitude.

Figure 7.4 draws both conclusions on one page, because each is easier to see than to read off a table of numbers spanning four decades. The left panel is the round-trip at true proportion, where the operation is a sliver too thin to render and the caption has to point at it. The right panel puts that sliver on a log axis beside the smallest paired difference the sweep could have detected, and the ordering there is the argument of this section: the instrument sits entirely above the quantity it would have to constrain.

The attribution deserves care, because the obvious reading is the wrong one. Nothing here implicates the borrow checker, bounds checking, or ownership. The single most likely source is already recorded in the forced-idiom log of Table 6.2: the per-shard container is each language's standard hash map, and those are different data structures. C++'s `std::unordered_map` is required by its iterator- and reference-stability guarantees to be node-based, so a lookup chases a pointer out of the bucket array into a separately allocated node, whereas Rust's `HashMap` is an open-addressed table that keeps entries inline and usually resolves a lookup within one cache line. The hash function is pinned to FNV-1a on both sides (Section 6.4) precisely so that hashing could not be the explanation, which leaves the container layout. That is a real and reportable difference between the two languages as they are actually used, since a developer reaching for the standard hash map gets whichever one their language ships. It is not evidence about Rust's safety model, and it should not be quoted as such. Confirming the mechanism would need the cache-miss counters Section 8.6 lists as the

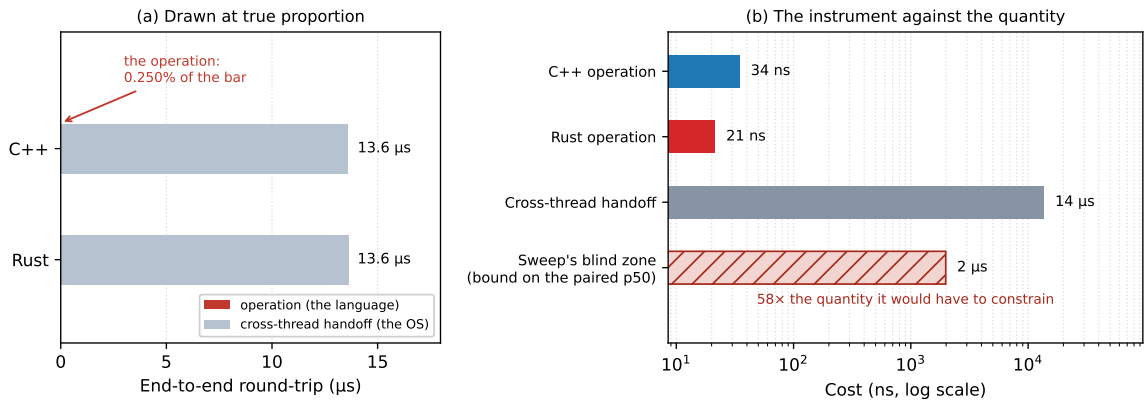


Figure 7.4: Where an end-to-end operation's cost goes, and why the sweep cannot bound a language effect. **(a)** The round-trip at true proportion: the operation the language performs is 0.250% of the bar, the remainder the cross-thread handoff. **(b)** The same quantities on a log axis against the smallest paired difference the sweep can resolve, shown as a blind zone. Both operation measurements fall inside it by a factor of 58, and (b) is also where the campaign's one resolved timing difference lives. Measured on the host of Section 7.1.

clearest instrumentation gap.

Three limits keep the finding the size it is. It is a single-threaded, uncontended microbenchmark, so it says what the operation costs and not what a loaded system does with it. It is measured on the same host as the sweep, so the absolute figures carry the same caveat, although the ratio is far more robust than either term. And it is one operation on one data structure: generalising from it to "Rust is 1.6× faster" would repeat exactly the error this thesis criticises in the average-case literature.

7.5.3 Latency, throughput, and saturation across the sweep

Figure 7.5 plots p99 latency against offered load, and the shape it shows is the one Figure 3.3 sketched. Through the tracking region the four variants climb together and gently, from roughly 240 μs at the lightest load to single-digit milliseconds by 335 kops/s, with confidence bands wide enough that no curve can be separated from another anywhere in that region. Then, in the single step of the rate ladder that follows, every variant's p99 rises by more than two orders of magnitude at once, from about 6 ms to several seconds. That is the overload cliff, and the sharpness of its onset is the finding: one increment of offered load separates a service meeting millisecond deadlines from one missing them by three orders of magnitude. A design with bounded, blocking queues has no graceful degradation to offer in between.

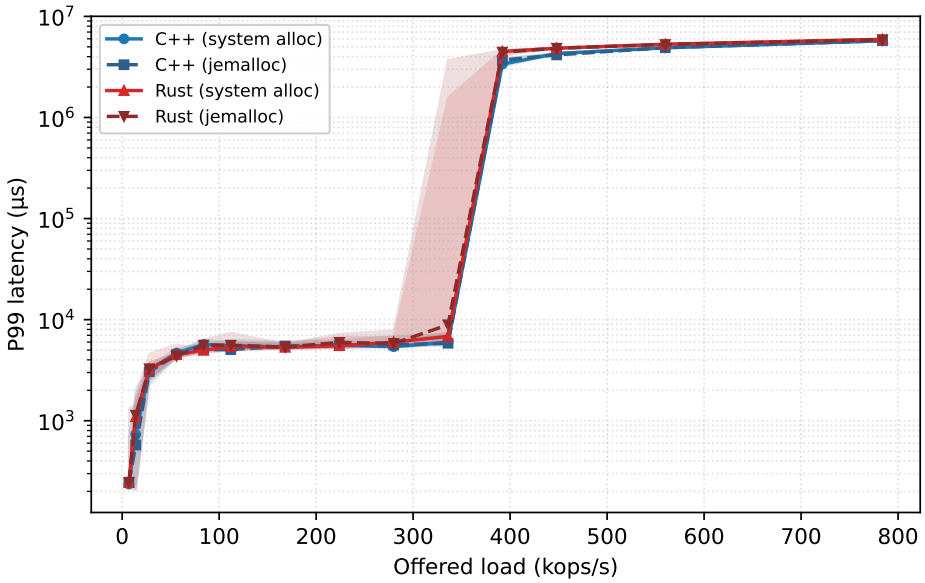


Figure 7.5: Tail latency (p99) versus offered load for the four build variants. Lines are medians over 20 repeats; shaded bands are bootstrap 95% CIs. Log-scaled y-axis. Measured on the host of Section 7.1.

Two properties of the far side of the cliff bound what those numbers mean, and they are the reason Section 5.5 tests no percentile there: latency past the knee is accumulated schedule backlog rather than service time, and because the measured phase is truncated the reported percentiles saturate near the width of that window, which makes every one of them above the knee a lower bound. What remains interpretable in this regime is how much work each variant still completes, and Section 7.5.4 takes that up.

Figure 7.6 shows achieved throughput against offered load, and it is the more informative of the two plots, because throughput is an average over a whole run and therefore far more robust than a percentile to interference landing on individual operations. Every variant tracks the ideal offered-equals-achieved line up to 336 kops/s and saturates at around 335 kops/s. Two things follow immediately. First, because achieved throughput matches offered load below the knee to within 0.1%, the operating point of Table 7.3 is genuinely a *sustained* load rather than a nominal one: the system is completing work as fast as it is offered, so the latencies reported there are not those of an already-queueing system. Second, the knee itself sits at the same place for all four variants, which is what the identical thread-per-shard structure predicts: where the design stops coping is a property of the design.

What happens *after* the knee is not shared. The four curves separate as soon as the

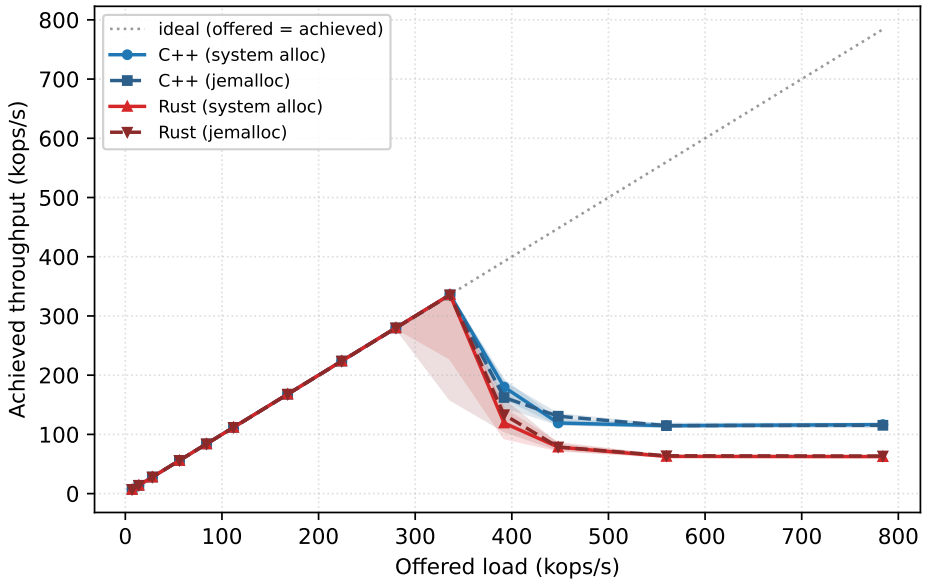


Figure 7.6: Achieved throughput versus offered load. The dotted line marks the ideal offered-equals-achieved regime. Throughput is counted over the measured phase only, so the warm-up iterations bias neither the numerator nor the denominator. Measured on the host of Section 7.1.

offered rate stops being met, and they separate by language rather than by allocator, which is the one thing the sweep was built to be able to detect and the one regime in which it can. Section 7.5.4 measures that separation rather than leaving it to the eye.

Figure 7.7 isolates the shape of the distribution at the 56 kops/s sustained load by plotting latency against percentile. The curves start together at the median and fan out as the percentile rises, which is the expected shape for a heavy-tailed distribution and is why a percentile-aware methodology is used throughout. The fanning should not be read as a language effect: as established above, the run-to-run dispersion at these percentiles is larger than the separation between the curves, so the figure shows where the measurement loses resolution rather than where the languages differ.

7.5.4 Behaviour past saturation (RQ1)

RQ1 asks how the two languages compare under increasing load *and overload*, and everything reported so far has been below the knee. This section is the other half. The operating point is 784 kops/s offered, defined as the deepest swept load at which every variant has stopped meeting the offered rate, and the metrics are the two that remain interpretable there (Section 5.5). Table 7.6 gives what each variant still sustains and

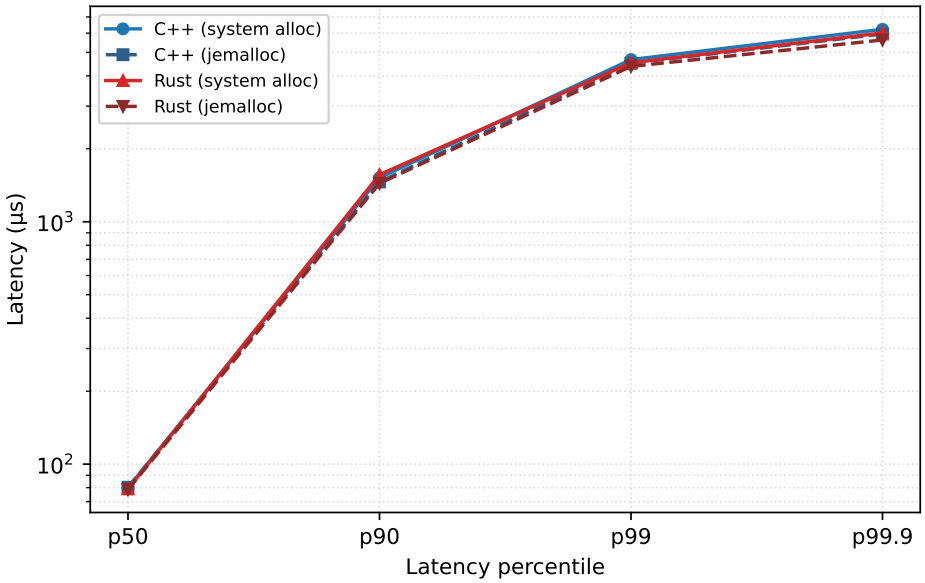


Figure 7.7: Latency by percentile at 56 kops/s sustained load; log-scaled y-axis. The four variants coincide at the median and separate as the percentile rises; on this host the separation is within run-to-run dispersion and is therefore not attributable to language or allocator. Measured on the host of Section 7.1.

Table 7.7 tests the comparison.

The languages are not equivalent in this regime, and the margin is not marginal. C++ sustains 116 kops/s where Rust sustains 63, a factor of 1.8. The paired difference is at least 50 kops/s under either allocator, it holds in 20/20 repeats, and its Holm-adjusted p sits at the corrected floor of the whole family of 14 tests. This is the largest and most reproducible language effect anywhere in the study, and it is the only one that speaks directly to the question RQ1 was written to ask.

The CPU column is what makes the result interpretable rather than merely large, because it rules out the reading that would otherwise be natural. A variant that completes less work might simply be doing less: throttled, blocked, idle. That is not what happens. Rust draws *more* total CPU than C++ here, reproducibly and under both allocators, having inverted the sign it carried at the representative load in Table 7.4. Divided through by the work that actually completed, its CPU cost per sustained operation is 1.9 times the other's. Nor is either variant near the capacity of the machine: both sit near six of the fourteen pinned cores. So the gap is not a capacity limit and not idleness. It is effort expended without completing operations, which is the signature of a synchronisation path that spins, or repeatedly parks and

wakes, where the other blocks once and sleeps.

Where that effort is spent is not measured here, but the candidates are not open-ended, and neither of them is the safety model. Past saturation the producers spend essentially all of their time in two places, and Table 6.2 already records both as points where the two standard libraries forced the implementations apart: the bounded blocking queue, hand-assembled from a mutex and two condition variables on one side against `sync_channel` on the other, and the completion handle, a `std::promise`/`std::future` pair against a freshly allocated channel per request. A difference of this size, appearing only once those primitives are contended, is where the forced-idiom log predicted a difference would appear if one did. Nothing in the borrow checker, in bounds checking, or in ownership is implicated, and the finding should not be quoted as though it were: what it compares is two standard libraries' blocking and wakeup machinery under contention. Confirming that attribution needs a contended-queue microbenchmark holding everything else fixed, which is the natural companion to the component decomposition of Section 7.5.2 and is listed as future work rather than claimed here.

Three things again bound what the finding licenses. It is one design: a shared-nothing service with one request outstanding per producer and a queue bound of 1 024, and the depth of the collapse depends on all three. It is measured with a harness whose producers block per request, so it describes what this shape of client experiences rather than what a pipelined one would. And the absolute figures belong to this host. What travels is the ordering and its cause: below saturation the language choice did not matter, above it the standard library's synchronisation primitives did, and the crossover is sharp enough that a comparison stopping at nominal load would have concluded the opposite.

That crossover also bears on the closest prior work, which reports a comparable degradation past a threshold request rate and attributes it to the asynchronous runtime [42]. Section 8.2 takes up what it means that the same shape appeared here with no such runtime on either side.

7.5.5 Resource use (RQ3)

Table 7.8 and Figure 7.8 report resource use at the same sustained load. Peak resident set size separates the four variants cleanly, and it does so whichever way the family is read: its intervals exclude zero, and its effect keeps its sign under both allocators.

Table 7.6: What each variant still sustains at 784 kops/s offered, well beyond the 335 kops/s knee: the median over 20 repeats above its bootstrap 95% confidence interval. Below saturation achieved throughput is pinned to the offered rate by construction and can express nothing about the languages; here it is the only thing that can. CPU utilisation exceeds 100% because it is summed across threads, so 600% is six busy cores of the fourteen pinned.

Variant	sustained throughput (kops/s)	CPU (%)
C++ (system alloc)	116.6	611.6
	[115.1, 117.9]	[610.0, 612.2]
C++ (jemalloc)	115.4	609.4
	[112.8, 117.9]	[607.8, 611.9]
Rust (system alloc)	62.4	632.7
	[61.7, 63.4]	[630.8, 634.2]
Rust (jemalloc)	63.5	633.1
	[62.6, 64.5]	[631.6, 634.5]

Table 7.7: Exact paired permutation test of C++ against Rust at 784 kops/s, the allocator held fixed, Holm-corrected across the same family of 14 tests as Table 7.4 rather than as a family of its own. "Repeats" counts how many of the 20 paired differences agree with the sign of the median, which a statistic computed from their mean cannot show. No latency percentile is tested here, for the reason fixed in advance in Section 5.5.

Metric	Allocator	Effect (Rust − C++)	Repeats	p	Holm
Sustained throughput (kops/s)	system	−54.0	20/20	0.0000019	0.000027
		[−55.8, −51.9]			
Sustained throughput (kops/s)	jemalloc	−52.2	20/20	0.0000019	0.000027
		[−53.1, −49.6]			
CPU utilisation (%)	system	+20.6	20/20	0.0000019	0.000027
		[+19.1, +23.3]			
CPU utilisation (%)	jemalloc	+23.6	20/20	0.0000019	0.000027
		[+20.8, +25.2]			

Two effects are present in it rather than one, and they are not of comparable size. Linking `jemalloc` costs 32.9–34.4 MiB in either language, while the Rust build costs 1.9–3.4 MiB more than the C++ build under either allocator, with a bootstrap interval of 1.8 to 3.5 MiB. So resource use follows both, but the allocator dominates by an order of magnitude: choosing between these two allocators moves roughly ten times as much memory as choosing between these two languages. That ordering is partly a consequence of the fairness controls rather than a fact about `jemalloc` in general, and saying so avoids overstating it. Both builds pin the arena count identically so that neither language's default could become an accidental advantage, and pinning it high is what makes the arena metadata a large fixed cost on both sides. A deployment tuning `jemalloc` for footprint rather than for comparability would not pay it.

The one row in the robust block of Table 7.4 that does not clear the correction is peak RSS under `jemalloc`, and the reason is a property of the statistic rather than of the software: the interval there is tight and nowhere near zero and 18 of the 20 paired repeats agree, but a handful of runs report a resident set some 30 MiB below their variant's median while being otherwise indistinguishable in throughput, wall time, and error count, and the permutation statistic is a mean of paired differences, which such a run dominates. This is the case the reading rule of Section 5.5 was written in advance to cover, and Section 8.6 takes up both the rule and the measurement defect behind it.

CPU utilisation behaves differently from either, and the difference is regime-dependent rather than noise. At this below-saturation operating point Rust draws marginally *less* CPU, by at most 3.4 percentage points, consistently under both allocators and clearing the correction under both. Section 7.5.4 finds the sign inverted past saturation and the magnitude much larger. Achieved throughput adds nothing here for the reason Section 5.5 gives: at a below-saturation point it is pinned to the offered load, and it is reported only to confirm that the point is sustained rather than nominal, which it is to within 0.1%.

Two qualifications keep the memory finding the size it actually is. It is a difference in fixed cost rather than in cost per context: at the 5 000 active UEs used in the sweep, the eight-byte layout difference measured in Section 6.4.1 accounts for roughly 40 kB, two orders of magnitude short of the gap, so the bulk of it is runtime and allocator-arena overhead that does not scale with the workload and would matter proportionally less, not more, in a larger deployment. And the measurement can establish the difference without explaining it: no hardware counters or allocation profiles were collected, so

Table 7.8: Resource use at a sustained offered load of 56 kops/s: achieved throughput, peak RSS, and total CPU utilisation. The median over 20 repeats above its bootstrap 95% confidence interval. Peak RSS is the one metric here whose intervals separate the four variants cleanly. Measured on the host of Section 7.1.

Variant	throughput (kops/s)	peak RSS (MiB)	CPU (%)
C++ (system alloc)	55.99	9.7	108.5
	[55.99, 55.99]	[9.6, 9.8]	[107.7, 109.8]
C++ (jemalloc)	55.94	42.6	108.0
	[55.93, 55.95]	[42.4, 42.6]	[106.9, 108.7]
Rust (system alloc)	55.99	11.6	105.0
	[55.99, 55.99]	[11.5, 11.6]	[104.0, 105.3]
Rust (jemalloc)	55.94	45.9	106.3
	[55.94, 55.95]	[45.8, 46.0]	[104.8, 107.2]

the split between arena pre-reservation, per-thread caches, and static footprint is not resolved here. Both points are revisited in Chapter 8.

The RQ3 finding is therefore neither the flat "no resource penalty" that overlapping intervals would suggest nor a general performance penalty. Rust costs a few megabytes more of fixed footprint, reproducibly and under both allocators but an order of magnitude less than the allocator choice itself; it is marginally cheaper in CPU while the service is coping and considerably more expensive once it is not; and on the bare operation of Section 7.5.2 it is the faster of the two.

7.5.6 Safety via fault injection (RQ2)

Instead of claiming that Rust has no bugs, the catalogue of seven defects set out in Table 3.1 is deliberately injected, each as a small self-contained program flavoured with the subsystem's domain so that it maps onto a plausible bug in the service. Its division into two groups is what the method of Section 5.3.2 turns on: five classes from inside the region Rust claims to govern, and two chosen precisely because safe Rust is expected *not* to prevent them. Without the second group the outcome would be decided by the choice of test cases, confirming the expected result but never able to disturb it. With it, the study measures a boundary rather than demonstrating a claim. The fault runner holds the expectation recorded for each case in advance and reports a violation in either direction, so a case that unexpectedly compiled, or unexpectedly failed to, would surface rather than being absorbed into the conclusion.

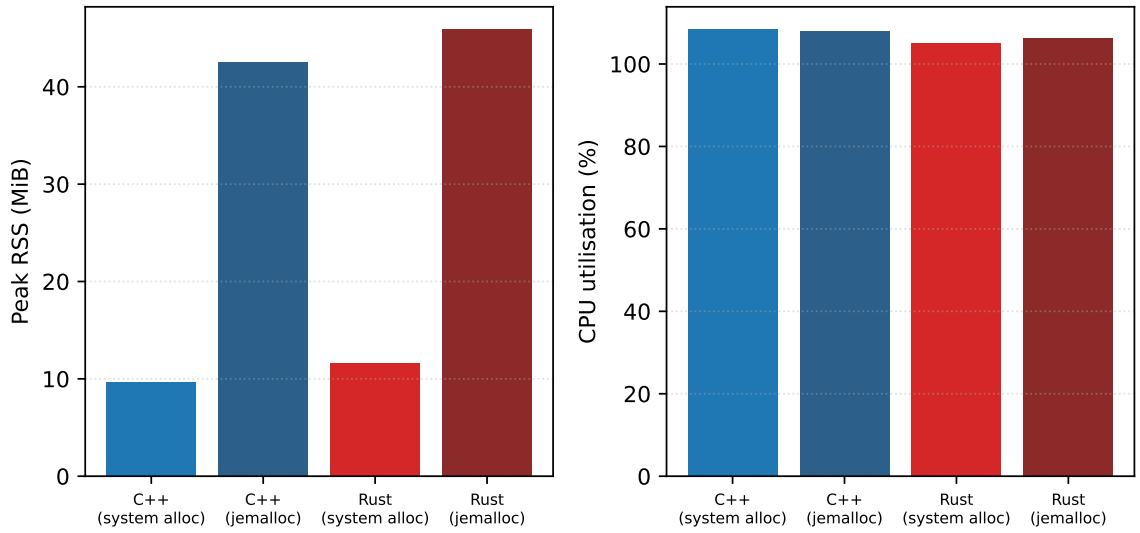


Figure 7.8: Peak resident set size (left) and CPU utilisation (right) per build variant at 56 kops/s. Measured on the host of Section 7.1.

Procedure. For each C++ defect the study builds and runs the program under both AddressSanitizer (ASan) and ThreadSanitizer (TSan) and records which tool, if any, detects it. No static analyser is run, for the reason set out in Section 5.3.2: its output is neither a compile-time rejection nor an observation of an execution, so it does not belong to any of the three outcomes the taxonomy records. The C++ columns of Table 7.9 therefore report the tooling this study ran rather than a ceiling on what C++ tooling could achieve. The analogous construct is written in safe Rust and compiled; where it compiles, it is then run under a bounded timeout and its observable behaviour recorded, since "it compiles" alone would not establish whether the defect survives into the running program. The catalogue, the runner, and the raw tool logs are published with the thesis.

Results. The outcomes are shown in Table 7.9, measured with `g++ 13.3` (`-std=c++23`, `-O0` so the optimizer cannot elide an injected defect) and `rustc 1.97` (edition 2021). That C++ toolchain is the local development machine's rather than the measurement host's `g++ 15.1.1`, and the catalogue was run there deliberately: whether a program compiles, and what a sanitiser reports when it runs, are properties of the toolchain rather than of the machine hosting it, which is the concrete content of calling this result environment-insensitive in Section 7.1. Within its scope the result is unambiguous: every one of the five memory-safety and data-race defects is a com-

pile error in safe Rust, so none can be written at all, whereas in C++ each is valid code whose defect manifests only at runtime, and only if a test actually exercises the buggy path. The memory-safety defects are caught by ASan; the concurrency defects are caught only by TSan and are invisible to ASan, underlining that no single C++ sanitizer covers the whole catalogue. Runtime detection further requires building a dedicated instrumented binary and paying the sanitizer's time and memory overhead, whereas the Rust errors are reported on every ordinary build regardless of test coverage. The parenthesised codes are the `rustc` diagnostics that fire (for example `E0499`, conflicting mutable borrows; `E0502`, a mutable borrow while a shared borrow is live; `E0382`, use of a moved value; `E0373`, a closure outliving borrowed data as it escapes to another thread).

The two boundary cases behave as predicted and delimit the claim. Both compile in safe Rust and both misbehave. The deadlock case hangs in both languages and had to be killed by the harness timeout; ThreadSanitizer does flag it in C++, reporting a lock-order inversion, while AddressSanitizer issues no report at all because nothing is corrupted. The lost-update case terminates in both languages with a silently wrong answer: against an expected total of 40 000 increments the safe-Rust program recorded 35 030, losing 4 970, and the C++ program under ThreadSanitizer recorded 25 448, losing 14 552.

The distinction these expose is precise, and it is the distinction the phrase "thread safety" routinely blurs. Rust's concurrency guarantee is the absence of *data races*, meaning unsynchronised concurrent access to the same location; it is not a guarantee about *lock-acquisition order* and not a guarantee about the *atomicity of a composed operation*. The lost update is the sharper of the two cases, because neither sanitiser reports it either: with every access properly locked there is no race for a race detector to find. It is therefore the one class in the catalogue that nothing in either language's toolchain addresses, and the one that fails most quietly, corrupting a counter rather than crashing. The consequence for practice is developed in Chapter 8.

Figure 7.9 presents the same outcomes as a colour-coded matrix, generated from the runner's machine-readable log (`faults/results/taxonomy.csv`), with a rule marking where the compile-time guarantee stops. Read as a whole it makes two things visible at once: above the rule the entire right-hand column is green while the C++ columns are a patchwork of runtime detections and outright misses, and below it the right-hand column turns red alongside them.

Table 7.9: Defect-class taxonomy: measured outcomes for the injected catalogue. C++ under `g++ 13.3` with ASan/TSan; Rust under `rustc 1.97`, edition 2021. "Compile error" means the safe-Rust program does not build, so the defect cannot be written. The lower block is the boundary group: defect classes chosen because safe Rust is *not* expected to prevent them, and does not.

Defect class	ASan	TSan	Safe-Rust outcome
Within the guarantee			
Use-after-free	detected	detected	compile error (E0499)
Double-free	detected	not detected	compile error (E0382)
Iterator invalidation	detected	detected	compile error (E0502)
Data race	not detected	detected	compile error (E0373†)
Unguarded shared mutation	not detected	detected	compile error (E0502†)
Boundary of the guarantee			
Lock-order deadlock	not detected‡	detected	compiles; deadlocks
Lost update	not detected	not detected	compiles; wrong result

† Several diagnostics fire for these cases; the first is shown (the data race also raises `E0382` and `E0502`, the unguarded mutation `E0524` and `E0597`). ‡ AddressSanitizer issues no report because nothing is corrupted: the instrumented binary simply deadlocks and is killed by the harness timeout.

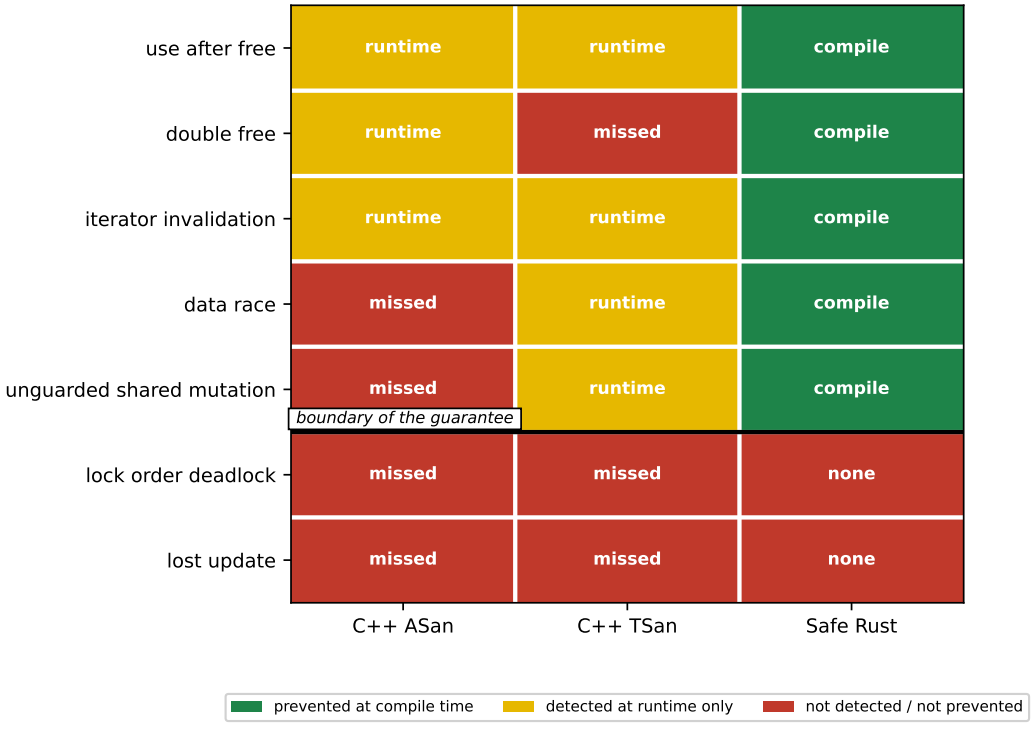


Figure 7.9: Defect-class outcome matrix, with the boundary group below the rule. Green cells are defects prevented at compile time (safe Rust); amber cells are defects that compile but are caught at runtime by the given sanitiser; red cells are defects the sanitiser does not detect or, in the right-hand column, that safe Rust does not prevent. Generated from the fault-injection runner logs.

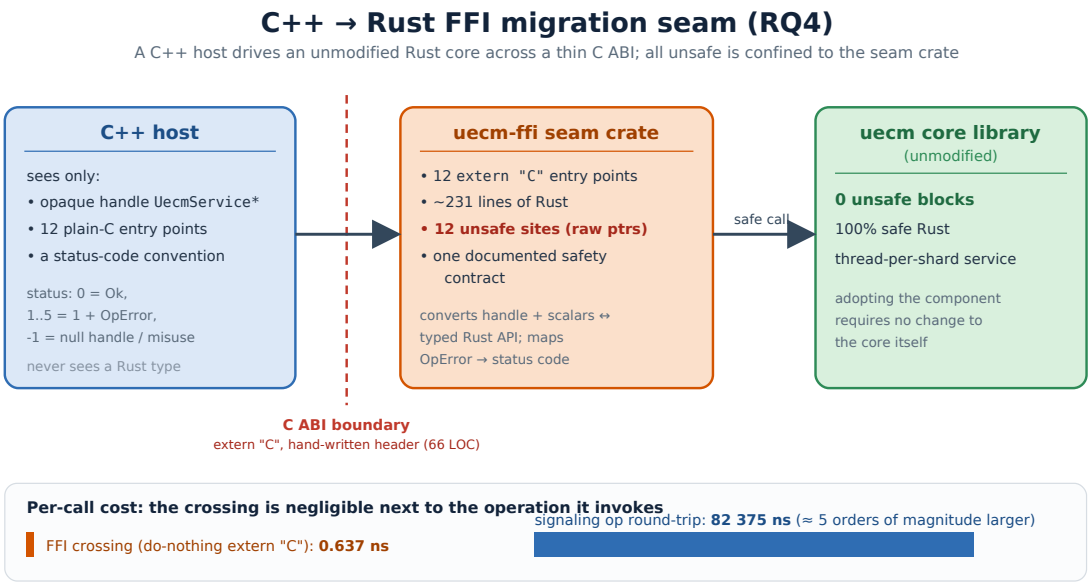


Figure 7.10: The C++-to-Rust FFI migration seam. The C++ host sees only an opaque handle, twelve plain-C entry points, and a status-code convention; a thin `extern "C"` seam crate concentrates all of the raw-pointer `unsafe` (twelve sites) and calls the unmodified Rust core, which remains free of `unsafe`. The do-nothing crossing costs a fraction of a nanosecond; measured against the same signaling operation called natively, the seam's cost per operation is bounded at 10.91%.

7.5.7 Migration (RQ4)

Incremental adoption is assessed with a working prototype that models the realistic direction of migration: an existing C++ host keeps a Rust component behind a C ABI. The Rust service is exposed through a thin `extern "C"` seam in a *separate* crate that depends on the unmodified core library, so adopting the component requires no changes to the component itself. The C++ host sees only an opaque handle and a set of plain-C entry points (a hand-written header, no generated bindings), and drives the full UE lifecycle across the boundary; the error model is transported faithfully (for example, a duplicate connection setup returns `InvalidState` through the seam). Figure 7.10 shows the resulting structure and where the `unsafe` sits.

The prototype yields concrete interop-cost figures, summarised in Table 7.10. The seam is small, twelve C-ABI entry points in roughly 230 lines of Rust plus a 66-line C header, with no third-party FFI dependency. The safety boundary is clean: the core library remains free of `unsafe`, and the raw-pointer handling the C ABI requires is confined to the seam crate (its twelve `unsafe` sites) under a single documented safety contract, so the C++ side never sees a Rust type.

The per-call cost is measured twice, because the two measurements answer different questions and only the second answers RQ4's. The first is the intrinsic crossing: a do-nothing `extern "C"` call costs 2.54 ns, an ordinary indirect call, and Figure 7.11 places it against the cost of a signaling operation. That bounds the crossing but says nothing about what the seam does to a real operation, because an empty call and a full round-trip are not the same experiment: the round-trip is dominated by a cross-thread handoff, and an argument that the handoff "would be paid identically anyway" is reasoning standing in for a measurement.

The measurement is therefore made directly. The same signaling operation, on the same service configuration and in the same loop, is invoked natively from Rust and then from C++ through the seam, interleaved over 15 repeats so the pair is matched. The paired difference is +192 ns per operation with a bootstrap interval of $[-954, +1341]$ ns and an exact paired p of 0.239: no cost is detected, and more usefully, the interval bounds any cost the measurement could have missed at 10.91% of the operation. The interleaving is not optional here. A single pass of each, run while preparing this section, made the seam appear *faster* than calling natively by a third, because the run-to-run spread of the handoff on a shared host is far wider than anything a C ABI could contribute.

The main engineering effort is therefore in *designing* a C-compatible API surface, an opaque handle, scalar-typed operations, and a status convention, rather than in any runtime overhead. The reverse direction (Rust calling into C++) is symmetric but requires a C-ABI wrapper on the C++ side and re-introduces `unsafe` at the Rust caller. The prototype and its measurements are published with the thesis, and inform the migration guidelines of Chapter 8.

Table 7.10: FFI migration prototype: seam size, safety boundary, and per-call interop cost (C++ host calling the Rust component across a C ABI, `-O2`). The core library is unmodified and `unsafe`-free; all interop is confined to the seam crate. The last three rows are the direct comparison: one signaling operation on the same service configuration and the same loop, invoked natively and then through the seam, so the difference isolates what the boundary adds rather than what the operation costs.

Metric	Value
C-ABI entry points	12
Seam size (Rust `lib.rs`)	231 LOC
Hand-written C header	66 LOC
`unsafe` in core library	0
`unsafe` in FFI seam	12
Raw FFI crossing	2.54 ns/call
Signaling op, native caller	12.3 µs/call
Signaling op, through the seam	12.7 µs/call
Seam cost per operation (paired)	+192 ns [−954, +1341]
as a fraction of the operation	+10.91% at the upper bound

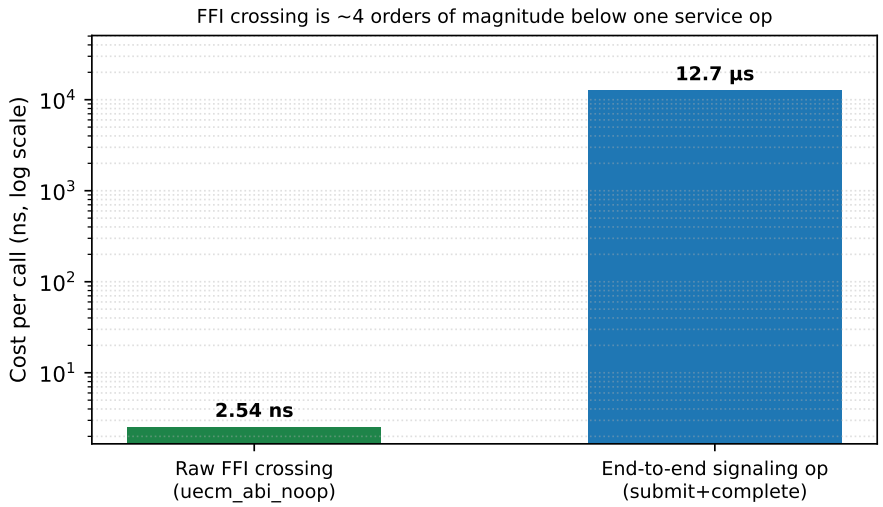


Figure 7.11: Per-call cost of crossing the C ABI (a do-nothing `extern "C"` call) versus the end-to-end cost of one signaling operation through the seam. Log-scaled y-axis: the FFI crossing is negligible next to the service's cross-thread round-trip.

8 Discussion

8.1 Summary of findings

Table 8.1 collects the headline finding for each research question together with the strength of the current evidence. The safety and migration results (RQ2, RQ4) do not depend on the measurement environment at all. The performance and resource results do, and the honest division is no longer between chapters but within RQ1: the saturation knee, the overload comparison, and the resource comparisons are resolved on the host of Section 7.1 and replicate across the allocator control, while the far tail below saturation remains unresolved for reasons that belong to the machine rather than to the analysis.

8.2 Discussion on the research questions

RQ1 (predictability). RQ1 has two halves, below saturation and beyond it, and the campaign answers them differently enough that reporting either alone would misrepresent the result.

Below saturation the languages are equivalent for any purpose a deployment would care about. The small paired difference at the median that does clear the correction is a fraction of an 80 μs operation that nothing running this service would notice, and the tail in this regime is a null this host cannot turn into an ordering. That is consistent with the wider evidence that Rust's safety checks are largely absorbed by modern out-of-order CPUs [45, 20]. It is also a weaker statement than it appears, for the reason Section 7.5.2 gives: the operation the languages execute is 0.250% of what the sweep measures, so an end-to-end null in a design dominated by a cross-thread wakeup is a statement about the design rather than about the languages. Measured apart, the operation differs by 1.6×.

Beyond saturation the answer changes in kind, and it is the one result in the study that answers RQ1 on RQ1's own terms. The C++ build sustains 1.8 times the throughput of the Rust one while drawing more CPU per completed operation, reproducibly and

Table 8.1: Summary of findings per research question and the current strength of evidence.

RQ	Headline finding	Evidence status
RQ1	The answer divides at saturation. Below it the variants agree to 2 μs (2.1%) at the median and no tail ordering is resolvable; the end-to-end number is only 0.250% operation, so it cannot bound a language effect in any case, while the operation measured apart is 1.6× faster in Rust. Past saturation they diverge decisively: C++ sustains 1.8× the throughput of Rust at 784 kops/s, in 20/20 repeats under both allocators.	Overload and operation: resolved. Far tail below saturation: unresolved on this host
RQ2	All five injected memory-safety and data-race classes are prevented at compile time in safe Rust, where in C++ each compiles and is caught, if at all, only at runtime and by no single sanitiser. The guarantee stops there: the two boundary classes compile in both languages.	Final
RQ3	A memory penalty that reproduces: 1.9–3.4 MiB more peak RSS in the Rust build under *both* allocators, an order of magnitude less than the allocator's own 32.9–34.4 MiB. CPU inverts across the regimes: Rust draws up to 3.4 percentage points less below saturation and 1.9× more per completed operation above it.	Memory: resolved, unexplained. CPU: resolved in both regimes, opposite signs
RQ4	The seam is small and keeps the core `unsafe`-free at compile time. Measured against the same operation called natively, no cost is detected and any tax is bounded at 10.91% of an operation; the real cost is API design. The core remains exposed at run time to memory errors in the C++ host it shares an address space with.	Final

at the corrected floor of the whole test family (Section 7.5.4). So the practical form of RQ1's answer is not that the language choice is irrelevant to predictability but that it is irrelevant *until the service stops coping*, at which point it decides how much work still gets done. A comparison that stopped at nominal load, as most in this literature do, would have concluded the opposite with the same apparatus.

That reframes the closest prior work rather than merely agreeing with it. Arvidsson and Kaller measure this quantity for 5G control-plane procedures and find Rust's 99th percentile degrading faster beyond a threshold request rate, which they trace to stackless `async/await` permitting a context switch immediately before an I/O operation begins [42]. Section 8.5 accepts that attribution, and this study was built to exclude the confound behind it: one worker per shard over a bounded blocking queue, no asynchronous runtime on either side. Chapter 4 committed in advance to what each outcome would mean, and the second branch is the one that fired. A degradation past a threshold rate appeared with the concurrency model held fixed, so the asynchronous runtime cannot be the only place such an effect originates. The prediction also named where to look first if that happened, bounds and overflow checking [50, 51], and that part of it was wrong: the effect appears only once the queue and the completion handle are contended, which points at the two standard libraries' blocking primitives, both of them already recorded in the forced-idiom log of Table 6.2 as places where the implementations were forced apart. Recording that the prediction resolved against its own first hypothesis is more useful to the next study than quietly substituting the explanation that survived.

RQ2 (safety). The result does not depend on the environment, and what makes it interesting is not the half that confirms the expectation but the half that bounds it. Within its scope the guarantee is absolute, and the difference from C++ is one of kind rather than of degree: a compiler that refuses to build a program rules the defect out of every execution, where a sanitiser rules it out only of the executions an instrumented test happens to take. That aligns with the CVE evidence that Rust memory-safety bugs invariably involve `unsafe` [32] and with the caution that unsafety can still propagate through dependencies [33], which is why keeping the core `unsafe`-free matters.

Where the guarantee stops matters just as much to a vendor deciding what Rust buys, and the two boundary classes are instructive for different reasons. The lock-order deadlock shows that Rust reasons about *aliasing*, not about *acquisition order*: a

`Mutex` guard is an ordinary value whose lifetime the borrow checker tracks perfectly, and tracking it perfectly says nothing about whether two threads take their locks in compatible orders. That this is a boundary of the language rather than an artefact of the injected example is confirmed by the tooling built around it, since static deadlock detection for Rust is an active research problem in its own right [52] and model checkers for Rust kernel code are motivated explicitly by the observation that ownership and borrowing leave data races, deadlocks, and livelocks reachable [53]. The lost update is sharper still, for the reason Section 7.5.6 gives: it escapes ThreadSanitizer for precisely the reason it escapes the borrow checker, since with every access correctly synchronised there is no unsynchronised access to find. The distinction it exposes, between a data race and a race condition, is the one the phrase "thread safety" routinely conflates, and Rust removes only the first.

The practical consequence is that adopting Rust retires one category of concurrency defect completely while leaving another untouched. The category left untouched is the one that fails silently, corrupting a counter or a state decision rather than crashing. A migration plan that treats "memory-safe" as "concurrency-safe" would therefore mis-budget its review and testing effort. That is why this boundary is reported as a first-class result rather than a caveat. Figure 8.1 states the whole result in one picture, with the two halves of the guarantee on either side of the line and the mechanism that does the enforcing, or the absence of one, named in each panel.

RQ3 (overhead). The answer splits by resource, and the splits do not all point the same way. On memory there is a penalty: the Rust build carries a few MiB more peak resident set than the C++ build, and keeps that sign under both allocators. Two things make it credible beyond the p-value. Peak RSS is a run-level maximum rather than a tail statistic, so it is insensitive to the scheduling jitter that destroys the p99. And the effect is an order of magnitude smaller than the allocator's own, which is the useful way to size it: a team choosing between these two languages moves less memory than a team choosing between these two allocators.

On CPU the answer depends on the load, and that dependence is the finding rather than a complication. Below saturation Rust draws marginally *less*, consistently under both allocators; past saturation the sign inverts and the magnitude grows (Section 7.5.4). A single number for "the CPU cost of Rust's safety model" would therefore be an average over two regimes that behave oppositely, and neither figure is the safety model in any

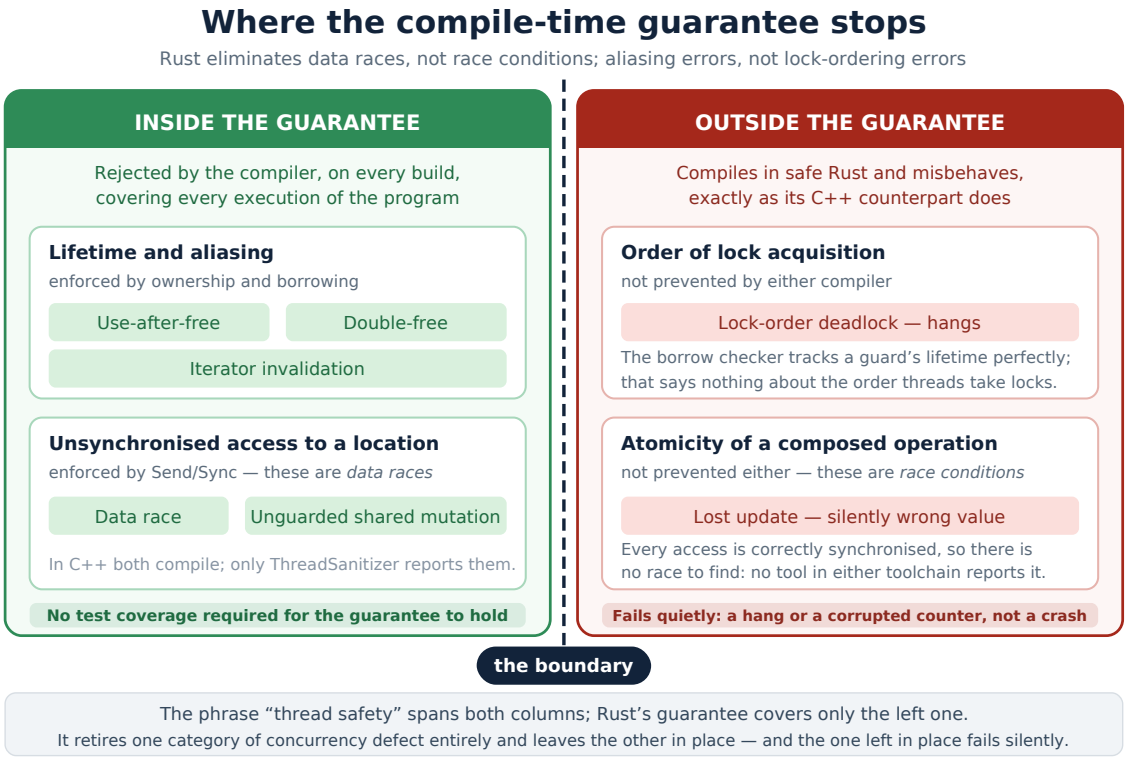


Figure 8.1: Where the compile-time guarantee stops. On the left are the classes safe Rust rejects outright, grouped by the *mechanism* that rejects them rather than by the defect families of Figure 3.2: ownership and borrowing for lifetime and aliasing errors, which is why iterator invalidation sits with them, and the `Send`/`Sync` discipline for unsynchronised access to a location. On the right are the two classes the catalogue included to find the edge, which neither compiler prevents. The phrase "thread safety" spans both sides of the line; the guarantee covers only the left.

case. On throughput the same split applies for the same reason: below saturation there is nothing to measure, because achieved throughput tracks offered load by construction, and above it there is the largest effect in the study.

Three things bound what the memory result licenses. It is small in absolute terms and, more importantly, it is fixed rather than per-context, so its share of a deployment's footprint falls as the deployment grows; the measured layout difference between the two `UeContext` types contributes almost none of it. It is unexplained: attributing it between allocator arena reservation, per-thread caches, and static footprint needs allocation profiles and hardware counters this study did not collect, which is the single clearest gap in the RQ3 instrumentation. And it should not be generalised into a claim about Rust's safety model, since nothing here isolates bounds checking or ownership-driven copying as the cause; the mechanisms that the literature has localised [50, 51] are candidates to test, not conclusions to import. What can be said is narrower and still useful to a vendor sizing a control-plane node: adopting Rust here costs a few megabytes of fixed footprint per process, and buys a faster core operation (Section 7.5.2) that this workload's structure never converts into throughput below saturation and, above it, converts into a deficit rather than a gain.

The sign of that result is worth defending, because the closest prior work in the domain reports the opposite one. Karlbäck and Orö measure peak resident set on a RAN workload and find Rust the *lowest* of the three languages and C++ the highest, with the gap widening as concurrency rises, a pattern they flag as unexplained [41]. The two findings are compatible once it is seen that they measure different quantities. Their C++ runs on Boost.Fibers, which allocates a stack per fiber, so ten thousand concurrent sequences cost ten thousand stacks; their Rust runs on Tokio, where a task is a heap-allocated future of a few hundred bytes. What their figures capture is therefore the per-context cost of the concurrency runtime, which is precisely why it grows with the number of contexts. The design measured here has no green threads on either side: a fixed thread-per-shard pool leaves no per-context allocation to differ, and what remains is the fixed footprint reported above, constant in the number of UEs rather than proportional to it. Their disagreement with this result is thus between two concurrency runtimes rather than between two languages, which is as direct a demonstration as the literature offers of why this study holds the concurrency model fixed. It also sharpens the caveat attached to the present number: it licenses nothing about Rust's memory behaviour under a task-per-UE design, which is a different experiment.

RQ4 (migration). The prototype shows that incremental adoption is technically cheap: a C++ host can call an unmodified Rust core across a thin C-ABI seam whose cost on a real operation is bounded at 10.91%, confining all `unsafe` to the seam. That bound comes from the matched comparison of Section 7.5.7 rather than from setting an empty crossing against a full round-trip; the two are not the same experiment, and on this host a single unpaired pass of each was enough to invert the apparent ordering. The dominant cost is therefore human, not machine: designing a C-compatible surface and, per adoption studies [54, 55], climbing Rust's learning curve. This reframes migration risk as an organisational rather than a performance concern, and the two largest documented adoptions into existing C and C++ codebases report the same ordering of costs [56, 57].

One boundary on that result has to be drawn explicitly, because the natural reading overstates it. "All `unsafe` confined to the seam" is a claim about what the compiler verifies, and it is true as measured: the core crate contains no `unsafe`, and every raw-pointer operation sits in twelve reviewable sites. It is not a claim that the core is protected at run time. Host and component share an address space, and a memory error on the C++ side can corrupt Rust-owned data regardless of how the Rust side was checked, which is why the remedies proposed in the literature isolate the two heaps with hardware support rather than relying on interface discipline [37, 38]. The defect classes that arise at such boundaries have been catalogued in their own right [39]. So what a migration of this shape buys is narrower than "the Rust part is safe": it is that the migrated component stops *contributing* new memory-safety defects, while remaining exposed to those of the host until the host is migrated too. That is still the right direction of travel, and it is a reason to sequence a migration so that the components sharing the most mutable state cross over together rather than one at a time.

8.3 Revisiting the hypotheses

Chapter 4 committed to three hypotheses before any measurement was taken. Stating their disposition explicitly, rather than letting the results speak for themselves, is what distinguishes a test from a description.

H1: supported below saturation, refuted above it, and too loosely worded throughout.
The hypothesis had two parts: that the languages are performance-equivalent

below saturation, and that any difference is confined to the far tail. The first is supported. The second is refuted, and not in the direction the wording anticipated: the difference lies not in the far tail but in *overload*, in the amount of work completed once the service stops coping (Section 7.5.4), which is a different quantity in a different regime. The hypothesis as written would have led the campaign to look in the wrong place, and that the result was found at all is due to the sweep being pushed past saturation on methodological grounds rather than in expectation of it. The far tail below saturation, which is what H1 actually named, remains unresolved on this host.

A second disposition is that the hypothesis is not well posed. "Performance-equivalent" was written as though it named one quantity, and Section 7.5.2 shows it names at least three that answer differently: at the level of the delivered operation below saturation the languages are equivalent, which is what a deployment at nominal load experiences; at the level of the code the compilers emit they are not, and the end-to-end measurement dilutes that difference by more than two orders of magnitude; and at the level of sustained work under overload they are not equivalent either, in the opposite direction. All three are true, and a hypothesis that cannot distinguish them is one this study should have stated more carefully before measuring.

H2: supported in its precise form, refuted in its loose one. The hypothesis was that safe Rust prevents at compile time "the memory-safety and concurrency defect classes that C++ leaves to best-effort detection". For the five injected classes of that kind it holds without exception. The fault-injection study still refutes the broader reading the wording invites: the two boundary classes are concurrency defects, they are exactly the sort C++ leaves to best-effort detection, and safe Rust does not prevent them either. H2 is therefore accepted for memory safety and data-race freedom and rejected for concurrency correctness at large, and the latter is the reading practitioners tend to assume.

H3: supported, and tested the right way on the second attempt. The hypothesis was that interoperability cost is negligible against the operations invoked. Measured directly (Section 7.5.7), no cost is detected and any per-operation tax is bounded at 10.91%. The bound, rather than the failure to detect a difference, is what supports the hypothesis. The first version of this measurement did not do that: it set an empty crossing against a full

round-trip and inferred the answer, which is a comparison between two different experiments and would have given the right conclusion for the wrong reason. The accompanying claim, that the real cost is engineering effort, is supported structurally rather than statistically: the seam is small and mechanical, but designing its C-compatible surface was the part of the prototype that required actual design decisions.

8.4 Engineering guidelines for incremental adoption

The results are meant to be usable, and a vendor weighing Rust for a control-plane component needs more than a set of measurements. The following guidelines are the practical form of the evidence above; each is tied to the finding that supports it, and the boundaries of that support are stated rather than glossed, because a guideline quoted without its limit is how the RQ2 result becomes "the migrated component is safe" and the RQ4 result becomes "migration is cheap", neither of which this study establishes. Figure 8.2 then sequences the first three into the order a first migration would actually run them.

Choose the first component for its interface, not its size. Because the boundary's run-time cost is bounded at 10.91% of an operation (RQ4), the criterion for a first Rust component is not how small or how hot it is but how narrow and how scalar its interface is. The seam here stayed at twelve entry points because the subsystem's API was already lifecycle-oriented, typed, and free of shared mutable structures crossing the boundary. A component whose interface passes rich C++ objects, templates, or callbacks will need those designed out first, and that redesign, not the FFI, is the schedule risk.

Put the ABI in its own crate and keep the core `unsafe`-free. Concentrating every raw-pointer operation in a separate seam crate depending on an unmodified core (Section 7.5.7) is worth adopting as a rule rather than treating as an accident of this prototype: it keeps the dependency-propagation problem Evans et al. describe [33] away from the safety-critical code, and makes the reviewable surface of the migration a fixed, countable thing. Two qualifications belong with it. It bounds what has to be

reviewed, not what can be *corrupted*: while the host remains C++ the two share a heap, so a team needing the stronger property must look at in-process isolation [37, 38] rather than at interface discipline alone. And the generated part of a seam is not above suspicion, since binding generators have their own defect history [40]; review it as code rather than trusting it as tooling output.

Do not re-budget concurrency review effort on the strength of memory safety. This follows directly from the boundary cases. Adopting Rust removes use-after-free, double-free, iterator invalidation, and data races from the defect budget entirely, and those are the classes that dominate the security statistics [8, 9]. It removes nothing from the deadlock and atomicity-violation budget. Lock-ordering discipline, invariants about which locks may be held simultaneously, and review for composed-but-not-atomic operations remain exactly as necessary as they were in C++, and they should stay in the code-review checklist and the design-review agenda after migration.

Prefer a shared-nothing structure to a shared-and-locked one, in either language. The strongest concurrency result in this study is not attributable to either language: the thread-per-shard design removes locks from the hot path altogether, which is why the boundary defects had to be constructed as standalone programs rather than found in the implementations. Where state can be partitioned by a natural key, as UE state can be partitioned by `ue_id`, partitioning it converts a class of concurrency defects into a structural impossibility for both languages, and it does so without needing either compiler's help.

Budget for the learning curve, and measure it. The dominant cost identified here is human, matching what adoption studies report [54]. This thesis cannot quantify that cost, because it is a single-author study, and the guideline is correspondingly modest: treat the first component as a training investment with a deliberately generous schedule, and instrument it, recording review time and defect density, so the second component can be planned on measured rather than assumed effort.

Measure the component, not the round-trip, and do not generalise either. The performance findings carry a methodological warning that generalises further than the numbers do. An end-to-end request/response measurement of a design like this

one is 0.250% language and the rest scheduler (Section 7.5.2), so below saturation it will return "no difference" between almost any two competent implementations, and reading that as evidence about the languages is a mistake this study made before it measured the components apart. A vendor running its own comparison should budget for both: the end-to-end measurement, which says what a deployment will feel, and the component one, which says what the language is doing.

Sweep past saturation, because that is where the languages stopped agreeing. This is the guideline the study would most like to have followed from the start rather than arrived at. Every conclusion it drew about performance while the sweep stopped at nominal load was a null, correct at that operating point and silent about the one where the two implementations differ by 1.8× (Section 7.5.4). The knee is not a detail at the edge of the graph: for a control plane it is the region that decides whether a signalling storm degrades service or ends it, and it is where a bounded, blocking design has no graceful behaviour left. A comparison that stops while the system is still meeting its offered load has measured the easy half of the question, and sweeping past the knee costs one more afternoon of machine time.

Beyond both of these, the benchmarking framework rather than its current output is the reusable artifact: the decision it should inform is "measure this on your component, and measure it past the point where your component stops coping", not "Rust is fast enough, proceed".

8.5 Related work

Chapter 2 set out the ground this work stands on. What follows positions the thesis against that literature rather than surveying it a second time, and then treats at length the two studies it has to answer to.

Four strands bear on the comparison. The first is cross-language performance measurement, which reports that compiled systems languages sit close together on time and energy [44, 34], that Rust's safety costs little at runtime [20, 58], and that this holds where timing is tightest: in embedded hypervisors [59], in production cloud infrastructure [18], and in packet processing within a few percent of a C reference [45, 60]. Where a cost is found, the literature has begun to localise it, to bounds, overflow, and division checks [50], to the bounds check specifically when set against hardware-

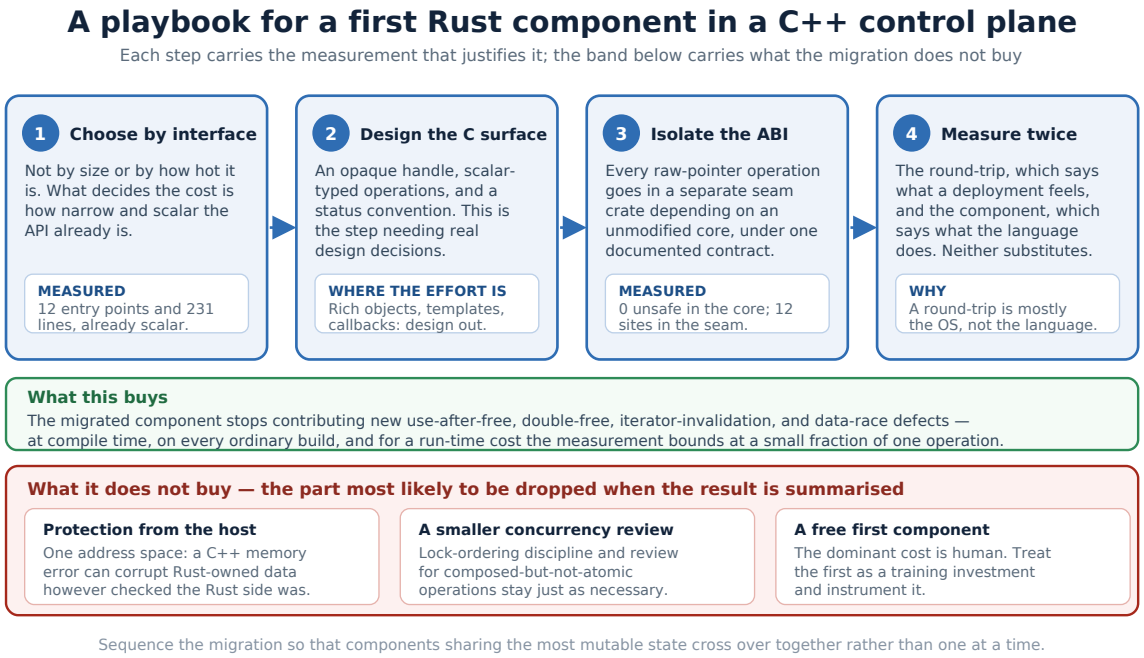


Figure 8.2: A playbook for a first Rust component in a C++ control plane, assembled from the RQ4 prototype and the RQ2 boundary. The upper row is the sequence, each step annotated with the quantity that justifies it; the green band is what the migration buys, the red band what it does not, which is the part most likely to be dropped when the result is summarised. Step 4 is there because measuring the round-trip alone answers a different question from the one being asked (Section 7.5.2).

enforced capabilities [51], and to recurring performance anti-patterns [61]. All of it is average-case, and none of it reports behaviour past a saturation point.

The second is the evidence for the safety claim itself: machine-checked foundations for the safe fragment [11, 29] and for the aliasing discipline `unsafe` code must respect [30]; confirmation that `unsafe` is used sparingly and that memory-safety CVEs invariably involve it [31, 21, 32]; ecosystem-scale auditing that finds defects at scale nonetheless [62, 63, 64]; and the caution that unsafety propagates through dependency chains, so most libraries cannot be fully checked [33]. Alongside these sit the industry and government reports motivating the shift [8, 9, 15, 16, 13] and the long record of memory corruption resisting runtime mitigation [25].

The third concerns the boundary, and it qualifies the adoption argument directly. Confining `unsafe` to a seam does not protect the Rust side, since foreign code in the same address space can corrupt Rust-owned data regardless, which is why the proposed remedies isolate the two heaps in hardware [37, 38]. The seam has its own defect classes [39], its tooling its own bugs [40], and lighter crossing models have been proposed for embedded use [65]. On reach, aliasing patterns expressible in C but not in safe Rust bound translation in principle [66], interoperation is the central open problem in embedded Rust [67], `unsafe` concentrates at exactly these boundaries [68], and the two largest documented adoptions into existing C and C++ report the resulting friction empirically [56, 57].

The fourth is the domain and the method. Predictability rests on the tail-latency and microsecond-scale literature [6, 7, 23, 69] and on the finding that thread-per-core structuring of the kind adopted here moves the tail in its own right [35]; the RAN context on O-RAN disaggregation [3]. Rust already has mobile-network precedent in the datapath [70] and, on the control-plane side, an open-source proof-of-concept gNB-CU with a separated CU-CP and CU-UP over E1, which establishes that the ASN.1 and SCTP protocol surface is no barrier in principle, but is an existence proof rather than a comparison: it reports no latency and has no C++ counterpart built to the same specification [71]. Control-plane latency is an active target in the core network too [72, 73]. The adoption evidence points at the learning curve rather than run-time cost [54, 55, 74], and the method here follows standard guidance on repeat-based performance evaluation [22] and on software fault injection [46].

Two systems studies come closest in method rather than in domain, because they too build the same component twice and attribute the difference: a comparison of

IPC mechanisms implementing one network function in both C++ and Rust [75], and RedLeaf, evaluated against C baselines constructed for the purpose [76]. Neither reports behaviour past a saturation point, which is the regime this thesis targets.

Two master's theses, both carried out at Ericsson, are the closest prior work by domain and deserve separate treatment.

Karlbäck and Orö compare Rust and Go against C++ for RAN applications on x86 and ARM, combining runtime measurements with development-efficiency metrics, and conclude that C++ remains the choice where performance is critical [41]. Their control deserves to be quoted rather than characterised, because it is the clearest statement in this literature of the confound the present study is built to remove. The authors record that they deliberately did not write the same program three times, making "an effort ... to utilize the different strengths of each language, and stay away from using exactly the same implementation" [41]. For the question they ask, which is what a team would obtain by writing idiomatic code in each language, that is the right choice. It is the wrong control for the question asked here. The concurrency runtime then varies with the language (Boost.Fibers over libevent, goroutines, and Tokio respectively), so no measured difference can be divided between the two. Section 8.2 returns to this with a concrete case, in which their resource results and the ones reported here have opposite signs and the disagreement resolves into a difference between runtimes rather than between languages.

Arvidsson and Kaller are closer still. They evaluate asynchronous programming frameworks, C++ with Boost.Fibers, Rust with Tokio and Monoio, Go, and Erlang, on 5G control-plane procedures, and report both median and 99th-percentile latency across a load sweep [42]. Their result is the most direct challenge in the literature to the expectation this thesis set out with. Boost.Fibers shows the lowest latency at every load, and while the Rust runtimes track it at the median, their 99th percentile rises more steeply beyond a certain request rate. That is a tail divergence in the right domain, and it would be dishonest to omit it. It is also, however, a comparison in which the concurrency runtime co-varies with the language: the authors attribute the effect to stackless `async/await` permitting a context switch immediately before an I/O operation is initiated, which is a property of Tokio and Monoio rather than of Rust.

That attribution does not have to be taken on trust, because the runtimes have since been examined directly in a real-time setting. Dissecting the execution model of asynchronous Rust against a thread-based C++ equivalent, Škoudlil et al. report that the

runtimes' scheduling policies are "complex and difficult to understand due to the use of abstraction layers" and that Tokio specifically is "designed to maximize throughput rather than time determinism"; restructuring the same application onto prioritised threads over a different runtime yields response times that match classical response-time analysis, which the idiomatic work-stealing configuration does not [36]. The tail behaviour at issue is therefore a property of the runtime, and one that changes when the runtime does. That confound is precisely what the frozen specification and the shared thread-per-shard model in this thesis are designed to remove. The two studies are therefore complementary rather than contradictory: theirs establishes that a tail difference is observable in this domain, and this one is constructed to determine whether the language is what causes it.

Not all of this evidence points the same way, and the thesis takes the qualifying findings seriously rather than citing only the favourable ones.

Even the positive networking result records a real cost. The idiomatic Rust driver issues substantially more instructions per packet than its C reference and is measurably slower in absolute terms, its safety checks being absorbed by out-of-order execution rather than being literally free [45]. The "zero-cost" framing [20] is therefore better read as workload-dependent than as universal. That is precisely why tail behaviour must be measured on the target workload rather than assumed.

The safety story is likewise not absolute. Empirical studies of real-world Rust find that memory- and thread-safety bugs still occur in practice, with blocking bugs and misused synchronisation among them [21, 64]. They also find that a substantial fraction of the ecosystem cannot be fully statically checked, because `unsafe` propagates through dependency chains [33].

These qualifications do not overturn the case for Rust, but they motivate two of the thesis's design choices. The first is measuring predictability and defect prevention directly on the subsystem instead of importing general-purpose, average-case conclusions. The second is keeping the safety-critical core `unsafe`-free, so that the dependency-propagation caveat does not apply to it.

Table 8.2 positions this thesis against the studies closest to it along the three axes on which it claims to differ: whether the comparison controls for design by building both systems from one specification, whether it reports tail behaviour under overload rather than average-case performance, and whether it measures the cost of adoption rather than describing it. Individually, none of the three axes is unoccupied: matched imple-

mentations exist in systems software [45, 75, 76], percentile latency has been reported for 5G control-plane procedures [42], and development effort has been measured for RAN applications [41]. No prior study combines the three for a stateful, concurrent control-plane workload, which is the narrower gap this work occupies.

8.6 Threats to validity and mitigations

Because every conclusion here attributes a difference to the implementation language, the credibility of the study rests on the controls that hold everything else equal and on an honest account of where those controls stop. The threats are grouped below by the kind of inference each endangers, and Table 8.3 maps them first, so that the residual risks can be read as a set rather than recovered one at a time from the prose that follows. Three of them are load-bearing enough to be worth naming here: the study cannot rule out single-author skill asymmetry, its end-to-end instrument is the wrong one for part of what RQ1 and RQ3 ask, and its strongest single finding, the overload divergence, rests on an attribution it did not measure directly. Figure 8.3 grades each row by how much of the threat its control actually removes, which the table's three columns cannot encode: the rows carrying no mitigation at all separate visually from those where something survives only at the margin.

Internal validity: is the language really the only difference? The central threat is implementation bias between the two versions. It is mitigated structurally rather than by assertion, by the controlled-variable strategy of Section 6.1 and Table 5.2: one frozen specification, the same allocator and matched optimisation levels, hash functions pinned on both sides so that neither standard library's default becomes an accidental advantage, divergences the languages force recorded in the forced-idiom log instead of smoothed over, and functional equivalence diffed transcript by transcript rather than assumed (Section 6.7). What remains uncontrolled is single-author skill asymmetry: one person wrote both implementations and is not equally expert in both, so a residual advantage to whichever language that person knows better cannot be excluded. The parity suite bounds this threat for *behaviour* but not for *performance*, and the honest statement is that it is mitigated, not eliminated.

Table 8.2: Positioning against the closest prior work. “Matched pair” means both systems are built from one specification so that the language is the only systematic difference; “tail under overload” means high percentiles are reported past the saturation point rather than average or nominal-load performance; “adoption cost” means the cost of incremental migration is measured rather than surveyed. “Partial” marks percentiles reported across a load sweep that stops at a latency requirement rather than continuing past saturation.

Study	Domain	Matched pair	Overload tail	Adoption cost
Pereira et al. [44]	General benchmarks	no	no	no
Bugden & Alahmar [34]	General benchmarks	no	no	no
Balasubramanian et al. [20]	OS / systems	no	no	no
Emmerich et al. [45]	Packet forwarding	yes	no	no
Li et al. [75]	Network function / IPC	yes	no	no
Narayanan et al. [76]	OS / device drivers	yes	no	no
Mo et al. [59]	Embedded hypervisor	no	no	no
Qazi et al. [70]	Cellular packet core	no	no	no
Qin et al. [21]	Rust ecosystem	n/a	no	no
Evans et al. [33]	Rust ecosystem	n/a	no	no
Fulton et al. [54]	Developer study	n/a	no	surveyed
Karlbäck & Orö [41]	RAN applications	no	no	dev. effort
Arvidsson & Kaller [42]	5G control plane	no	partial	no
This thesis	RAN control plane	yes	yes	measured

Table 8.3: Threats to validity by the kind of inference each endangers, the control applied, and what that control leaves behind. Only the residuals in the last column are live risks; the middle column is what has already been done about each. The one row for which no control was available at all, the missing hardware counters behind the RQ3 result, says so rather than softening it.

Threat	Control applied	Residual risk
Internal: is the language really the only difference?		
Implementation bias between the two versions	Frozen specification, shared allocator and optimisation level, pinned hash functions, forced-idiom log, transcript-level parity testing	Single-author skill asymmetry. Parity bounds behaviour, not performance
Construct: do the metrics measure what the questions ask?		
The sweep measures the design, not the language	The operation is measured apart from the handoff and both are reported (§7.5.2)	One operation on one data structure is narrower than the sweep it supplements
The overload result is attributed but not localised	The candidates are narrowed to two entries already in the forced-idiom log (§7.5.4)	No contended-queue microbenchmark was run, so the mechanism is argued rather than measured
A defect catalogue is a sample the author chose	Two classes drawn from outside the guarantee, with expectations recorded in advance	Seven programs remain seven programs
RQ3 captures resource cost but not its origin	None available on this host	The memory result is established but unexplained
RQ4 measures the seam, not the safety of the whole	The distinction is stated wherever the result is quoted	One address space: a compile-time property of the Rust side bounds nothing at run time
Conclusion: are the differences real or noise?		
Run-to-run non-determinism	20 interleaved repeats, bootstrap intervals, exact paired permutation test, Holm correction, floors quoted	Pairing cancels only what a block shared, so the far tail below saturation stays unresolved
A corrected p-value that does not replicate	The allocator control runs every comparison twice; replication is followed over the p-value	It has cost a result before and would again; here the tail rows are the ones it withholds
A contaminated run defeating a mean-based statistic	The reading rule of §5.5, fixed in advance, plus the per-repeat agreement count	Peak RSS under `jemalloc` is read from its interval against its own p-value
External: how far do the results travel?		
One subsystem, one workload, one machine	RQ2 and RQ4 rest on the compilers and the C ABI rather than on the host	The performance results do not travel, and the overload result travels least of all
A virtualised measurement host	Interleaving, pairing, and 20 repeats; the guest's limits recorded in the data (§7.1)	Frequency unobservable and CPU steal non-zero, neither removable from inside the guest

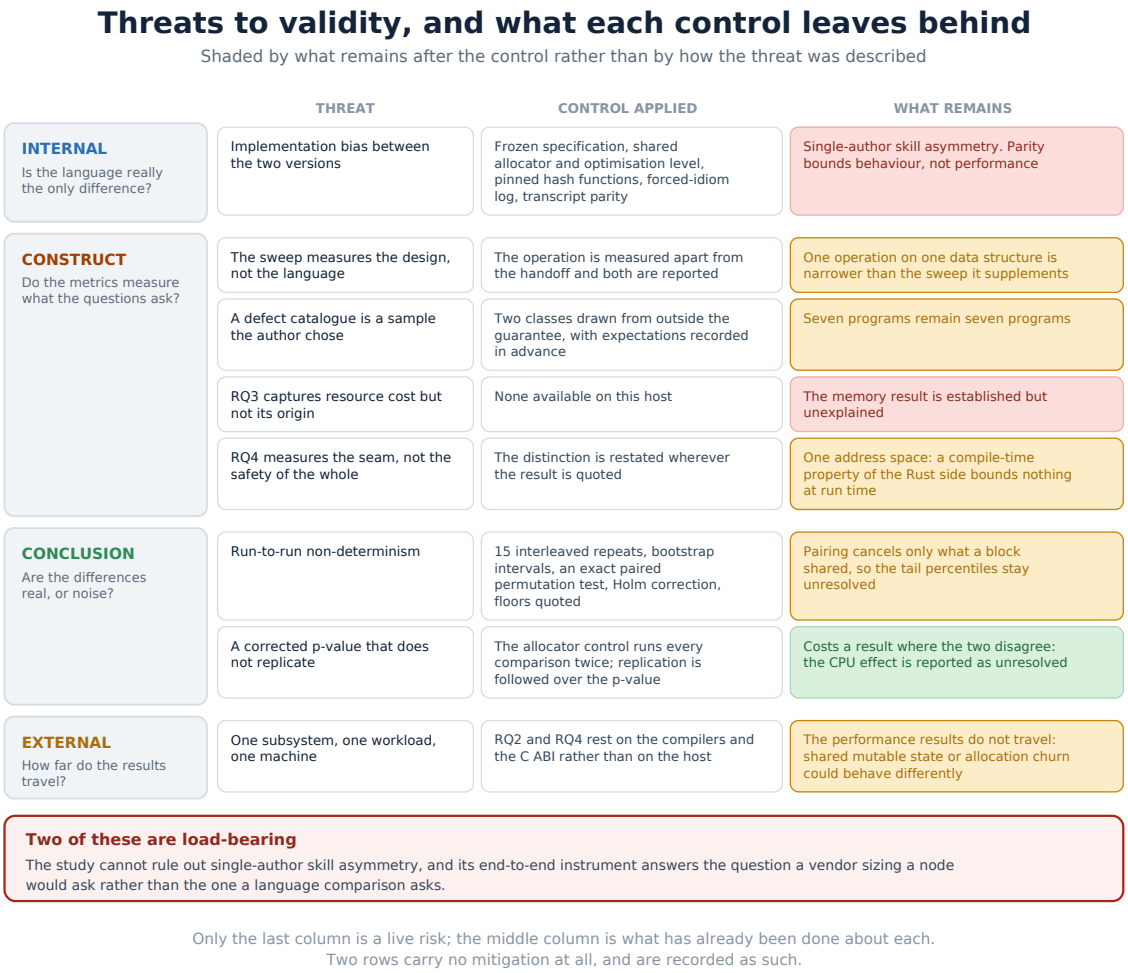


Figure 8.3: Threats to validity, shaded by what remains after the control rather than by how the threat was described. Amber marks a residual risk the control bounds without removing; red one it does not reach at all; green a control that does its job and is paid for. Reading the right-hand column alone gives the live risks as a set, and the red rows are the load-bearing ones: single-author skill asymmetry, the absence of hardware counters, and the unmeasured overload mechanism.

Construct validity: do the metrics measure what the questions ask? RQ1 asks about predictability and is answered with high percentiles, the maximum, and jitter rather than the mean, which is the standard way of measuring it in this literature [6]. The residual risk is that p99.9 on a synthetic workload is not the same quantity as a missed procedure deadline in a live network. A second and larger construct gap in RQ1 was found rather than anticipated. The end-to-end latency the sweep reports is 0.250% operation and the rest cross-thread handoff (Section 7.5.2), so it measures the delivered cost of the design faithfully and the cost of the language barely at all. Which of those RQ1 is asking about depends on the reader: a vendor sizing a node wants the first, a language comparison wants the second. The study now reports both and keeps them apart, but the question as originally worded did not distinguish them.

RQ2 asks which defect classes a language prevents and is answered by a catalogue of seven hand-written programs. A catalogue is a sample, and a sample chosen by the author could flatter either language. Including the two boundary classes is the direct mitigation, since it forces the catalogue to contain cases the expected conclusion does not cover. Seven programs, however, remain seven programs.

RQ3 measures peak RSS and aggregate CPU, which capture resource cost but not its origin. No hardware counters, allocation profiles, or cache statistics were collected, so the study can say that no language-level penalty appeared without being able to say why. The mechanisms most likely responsible have been isolated elsewhere [50, 51], and confirming which of them applies here would need counter-level instrumentation this study did not collect.

RQ4 measures the size of the seam, the `unsafe` it concentrates, and the per-call cost. These are the right quantities for the question as posed. They are not a measure of how safe the composed system is. Because host and component share an address space, a compile-time property of the Rust side does not bound what the C++ side can corrupt at run time [37, 38]. The construct measured here is therefore the cost and reviewability of the boundary rather than the safety of the whole.

Conclusion validity: are the differences real or noise? Every configuration is measured over 20 interleaved repeats and reported as a median with a nonparametric bootstrap 95% confidence interval, in line with established guidance [22]; latency is captured against an intended send schedule so that coordinated omission does not hide the tail [24, 47]; and comparisons are tested rather than left to interval overlap, the

interleaved design being exploited as a matched one (Section 5.5, Table 7.4).

The repeat count is chosen so that the design is not the binding constraint. An exact test on 20 pairs floors at 0.0000019, and at 0.000027 after correcting 14 comparisons; both sit far below the conventional threshold, so a null result here reflects the data rather than the arithmetic. An earlier pass at seven repeats did not have that property, and its corrected floor above 0.05 made every null in the table uninterpretable.

Four limitations remain. First, the measurements were taken inside a virtual machine that does not expose frequency or turbo control and that records non-zero CPU steal (Section 7.1), so the far tail contains a contribution from the hypervisor that cannot be separated from the software under test. This is why the far tail below saturation is reported as unresolved rather than as a null. It interacts with the pairing in a way worth naming: pairing removes the interference that is *common* to a repeat, which is why run-level statistics such as peak RSS, CPU, and sustained throughput come through cleanly, but it cannot remove interference that lands on one run and not its partner, which is exactly the regime the tail percentiles live in. That is also why the overload comparison is made on throughput rather than on latency: a run-level average is the statistic pairing can rescue on a host like this one.

Second, a corrected p-value below 0.05 is necessary here but visibly not sufficient, and the converse also holds. On an earlier campaign, before the two `jemalloc` builds were held to one configuration, CPU utilisation cleared the Holm correction under one allocator and reversed sign under the other; no threshold would have caught that, and it was visible only because the design runs every comparison twice. In the present campaign the control withholds nothing that a threshold would have promoted, but the tail rows still fail it, and the rule is applied to them for the same reason it was applied to CPU before. In the opposite direction, peak resident set size under `jemalloc` has an interval far from zero and 18 of 20 repeats in agreement while its p-value sits near unity, because a handful of runs report a resident set some 30 MiB below their variant's median and a statistic computed from the mean of paired differences cannot survive them. That row is read from its interval, under the rule fixed in Section 5.5 rather than chosen once the disagreement appeared. The bimodality is a defect in the measurement, not in either allocator: it affects both languages in equal numbers and appears at every load level, and diagnosing it, rather than averaging over it, is the honest remedy and is left as instrumentation work.

Third, the end-to-end measurement is the wrong instrument for part of what RQ1

and RQ3 ask, and no amount of statistical care fixes that. Section 7.5.2 shows the quantity a language could affect to be 0.250% of the quantity being measured below saturation, so a null result end to end is close to guaranteed there regardless of the languages. This is a construct limitation rather than a conclusion-validity one, but it belongs here because it is what a reader would otherwise take the null results to mean. Two remedies are used, and they are complementary: the operation is measured apart from the handoff, and the sweep is pushed into the regime where the design stops absorbing the difference. The residual risk is that a microbenchmark of one operation is a narrower object than the sweep it supplements.

Fourth, the strongest finding in the study is the least directly instrumented. The overload divergence itself is measured, replicated across the allocator control, and consistent in every repeat; the *explanation* offered for it is not. Attributing it to the two standard libraries' blocking queue and completion primitives rests on where the producers must be spending their time in that regime and on those two constructs already appearing in the forced-idiom log, which is an argument rather than a measurement. A contended-queue microbenchmark holding the surrounding service fixed would settle it, and until one is run the attribution should be read as the most plausible candidate rather than as a result. What does not depend on it is the finding: the divergence is there whatever causes it.

External validity: how far do the results travel? The study covers one subsystem, one workload, one concurrency structure, and one machine, and each of those bounds the reach of the conclusions differently. The safety and migration results (RQ2, RQ4) generalise most readily, because they are properties of the compilers and of the C ABI rather than of this subsystem: any C++ codebase writing these defect classes gets the same compile-time outcome, and any narrow, scalar-typed interface gets a similarly cheap seam. The performance results generalise least. UECM was deliberately chosen to be small and partitionable, and its shared-nothing design removes lock contention from the hot path; a subsystem with genuinely shared mutable state, deep call graphs, or heavy allocation churn could exercise the two languages' abstraction and allocation costs quite differently. The workload is synthetic and spec-justified rather than replayed from production traffic, so its operation mix is plausible rather than validated. And nothing here speaks to PHY or L2 processing, to the data-heavy user plane, or to the operational concerns (upgrade, observability, multi-tenancy) that a production deployment would raise. The defensible claim is therefore narrow and carries a condition:

for stateful, concurrent, partitionable control-plane services of this kind, and while they are meeting their offered load, the language choice is not a performance decision. Once they are not, it is one, and the overload result is the least transferable finding in the study rather than the most, because it depends on the queue bound, on one request being outstanding per producer, and on two particular standard-library primitives. Extending either claim beyond this class requires rerunning the framework, which is why the framework rather than the numbers is offered as the reusable contribution.

A final limit on reach is temporal rather than structural, and it applies to the comparison itself rather than to any one result: one of the two objects being compared is moving. The C++ side is pinned to C++23 as specified and shipped, and to the compiler versions recorded in Section 7.1, while the standards committee is actively pursuing enforced safety profiles (Section 2.3). This does not qualify anything measured here, because none of that work is standardised or generally implemented, and the concurrency profile that would reach the boundary classes of RQ2 is placed after an initial set that is itself still to be specified [28]. It does mean the RQ2 result should be read as a statement about these languages as they stand rather than as a permanent one. An implemented and enforced lifetime profile would narrow the memory-safety half of the catalogue; an implemented concurrency profile would bear on the half deliberately placed outside Rust's guarantee. This is again a case where the framework is what survives: the catalogue is designed to be re-run against a compiler that enforces profiles, and the answer would then be measured rather than predicted.

8.7 Ethical and professional reflection

The conduct of the study raised no data-protection questions, for the reasons set out in Chapter 5: all traffic is synthetic, the contexts carry fabricated identifiers, and the subsystem is modelled on rather than copied from production code. The *use* of the study is a different matter and deserves stating explicitly, because a language-selection recommendation in this domain is not a neutral technical output.

Two responsibilities follow. The first concerns over-claiming. A vendor could reasonably read the RQ2 result as licence to describe a migrated component as "safe", and the boundary cases are the reason that reading would be wrong: a deadlocked control plane is an outage, and a silently lost counter update can misinform a mobility decision, and neither is prevented by anything demonstrated here. Reporting the boundary as

prominently as the guarantee is a professional obligation, not a stylistic choice, precisely because the audience for this work is making a decision that affects networks people depend on. The second concerns the weight placed on evidence the host cannot fully support. Tail-latency numbers measured inside a virtual machine with non-zero CPU steal are not a sound basis for a production commitment, and saying so in the text is only effective if the qualification travels with the numbers when they are quoted elsewhere; that is part of why the summary table records an evidence status per research question rather than a finding alone. Beyond the immediate results, a migration recommendation also carries organisational consequences, retraining, hiring, and a period of reduced velocity, that fall on people rather than on systems, and the honest framing of the RQ4 finding is that the cost was relocated from run time to human effort rather than removed.

9 Conclusion and Future Work

This thesis compared modern C++ and Rust on a UE Context Management service modelled on a real Nokia gNB-CU-CP component, along four dimensions: tail-latency predictability, compile-time defect prevention, safety-model overhead, and incremental-adoption cost. Built against a single frozen specification and driven through identical code paths, the two implementations let language effects be isolated from design and skill differences.

Three conclusions follow from the evidence. First, on safety the difference is one of *kind*, and it has a definite edge. Safe Rust rejects all five injected memory-safety and data-race classes at compile time, turning runtime hazards that C++ can only hope to catch with sanitiser-instrumented tests into errors that cannot be written at all; but the two boundary classes, a lock-order deadlock and a lost update, compile and misbehave in safe Rust exactly as they do in C++. The guarantee is therefore best described precisely: Rust eliminates data races, not race conditions, and aliasing errors, not lock-ordering errors. Second, on migration the cost is dominated by API design and the learning curve rather than by any run-time tax: an unmodified Rust core can be driven from C++ across a thin, `unsafe`-isolated seam whose cost on a real operation, measured against the same operation called natively, is bounded at 10.91%.

Third, on performance the evidence divides twice before it divides by resource: first by *what is being measured*, and then by *whether the service is coping*. Getting both divisions right was the study's most consequential correction to itself, and each was found rather than anticipated.

The first division is one of instrument. An end-to-end operation in this design is 0.250% the work the language performs and the rest a cross-thread handoff the operating system performs, so an end-to-end null cannot bound a language effect at all: the interval it supports is roughly two orders of magnitude wider than the work it would have to constrain. Measured on the operation alone the languages are not equivalent, and the faster of the two is Rust, by 1.6×, for reasons that trace to the standard library's choice of hash-map layout rather than to anything about safety.

The second division is one of regime, and it is the more consequential of the two. Below saturation the two languages are equivalent at the median, which is a real result about

deployments of this shape, and their far tail remains unresolvable on a virtualised host, which is an admission about the instrument. Past saturation they are not equivalent: the C++ build sustains 1.8× the throughput of the Rust build, in 20/20 interleaved repeats under both allocators, while drawing 1.9× the CPU per completed operation. The likely cause is again a standard library rather than a safety model, this time the blocking queue and completion primitives; confirming that needs a contended-queue microbenchmark this study did not run. On memory the Rust build carries a few MiB more fixed footprint under both allocators, an order of magnitude less than the allocator choice itself costs.

Taken together these support a narrow but decision-grade claim, and it is narrower than the one this thesis expected to make. For stateful, concurrent, partitionable control-plane services of this kind, and *at nominal load*, the choice between modern C++ and Rust is not a performance decision; it is a decision about which classes of defect an organisation wants to make unwriteable, and about whether it is prepared to pay the engineering and learning cost of getting them. Under overload it becomes a performance decision again, and one that currently favours C++, for reasons that lie in the concurrency plumbing rather than in either language's guarantees and that a service required to degrade gracefully cannot treat as incidental. What this thesis adds is therefore not a verdict but an instrument: a frozen specification, a parity-tested pair of implementations, a reproducible measurement framework that reaches past the saturation knee, and a defect catalogue that maps both what the guarantee covers and where it stops. Figure 9.1 collects the three conclusions, the artifacts offered in place of a verdict, and the work that remains.

Several directions extend the work, each of them the direct remedy to a limitation named above. Two are immediate. The first is a contended-queue microbenchmark isolating the bounded queue and the completion handle in both languages, which is what would turn the overload result of Section 7.5.4 from a measured divergence with an argued cause into an explained one. The second is executing the bare-metal campaign, whose target is now much narrower than it was: the repeat count is not the binding constraint, the saturation knee and the overload comparison are already resolved, and what bare metal would buy specifically is the far tail below saturation. Beyond those:

- Replay captured, anonymised production traces in place of the synthetic workload, so that the operation mix is validated rather than merely spec-justified.

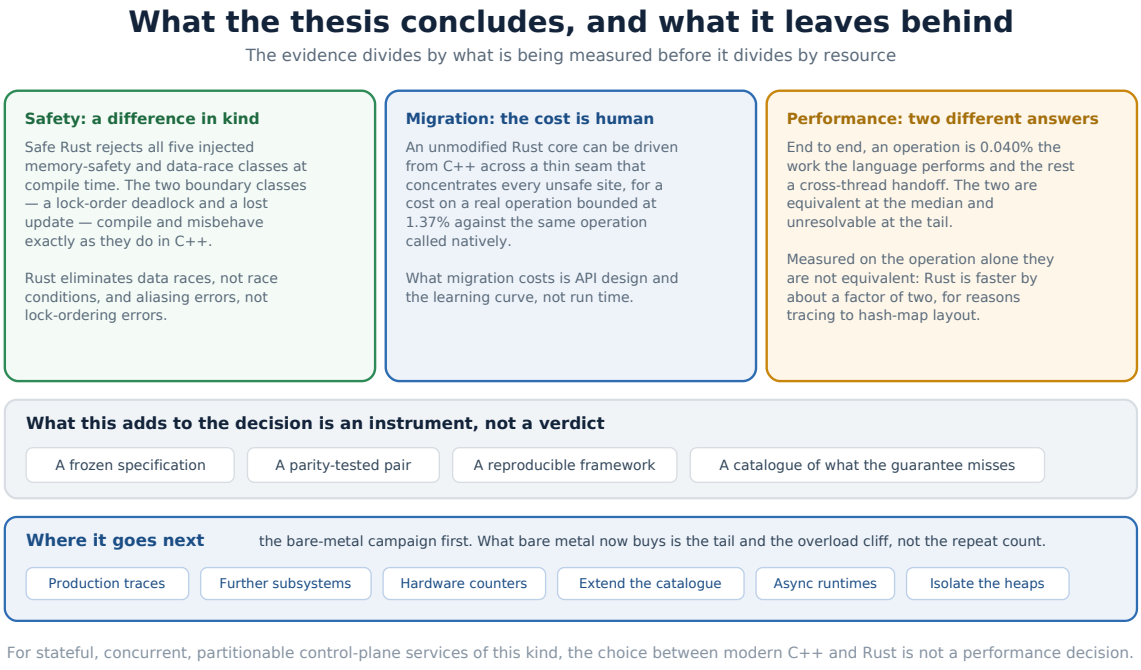


Figure 9.1: What the thesis concludes, and what it leaves behind. The three cards are the conclusions in the order the evidence supports them, the performance card divided twice as the evidence divides it. The grey band is the instrument offered in place of a verdict, the blue band the work that remains, with the contended-queue microbenchmark first because it would explain the largest finding rather than merely confirm it.

- Scale the comparison to further control-plane subsystems, or to the data-heavy L2/PHY path, where shared mutable state and allocation churn would exercise the two languages differently than a shared-nothing design does.

- Instrument the comparison with hardware performance counters and allocation profiles. This is what would turn the memory result of Section 7.5.5 from an observation into an explanation, and what would confirm or refute the cache-locality account offered in Section 7.5.2 for the operation-level gap. It is the clearest instrumentation gap in the study as it stands.

- Extend the fault catalogue further along the boundary mapped here, towards leaks, panics, and misuse of `unsafe`.

- Evaluate Rust async runtimes against the thread-per-shard model used here. That would separate the tail divergence Arvidsson and Kaller attribute to stackless `async/await` [42] from the language itself, under the controls this study imposes.

- Measure what in-process isolation of the two heaps costs at the seam, since that is what an `unsafe`-free core would need to be protected as well as verified [37, 38].

- Track defect density and review effort in a longer-term case study, as Rust is adopted incrementally in a production RAN codebase.

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Appendix A Fault-injection example

As an illustration of the fault-injection study (Chapter 7, RQ2), the use-after-free defect is shown in both languages. In C++ the alias into the context map survives the `erase` that frees it, so the following program compiles cleanly and is flagged only at runtime by AddressSanitizer:

```
std::unordered_map<unsigned long, UeContext> ctxs;
ctxs[1001] = UeContext{1001, 0};
UeContext* p = &ctxs.at(1001);  // alias into the container
ctxs.erase(1001);               // frees the node that p refers to
p->messages_handled += 1;       // use-after-free
```

The analogous safe-Rust program does not compile: the borrow checker forbids mutating the map (`remove`) while the reference `p` into it is still live, so the defect cannot be written in the first place (error `E0499`):

```
let mut ctxs: HashMap<u64, UeContext> = HashMap::new();
ctxs.insert(1001, UeContext { messages_handled: 0 });
let p = ctxs.get_mut(&1001).unwrap(); // mutable borrow starts
ctxs.remove(&1001);                   // borrow while p live -> rejected
p.messages_handled += 1;              // p used after the remove
```